How the radiation field sets the molecular content in MOCASSIN-PDR

The `h2-model` branch: physics, prescriptions, numerical strategy and code locations

Compiled from the source tree of MOCASSIN-PDR

branch `h2-model`, commit `8fc1512` (2026-10-15), worktree `~/software/mocassin_h2`

(document prepared with Claude Code from a reading of the source; no run was made for it)

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Abstract

This document describes, process by process, how the radiation field computed by MOCASSIN's three-dimensional Monte Carlo (MC) photoionisation and dust radiative transfer determines the molecular content of a model in the `h2-model` branch of MOCASSIN-PDR, in which that transfer is coupled to the 33-species REDUCED steady-state chemistry of 3D-PDR and to a 302-level model of the ground electronic state of H_2 whose Solomon photodissociation rate replaces the network's single-line prescription. The photorates are evaluated on each cell's own MC field: its never-scattered part arrives along the straight lines from the sources, its scattered part is distributed over scattering sites by the formal solution of the transfer equation, and the dust attenuation, already in the MC field, is not applied again. The absorption in the H_2 and CO lines, which the MC does not transport, is applied through column densities integrated along those paths, source-ward for the direct beam and through each site for the diffuse light; the level model shields each of its 32 453 lines of the Lyman, Werner, B'–X and D–X systems with Voigt profiles averaged through the cell's own column, with line overlap, trapping and formation pumping. For every step the document gives the physics, the prescription with its reference, the numerical strategy and the file and line of the implementation, in the order in which the code executes them: the MC estimator of the field and its direct and diffuse parts; the dust quantities and the ray geometry; the H_2 level model; the photoreaction rates; the steady-state chemical solve; and the feedback of the molecules on MOCASSIN's opacity, ionisation and thermal balance. Measured numbers from the project's logs size the main approximations: on the H II-region fixture the scattered field carries 75–100% of the FUV field beyond $A_V \approx 0.5$; the H_2 lines and the H/H_2 transition are converged at the default packet count, while the carbon transition and CO are not; and the damping width adopted for the H_2 lines moves the transition by 0.1–0.2 mag. The physical assumptions are collected in Section 11 and the known limitations, including several found while preparing this document, in Section 12. Appendix A is an index of code locations.

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1 Scope, conventions and overview

1.1 What is described

MOCASSIN-PDR couples three pieces of physics:

1. **MOCASSIN** (Ercolano et al., 2003, 2005, 2008): a three-dimensional Monte Carlo (MC) code for photoionisation and dust radiative transfer on a Cartesian grid of homogeneous cells. It owns the radiation field at every frequency, the dust (optical properties, temperatures, charge), the ionisation state of every element, the electron density and the gas temperature.

2. **3D-PDR** (Bisbas et al., 2012), vendored under `source/pdr/` (upstream commit `fabcdc6`, imported 2026-08-28; `source/pdr/UPSTREAM.md`): the REDUCED 33-species chemical network solved to steady state, its photoreaction prescriptions, its H_2 formation rate and its escape-probability level-population solver for the coolants. 3D-PDR's own treatment of the radiation field (an external field attenuated along rays to the cloud boundary) and its own thermal balance are not used. MOCASSIN drives the vendored routines through the bridge module `source/pdr_mod.f90`.

3. **The H_2 level model** of the `h2-model` branch (`source/h2_mod.f90`, driven by `h2LevelModel` in `pdr_mod.f90`): the 302 rovibrational levels of H_2 X in statistical equilibrium under collisions, quadrupole decay, ultraviolet pumping through the excited electronic states, Solomon dissociation and formation on grains. It is a re-implementation from the data and the conventions of Cloudy's large H_2 model (Chatzikos et al., 2023; Ferland et al., 2017; Shaw et al., 2005). Its Solomon rate is the H_2 photodissociation rate the chemistry uses.

By *molecular content* we mean the abundances of the 33 species in every cell and, for H_2, the populations of its 302 levels. The document follows the path by which the MC radiation field determines them, and the feedback of the molecules on the field, the ionisation and the temperatures.

The text follows the source tree at `~/software/mocassin_h2` (git worktree, branch `h2-model`, HEAD `8fc15120dea2`, 2026-10-15). Every `file:line` reference is to that commit; files are under `source/` unless a path says otherwise. The branch has moved on since (its tip was `78a74bd` on 2026-10-16). The only source change on the path described here is D-414 (commit `df9a626`, merged into `h2-model` as `901f1e8`), which makes the mean grain temperature a mean over the size bins that hold grains (Section 4.4); it changes nothing where every grain species fills the size grid, as in all the fixtures. D-417 registers a temperature per grain material for H_2 formation for future development, without a code change. DEVLOG entries are cited as D-nnn (`DEVLOG.md`), branch entries as H-nnn (`DEVLOG-H2.md`) and the limitations of `doc/pdr-status.md` as L-nn. Measured numbers quoted without a log entry are read from the run logs of the two production fixtures: `hiipdr`, an H II region and PDR around a 40 kK star in gas of $n_{\rm H} = 100\,{\rm cm}^{-3}$ with silicate grains on 31^3 cells (`runs/d405-hii31-d0/run.log`), and `pn150k`, a 150 kK planetary nebula with $n_{\rm H} = 10^3\,{\rm cm}^{-3}$ and graphite grains on 41^3 cells (`runs/d405-pn41-d0/run.log`). Both are uniform spheres around one source, computed in one octant (`symmetricXYZ`).

Build flavour. The description is of the ray-traced build,

```
make PDR=1 PDRGEOM=3D mocassin
```

which compiles with the flags

```
-DPDR -DRAYTHEIA -DRAYTHEIA_MO
-DTHERMALBALANCE -DREDUCED -DH2FORM_CT02 -DCHEMSTEADY -DCHEMRETRY
-DFORCECONVERGENCE -DNGACCEL -DNGCYCLE -DNGRELAX -DILLINOIS -DREBRACKET -DSOBOLEV
```

(`Makefile:82-89`): the first line selects the ray-traced geometry, the others are fixed for every PDR build. Of the fixed flags, `-DREDUCED`, `-DH2FORM_CT02`, `-DCHEMSTEADY`, `-DCHEMRETRY` and `-DSOBOLEV` change what the coupled code computes (the network, the H_2 formation rate, the steady-state integration and the escape-probability form); `-DTHERMALBALANCE`, `-DFORCECONVERGENCE`, `-DNGACCEL`, `-DNGCYCLE`, `-DNGRELAX`, `-DILLINOIS` and `-DREBRACKET` act only in 3D-PDR's own iteration and temperature routines, which the bridge does not call. Two other PDR flavours exist and are not described except where noted: the 1D-slab build (`-DONEDIMENSIONAL`), whose chemistry still receives 3D-PDR's prescription $\chi_0\, e^{-\gamma A_V}$ on twelve rays and in which the level model is a stub that leaves the network's H_2 rate in place (`pdr_mod.f90:6015-6027`); and the warm start (`make PDR=1 mocassinWarm`). The level model also skips plane-parallel illumination and nested grids (`pdr_mod.f90:5256-5259`), so the description applies to one grid illuminated by point sources.

1.2 Notation and units

Table 1 lists the symbols used in more than one section. cgs units are used throughout; frequencies are in rydbergs in the code ($1\,\mathrm{Ryd} = 13.6057\,\mathrm{eV}$, $\nu_{1\,\mathrm{Ryd}} = 3.28984\times10^{15}\,\mathrm{Hz}$). J_ν is the angle-averaged specific intensity (the mean intensity), so that $u_\nu = 4\pi J_\nu/c$ is the spectral energy density.

Table 1: Notation and units.

Symbol	Meaning
n_H	density of hydrogen nuclei, `grid%Hden` [$\mathrm{cm^{-3}}$]
$x_i = n_i/n_\mathrm{H}$	fractional abundance of species i, `grid%pdrAbun` [dimensionless]
J_ν	mean intensity [$\mathrm{erg\,s^{-1}\,cm^{-2}\,Hz^{-1}\,sr^{-1}}$]
k, ν_k, $\Delta\nu_k$	index, tabulated frequency and width of a bin of the MC frequency grid (`nuArray`, `widFlx`)
J_k^ste	MC estimator of the *total* field in bin k, $4\pi\int_{\mathrm{bin}\,k} J_\nu\,\mathrm{d}\nu$ [$\mathrm{erg\,s^{-1}\,cm^{-2}}$], `grid%Jste`
J_k^dif	the part of J_k^ste carried by scattered or re-emitted packets, `grid%Jdif` [same units]
u_D	$8.94\times10^{-14}\,\mathrm{erg\,cm^{-3}}$, the 6–13.6 eV energy density of the Draine (1978) interstellar field (`constants_mod.f90:29`); $c\,u_\mathrm{D} = 2.68\times10^{-3}\,\mathrm{erg\,s^{-1}\,cm^{-2}}$
χ	band-integrated (6–13.6 eV) field in Draine units, a dimensionless energy density, $\chi = \sum_{k\in\mathrm{FUV}} J_k^\mathrm{ste}/(c\,u_\mathrm{D})$; $\chi = 1$ is $G_0 = 1.69$ in Habing units of the same band (Section 3.3)
χ_0	geometrically diluted, unattenuated field of the sources, same units (Equation 4)
χ_dir, χ_dif	direct (never scattered or re-emitted) and diffuse parts of χ
s_D	share of the Draine field's 6–13.6 eV energy in the Lyman–Werner (LW) window 11.2–13.6 eV
f_dir, f_dif	LW window factors of the direct and diffuse parts (Section 3.5)
η_{lu}	photon occupation number at the line $l \to u$, $J_\nu c^2/(2h\nu^3)$ [dimensionless]
κ_ν^abs, κ_ν^sca	dust absorption and scattering coefficients [$\mathrm{cm^{-1}}$], `absOpac`, `scaOpac`; κ_V their sum at 5500 Å
A_V	visual extinction [mag]; $1.0857 = 2.5\log_{10} e$ mag per unit optical depth (`magPerTau`)

Symbol	Meaning
$R_{\rm FUV}$	ratio of the band-mean FUV dust extinction coefficient to κ_V, per cell (Section 4)
N_X	column density of species X [$\mathrm{cm^{-2}}$]; N_l the column of H_2 in level l
b	Doppler parameter, $b^2 = 2kT/m_{\rm H_2} + v_{\rm turb}^2$ [$\mathrm{cm\,s^{-1}}$]
$\tau_0,\ a$	line-centre optical depth and Voigt damping parameter of an H_2 line
$S(\tau_0, a)$, $W(\tau_0, a)$	single-line self-shielding function (profile-weighted transmission of a continuum in a Voigt line) and its integral over τ_0 (curve of growth)
$k_{\rm Sol}$	Solomon photodissociation rate per H_2 molecule [$\mathrm{s^{-1}}$], `grid%h2Sol`
$R_{\rm f}$	H_2 grain formation coefficient [$\mathrm{cm^3\,s^{-1}}$]; formation rate per volume $R_{\rm f}\, n_{\rm H}\, n({\rm H})$ [$\mathrm{cm^{-3}\,s^{-1}}$]
ζ	cosmic-ray ionisation rate, `pdrZeta` [$\mathrm{s^{-1}}$]
T_e, T_d	gas (electron) temperature and mean grain temperature [K]; $T_{\rm CMB} = 2.725\,\mathrm{K}$

1.3 Who owns what

The coupling is designed so that every quantity has a single writer (`CLAUDE.md`, "one reality per quantity"). Table 2 lists the quantities that lie on the path from the field to the molecules.

Table 2: Ownership of the quantities on the path from the radiation field to the molecular content.

Quantity	Owner	Written in	Read by
J_ν in every bin (total and diffuse)	MOCASSIN MC	`pathSegment`, `photon_mod.f90:1900-1924, 2236-2250`	everything below
dust opacity, albedo, T_d	MOCASSIN	`dust_mod.f90`, `getDustT` (`update_mod.f90`)	$R_{\rm FUV}$, H_2 formation, level model
T_e	MOCASSIN `thermBalance`	`update_mod.f90:1734`	chemistry, level model
ionisation stages of H, He, C, O, Mg (and all other elements), n_e	MOCASSIN `ionBalance`	`update_mod.f90:2527`	imposed on the chemistry
atom photoionisation and recombination rates of H, He, C, O, Mg; $H_2 \rightarrow H_2^+$	MOCASSIN	`update_mod.f90: 2840-2878`	chemistry (replace network channels)
A_V and the source-ward columns $N(H_2), N(C), N(CO)$	bridge geometry	`computeAVradial`, `pdr_mod.f90:1859`	chemistry
scattering sites and their two-leg columns	bridge geometry	`traceDiffuseSites`, `pdr_mod.f90:2136`	chemistry, level model
H_2 level populations, $k_{\rm Sol}$, H_2 heating	level model	`h2LevelModel`, `pdr_mod.f90:5201`	chemistry, thermal balance
abundances of the 33 species	chemistry	`pdrStep`, `pdr_mod.f90: 3180`	MOCASSIN, level model, columns

Quantity	Owner	Written in	Read by
fraction of each element not in molecules (`elemAvail`)	chemistry	`setElemAvail`, `pdr_mod.f90:5120`	`ionBalance`
$n(H_2)$ for the gas opacity (`H2den`)	chemistry	`pdr_mod.f90:4573`	`addOpacity`
cooling by CO	chemistry (3D-PDR level populations, tabulated in T)	`pdrStep`	`thermBalance`
cooling by C^+, C, O	MOCASSIN's own atoms, with the chemistry's neutral colliders and escape probabilities (D-363)	`emission_mod.f90:1861-1887`	`thermBalance`, line fluxes
heating terms MOCASSIN lacks (`pdrHeatXtra`)	chemistry and level model	`pdr_mod.f90:4499-4546`	`thermBalance`

1.4 Overview of the chain

Figure 1 shows the chain inside one MC iteration. The field reaches the molecules through five channels: (i) directly, through the photodissociation and photoionisation rates, which read the local MC field (Sections 3 and 7); (ii) through the gas temperature, which sets the rate coefficients, the collisional excitation of H_2 and, through the thermal speed and the sticking of H atoms, the H_2 formation rate; (iii) through the dust temperature, which sets the efficiency of H_2 formation on grains (Section 4.4); (iv) through MOCASSIN's ionisation balance, which fixes the C^+/C ratio and the electron density the chemistry works with; and (v) through the molecules themselves, whose columns, integrated along the paths the photons took, shield the H_2 and CO lines at the next iteration (Section 5). The molecules act back on the transfer only through the H_2 photoionisation continuum and, indirectly, through the ionisation and energy balances (Section 9); no molecular line opacity enters the MC transfer.

2 The iteration in which the chain runs

All of the coupling lives inside MOCASSIN's MC iteration, the recursive subroutine `iterateMC` (`iteration_mod.f90:56-3345`), which advances the iteration counter and calls itself until the stop test of Section 10 passes or the maximum number of iterations is reached. The PDR part switches on at iteration `pdrStartIter` (default 1, `common_mod.f90:893`); a larger value lets MOCASSIN first converge an ionisation structure, a gas and a dust temperature without molecules. From then on every iteration executes the steps of Table 3, in that order.

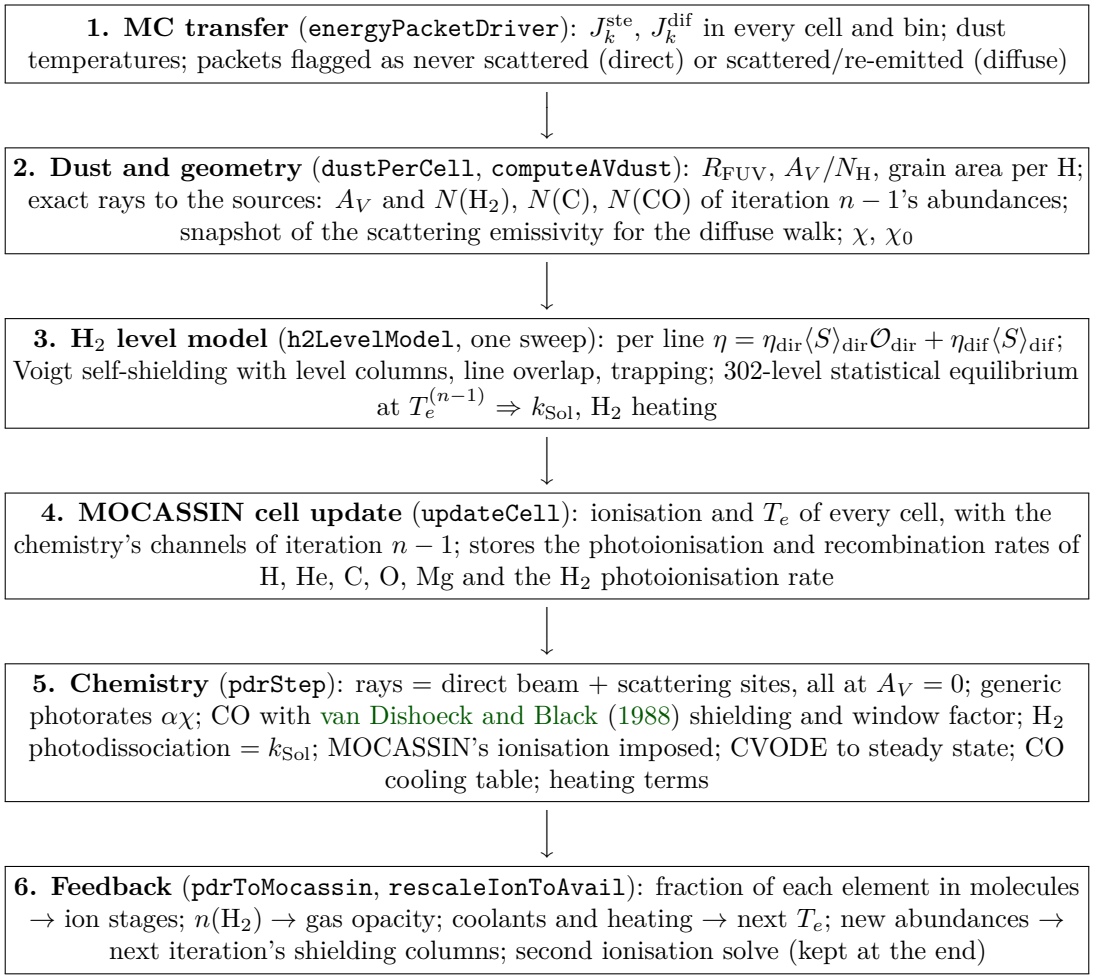


Figure 1: The chain from the radiation field to the molecular content inside one MC iteration n of the `h2-model` branch (3D build), in the order of execution. Inputs marked "iteration $n-1$" or with the superscript $(n-1)$ are one iteration old (Section 2.1); the lags vanish at a fixed point of the iteration. The steps and their line numbers are listed in Table 3.

Table 3: One MC iteration with the PDR coupling active (3D build). Lines are in `iteration_mod.f90` unless stated.

#	Step	Location	What it does
1	MC transfer	`energyPacketDriver`, :611–613 (sources), :655–660 (optional diffuse source)	propagates the energy packets of every source; an absorbed packet is re-emitted on the spot by the local gas or dust and followed until it escapes; accumulates $J^{\rm ste}$ and $J^{\rm dif}$ (Section 3)
2	estimator units	:924, :940	multiplies by 10^{-9} (packet luminosity in $10^{36}\,{\rm erg\,s^{-1}}$, cell volume in $10^{45}\,{\rm cm^3}$, path lengths in cm) and divides by 8 for a `symmetricXYZ` octant
3	dust per cell	`calibratePDRdust` :986 (first time), `dustPerCell` :993	$R_{\rm FUV}$, $A_V/N_{\rm H}$, grain area per H, model-wide fallbacks (Section 4)
4	columns and A_V	`computeAVdust` :995	`snapDiffuseSources` (snapshot for the diffuse walk), `computeAVradial` (A_V and source-ward columns) (Section 5)
5	field summary	`mocassinToPDR` :1023	χ (local) and χ_0 (geometric, unattenuated) per cell
6	seed (first PDR iteration only)	:1027–1029, :1037	`seedMolecules`, then steps 4–5 again on the seeded abundances, then `pdrToMocassin` (Section 8.2)
7	H_2 level model	`h2LevelModel(grid,1)` :1043	one sweep over all cells on this iteration's field (Section 6)
8	MOCASSIN cell update	`updateCell` :1261	ionisation and thermal balance of every cell; stores the atomic photoionisation and recombination rates the chemistry will use
9	chemistry	`pdrStep` :1986, cells dealt round-robin to the ranks (:1980–1986), then `MPI_Allreduce` of abundances, heating, cooling and rates	photorates, steady-state solve, heating and cooling for the thermal balance (Sections 7–8)
10	handback	`pdrToMocassin` :2154	fraction of each element not in molecules (`elemAvail`)
11	second ionisation solve	`ionSecondSolve` called at :2168 (routine :3361–3434); its control pass, before the chemistry, at :1274	`ionBalance` solved again on the channels and molecular content this iteration's chemistry left, then put back (D-404); the result is set aside for step 15
12	damping, hydrogen budget	:2424 (only if `iterDamp`>0; default 0), `rescaleIonToAvail` :2474	ion stages rescaled to `elemAvail` so that $x(\mathrm{H}^0)+x(\mathrm{H}^+)+2x(\mathrm{H}_2)=1$

#	Step	Location	What it does
13	stop test	:3243	integrated-quantity test (`convTotals`, Section 10)
14	next iteration	recursive call of `iterateMC`	the opacity of the next transfer is built from this iteration's ionisation, dust and $n(H_2)$
15	finish	`ionKeepSecond` :3249/:3282 (routine :3455–3521), `h2LevelModel(grid, 12)` :3254/:3283, outputs	the last second solve replaces the first in the written model; level model swept to convergence on the final model; `h2Cells.out`, `lineFluxH2.out`, `h2Pop.bin`

2.1 What each step reads, and from when

The coupling is a fixed-point iteration in which some inputs are one iteration old. Within iteration n:

- the **radiation field** used by steps 3–9 is iteration n's, transported through the opacity built from iteration $n-1$'s ionisation, dust and $n(H_2)$;
- the **shielding columns** (steps 4 and 7) are integrated over the abundances left by iteration $n-1$'s chemistry. The snapshot is taken before any cell of iteration n is solved, so the columns do not depend on the rank count or on the order in which the cells are solved;
- the **level model** (step 7) runs before `updateCell`: it sees iteration $n-1$'s T_e, collider abundances and $n(H_2)$, and iteration n's field; the other cells' level populations in its columns are those of its previous sweep (a Jacobi iteration);
- **MOCASSIN's ionisation balance** (step 8) takes the chemistry's channels to and from molecules, and normalises the ion stages to the molecular content, of iteration $n-1$ (L-47); step 11 measures what the delay costs and, at the end of the run, removes it from the written ions;
- the **chemistry** (step 9) sees iteration n's T_e, n_e, ionisation and atomic rates (step 8), iteration n's field, the step-7 Solomon rates and the columns of step 4;
- the **coolant line escape** in step 9 integrates the optical depth over the level populations of the other cells as stored on the rank; these are iteration $n-1$'s except for cells solved earlier in the same sweep on the same rank, so this one input does depend weakly on the order of the sweep and the rank count;
- the **heating terms** that MOCASSIN takes from the chemistry and the level model (`pdrHeatXtra`) and its **CO cooling table** enter step 8 as step 9 of iteration $n-1$ left them, and the heating terms are held fixed during the temperature search (Section 9.3). The cosmic-ray, turbulent and chemical terms were evaluated at $T_e^{(n-1)}$; the H_2 terms come from the level model of iteration $n-1$, which ran at $T_e^{(n-2)}$.

Inside one iteration the chemistry is solved once per cell: every input of the solve is frozen during the iteration, so repeating it would reproduce the same answer (D-354, which retired the inner coupling loop after finding that a column rescale between passes was the only thing that changed and that it oscillated). The damping across iterations (`iterDamp`) is off by default (D-406): with it, the written model was a blend of solves that trailed the undamped one by three to six iterations and, where a cell has two solutions, wrote states no solve gave (L-49).

At a fixed point of the iteration every lagged input equals its current value, so the lags determine the path to the solution, not the solution. Two things qualify this. First, the MC field is redrawn at every iteration, so the fixed point is reached only in a statistical sense: per-cell quantities fluctuate about it with the MC noise of Section 10.3, and because the photorates are non-linear in the field and the shielding is non-linear in the columns, the noise can bias cell means (L-26, D-408). Second, the coupled system has cells with two stable states (the carbon ionisation turn, D-405), in which the lagged inputs decide which state a cell is in at a given iteration.

3 The Monte Carlo radiation field

3.1 Transport and the path-length estimator

MOCASSIN transports the radiation of each source s of luminosity L_s as N_s energy packets of equal luminosity $\dot{\varepsilon}_s = L_s/N_s$ (`deltaE`; Ercolano et al. 2003). A packet's frequency bin is drawn from the source's cumulative emitted-energy distribution (`inSpectrumProbDen`; an energy, not a photon-number, cumulative, as packets of equal energy require). The packet travels in a straight line until it interacts, the optical depth to the interaction, $-\ln(1-\mathcal{R})$ with $\mathcal{R}$ uniform in $[0,1)$, being accumulated with the total opacity of the cells it crosses: the gas continua (bound–free absorption by every ion present, including C I and the other neutral metals in the FUV, the H_2 photoionisation continuum above 15.4 eV, free–free) plus dust absorption and scattering. At an interaction the packet is

- scattered by dust with probability $\kappa_\nu^{\rm sca}/\kappa_\nu$, with κ_ν the total opacity (`photon_mod.f90:2025-2035`): coherently, keeping its frequency, into a new direction drawn from the Henyey and Greenstein (1941) phase function with the bin's asymmetry parameter g_ν (`hg`, `:2154-2165`), or isotropically if isotropic scattering is selected;
- otherwise absorbed, by the dust with probability $\kappa_\nu^{\rm abs}/(\kappa_\nu - \kappa_\nu^{\rm sca})$ and by the gas otherwise (D-233; `:2066-2103`), and re-emitted on the spot as a new packet of the same energy at a frequency drawn from the local dust (thermal) or gas (hydrogen and helium diffuse continua and lines; L-51) emissivity.

Packets carry a weight w (`enPacket%weight`) that is unity unless the optional composite path-length biasing of Baes et al. (2016) is switched on (`pathStretch`, off by default, `common_mod.f90:1099-1103`).

The mean intensity is measured with the path-length estimator of Lucy (1999, eq. 12). A packet of luminosity $\dot{\varepsilon}_s$ that travels a length ℓ inside a cell of volume V adds $\dot{\varepsilon}_s \ell/c$ to the time-averaged radiant energy in the cell, and $u = 4\pi J/c$, so the integral of the mean intensity over bin k is

$$J_k^{\rm ste} \equiv 4\pi \int_{{\rm bin}\,k} J_\nu \,{\rm d}\nu = \frac{1}{V} \sum_{{\rm segments}\in k} \dot{\varepsilon}_s \, w \, \ell \qquad [{\rm erg\,s^{-1}\,cm^{-2}}], \tag{1}$$

the sum running over every path segment of every packet (stellar or not, of every source) in the cell and in the bin, whether the segment ends at the cell's face (`photon_mod.f90:2236-2250`) or at an interaction inside it (`:1900-1924`). $J_k^{\rm ste}$ is a sum over the bin, *not* a quantity per unit frequency; the band energy density is $u_{\rm band} = \sum_k J_k^{\rm ste}/c$. The accumulation is in MOCASSIN's internal units (packet luminosity in $10^{36}\,{\rm erg\,s^{-1}}$, cell volume in $10^{45}\,{\rm cm^3}$, lengths in cm); the factor 10^{-9} that converts to cgs, and the division by 8 for a `symmetricXYZ` octant (into which every packet of a single central source is launched), are applied after the transport (`iteration_mod.f90:924`, `:940`). The default frequency grid has `nbins` = 600 bins (`set_input_mod.f90:77`). The estimator is unbiased and its relative variance in a cell and bin falls as the inverse of the number of segments recorded there, so it degrades with depth as the packets thin out (Section 10.3).

3.2 Direct and diffuse parts

A packet leaves a source with `lgStellar` = `.true.`; it is re-initialised as non-stellar whenever it is scattered (`photon_mod.f90:2154`) or re-emitted. In a PDR run every packet adds to $J^{\rm ste}$, and the non-stellar ones are mirrored into $J^{\rm dif}$ (`lgJdifLive` is forced true for any PDR run, `set_input_mod.f90:775-778`; accumulation at `photon_mod.f90:1920-1922` and `:2247-2249`). Hence

$$\begin{aligned} J_k^{\rm ste} &= \text{total field}, \\ J_k^{\rm dif} &= \text{scattered} + \text{re-emitted}, \\ J_k^{\rm ste} - J_k^{\rm dif} &= \text{the never-scattered source photons}. \end{aligned} \tag{2}$$

The last is the direct beam of the sources, averaged over the cell and attenuated along its path by dust absorption *and* scattering and by the gas continua; it arrives at the cell along the straight line from each source. In the 6–13.6 eV band the diffuse part is dust-scattered starlight of any order of scattering plus the gas emission that falls in the band (the two-photon continuum of hydrogen and the line packets not destroyed by dust, D-376); thermal dust emission contributes nothing there. Two properties of the split matter for what follows. First, the diffuse part dominates the PDR: on `hiipdr` the diffuse fraction $\chi_{\rm dif}/\chi$ rises from 0.35 at $A_V < 0.2$ to 0.75 at $A_V = 0.5$–0.8, 0.93 at 1.2–1.6 and above 0.99 beyond $A_V = 2.1$ (D-344), so the deep PDR sees almost no direct starlight. Second, because scattering keeps photons in the cloud, the total field decays with depth much more slowly than pure extinction: on the `V1` slab the transported FUV field fell by $e^{-1.86}$ per magnitude of A_V, against $e^{-3.02}$ for absorption and scattering removing the photons alike (`pdr_mod.f90:1836-1844`, D-086, D-087). (With the debug switch `lgDebug` the diffuse packets go into $J^{\rm dif}$ alone; every consumer adds them back in that case.)

3.3 Bands and the Draine normalisation

- **FUV band**: $\nu \in [0.44066, 1.0]$ Ryd, i.e. 6.0–13.6 eV (`constants_mod.f90:20-21`). The bins inside it, `FUVlowP`…`FUVhighP`, run from the first bin with $\nu_k > 0.44066$ Ryd to the last with $\nu_k \le 1$ Ryd, located once at start-up (`continuum_mod.f90:92-94`). On the fixtures the band is 37 (`hiipdr`) and 39 (`pn150k`) of the 600 bins, starting at $\nu_k = 6.17$ and 6.02 eV; the bins are $\Delta\nu/\nu \approx 2\%$ wide. (The comment at `constants_mod.f90:18-20` calls the band 912–2400 Å; the value 0.44066 Ryd is 2068 Å, i.e. 6.0 eV, which is the band the code uses.)

- **Lyman–Werner (LW) window**: $\nu_k \in [0.8232, 1.0)$ Ryd, 11.2–13.6 eV (`constants_mod.f90:25`), about 9 bins: the range in which H_2 absorbs in its Lyman and Werner bands and is photodissociated, in which CO is photodissociated, and above whose lower edge (11.26 eV) C I is photoionised.

- V **band**: the bin containing 5500 Å (`VbandP`, `constants_mod.f90:32`), used to define A_V.

- **Draine unit**: $u_{\rm D} = 8.94 \times 10^{-14}\,{\rm erg\,cm^{-3}}$ (`constants_mod.f90:29`), the 6–13.6 eV energy density $(4\pi/c)\int E\,F_{\rm D}(E)\,{\rm d}E$ (with E converted to erg in the integrand) of the interstellar field of Draine (1978), whose photon intensity $F_{\rm D}$ is Equation (7); integrating the polynomial gives 8.940×10^{-14}, and the code's own quadrature on its bins 8.938×10^{-14} (`pdr_mod.f90:2280-2286`). In photon units the same field carries $4\pi \int F_{\rm D}\,{\rm d}E = 1.94 \times 10^8$ photons cm^{-2} s^{-1} over 6–13.6 eV and 2.06×10^7 in the LW window.

- **Relation to Habing units.** The Habing (1968) field is usually normalised by its 6–13.6 eV flux, $H_0 = 1.6 \times 10^{-3}\,{\rm erg\,s^{-1}\,cm^{-2}}$ (as in Bisbas et al., 2015), an energy density $H_0/c = 5.3 \times 10^{-14}\,{\rm erg\,cm^{-3}}$. In that band $\chi = 1$ is $G_0 = c\,u_{\rm D}/H_0 = 1.68$ (1.69 with the value $5.29 \times 10^{-14}\,{\rm erg\,cm^{-3}}$ quoted at `constants_mod.f90:26-28`); 3D-PDR's heating routine converts with 1.68 (`pdr/heatingfunctions.F90:57`), in terms this coupling does not use (Section 9.3). The factor 1.71 compares the two fields at 1000 Å instead: Habing (1968) gives

$\lambda u_\lambda = 4\times10^{-14}\,\mathrm{erg\,cm^{-3}}$ there, Equation (7) gives 6.84×10^{-14}. Draine and Bertoldi (1996) normalise their field parameter (also written χ) to Habing's at 1000 Å, so that the Draine field has $\chi_{\rm DB} = 1.71$; the PDR code comparison adopts $\chi = 1.71\,G_0$ (Röllig et al., 2007, their Table 5), and TORUS-3DPDR divides its MC field by $1.71H_0$ (Bisbas et al., 2015, their Eq. 12), a Draine unit 2.1% larger than $c\,u_{\rm D}$. A band of 912–2400 Å, as in TORUS-3DPDR, holds 13% more of a Draine-shaped field than 6–13.6 eV. As in the code comparison, χ here is a 4π average (an energy density), whatever the angular distribution of the field.

3.4 Band-integrated field strengths

In `mocassinToPDR` (`pdr_mod.f90:2538-2677`):

$$\chi = \frac{1}{c\,u_{\rm D}} \sum_{k=\mathrm{FUVlowP}}^{\mathrm{FUVhighP}} J_k^{\rm ste}, \qquad \text{(local, attenuated; } \texttt{chiFUV}\text{, :2594–2600)} \tag{3}$$

$$\chi_0 = \sum_s \frac{L_{\mathrm{FUV},s}}{4\pi r_s^2\, c\,u_{\rm D}}, \qquad \text{(geometric, unattenuated; } \texttt{chiFUV0}\text{, :2604–2616)} \tag{4}$$

where $L_{\mathrm{FUV},s}$ is the luminosity of source s in the FUV bins, taken from the same cumulative distribution the packets are drawn from (:2558–2562), and r_s the distance of the cell centre from the source. χ_0 is the energy density of the undimmed beams, $u = F/c$ for a beam of flux F; it is used only by the 1D build and by diagnostics. χ is passed to 3D-PDR's heating routine, where only the photoelectric terms read it, and those are not used (Section 9.3); it also enters the seed (Section 8.2). The photorates use the split of Section 3.2, evaluated per cell in `pdrStep` (`pdr_mod.f90:3401-3417`):

$$\chi_{\rm dir} = \frac{1}{c\,u_{\rm D}} \sum_{k\in\mathrm{FUV}} (J_k^{\rm ste} - J_k^{\rm dif}), \qquad \chi_{\rm dif} = \frac{1}{c\,u_{\rm D}} \sum_{k\in\mathrm{FUV}} J_k^{\rm dif}, \tag{5}$$

so that $\chi_{\rm dir} + \chi_{\rm dif} = \chi$; $\chi_{\rm dir}$ is clipped at zero against the rounding of the two single-precision sums.

3.5 Lyman–Werner window factors

3D-PDR's photorate coefficients α are rates per unit Draine field, computed for the Draine spectral shape, and are applied to the 6–13.6 eV band strength. For the four reactions driven only in the LW window (H_2, HD and CO photodissociation, C I photoionisation) a field whose shape differs from Draine's in the band drives them at a different rate for the same χ. The bridge therefore multiplies each such rate by the ratio of the window's share of the band energy in the MC field to the same share in the Draine field, separately for the direct and the diffuse part (D-349, D-350):

$$f_{\rm dir} = \frac{1}{s_{\rm D}} \frac{\sum_{k\in\mathrm{LW}}(J_k^{\rm ste} - J_k^{\rm dif})}{\sum_{k\in\mathrm{FUV}}(J_k^{\rm ste} - J_k^{\rm dif})}, \qquad f_{\rm dif} = \frac{1}{s_{\rm D}} \frac{\sum_{k\in\mathrm{LW}} J_k^{\rm dif}}{\sum_{k\in\mathrm{FUV}} J_k^{\rm dif}} \tag{6}$$

(`pdr_mod.f90:3418-3420`). Here $s_{\rm D}$ is the LW share of the Draine field's 6–13.6 eV energy, computed once by `draineWindowShare` (`pdr_mod.f90:2288-2314`) from the photon intensity of Draine (1978, eq. 11, given there as valid over 5–13.6 eV)

$$F_{\rm D}(E) = 1.658\times10^{6}E - 2.152\times10^{5}E^2 + 6.919\times10^{3}E^3 \quad [\mathrm{photons\,cm^{-2}\,s^{-1}\,sr^{-1}\,eV^{-1}}], \quad E \text{ in eV}, \tag{7}$$

weighted by $E\,\Delta E_k$ on MOCASSIN's own bins, with the same bin membership as the MC sums. The continuous band gives $s_{\rm D} = 0.148$; on the bins it is 0.155 (`hiipdr`) and 0.164 (`pn150k`), printed as `[D350]` in the run logs. Using the binned value makes $f = 1$ exactly for an MC field of Draine shape, whatever the bins. The correction rests on three approximations. (i) It rescales the energy in the window but not the spectral shape inside it, which matters because the window is emptied

from its blue end first (dust extinction rises towards 912 Å and C I absorbs above 11.26 eV). (ii) It is a ratio of energy shares, whereas the rates count photons; the two differ by the ratio of the mean photon energies in the window, which lie between 11.2 and 13.6 eV, so by at most ∼20%, and much less where the shape in the window is close to Draine's. (iii) One factor serves all the sites of the diffuse field (Section 5.4). The size of what it corrects is large: the MC field's window share falls from 0.299 at $A_V = 0.07$ to 0.152, 0.094, 0.041 and 0.008 at $A_V = 0.74$, 1.13, 1.70 and 2.50 on `hiipdr` (D-349), so f runs from 1.9 near the star to 0.05 deep in the PDR. The factors are handed to the vendored code as the per-ray array `ray_lw_factor` (`pdr/calc_reac_rates.F90:64-74`; a local modification listed in `pdr/UPSTREAM.md`). In the `h2-model` branch only the CO rate actually uses them: the H_2 rate is replaced by the level model, C photoionisation by MOCASSIN's own rate, and the network has no HD (Section 7). The other photoreactions keep the Draine shape (Section 7.2).

3.6 Specific intensity and occupation number at each H_2 line

The level model needs the field at each of the 32 453 pumping lines rather than a band integral. In `solveCell` (`pdr_mod.f90:5670-5695`) each bin's direct and diffuse mean intensities are

$$J^{\rm dir}_{\nu,k} = \frac{J^{\rm ste}_k - J^{\rm dif}_k}{4\pi\,\Delta\nu_k}, \qquad J^{\rm dif}_{\nu,k} = \frac{J^{\rm dif}_k}{4\pi\,\Delta\nu_k}, \qquad \Delta\nu_k = \mathrm{widFlx}_k \times 3.28984 \times 10^{15}\,\mathrm{Hz}, \tag{8}$$

in $\mathrm{erg\,s^{-1}\,cm^{-2}\,Hz^{-1}\,sr^{-1}}$, and every line lu of wavenumber $\tilde\nu_{lu}$ takes the bin that contains it, $\nu_k \le \nu_{lu} < \nu_{k+1}$ (`tBin`, `:5310-5320`, the rule MOCASSIN uses to place its own line packets). The photon occupation numbers are

$$\eta^{\rm dir}_{lu} = J^{\rm dir}_{\nu,k(lu)}\,\frac{c^2}{2h\nu^3_{lu}}, \qquad \eta^{\rm dif}_{lu} = J^{\rm dif}_{\nu,k(lu)}\,\frac{c^2}{2h\nu^3_{lu}}, \qquad \nu_{lu} = c\,\tilde\nu_{lu}. \tag{9}$$

Lines outside the frequency grid get $\eta = 0$. Lines from vibrationally excited levels that lie below 11.2 eV read their own bins as well, inside or outside the 6–13.6 eV band. The field at a line is the bin average of the continuum, 2% wide: the MC field contains no H_2 (or H I Lyman-series) line opacity, and the line shielding is applied afterwards through column densities (Section 6). This separation of continuum and line attenuation is exact for a line whose absorption does not change the continuum it sits on. It is not exact for the continuum itself: photons that H_2 lines absorb remain in the MC field, where they go on heating the dust, ejecting photoelectrons, photoionising C I and being scattered. The fraction of the band so affected is at most the window's share and falls to the line cores once the lines saturate; its effect has not been measured.

4 Dust quantities seen by the chemistry and the level model

MOCASSIN computes the dust absorption and scattering coefficients $\kappa^{\rm abs}_\nu$, $\kappa^{\rm sca}_\nu$ [$\mathrm{cm^{-1}}$] and the asymmetry parameter g_ν of every cell from the grain species and the size distribution of the input (Ercolano et al., 2005): per-grain cross sections from Mie theory on optical constants (Bohren and Huffman, 1983) or from tabulated efficiencies, weighted by each species' number abundance and the size distribution and multiplied by the grain number density N_d that follows from the dust-to-gas mass ratio (`doc/dust.md`). It also solves the radiative-equilibrium temperature of every species and size in every cell from the local MC field. The bridge derives from these, per cell, the quantities below. None is an input parameter: they follow the model's own grains (`CLAUDE.md`, "no hardcoded values"). Table 4 gives their values on the two fixtures.

Table 4: Dust quantities measured on the two production fixtures (run logs, lines `[D073]`, `[D109]`, `[D195]`; values at the calibration cell, uniform across each model).

Quantity	`hiipdr` (silicate)	`pn150k` (graphite)
$A_V/N_{\rm H}$ [mag cm^2]	5.18×10^{-22}	3.58×10^{-22}
$N_{\rm H}/A_V$ [cm^{-2} mag^{-1}]	1.93×10^{21}	2.79×10^{21}
$R_{\rm FUV}$	4.31	2.98
FUV albedo, energy-weighted over 6–13.6 eV	0.45	0.37
FUV asymmetry parameter g, scattering-weighted	0.61	0.29
grain area per H, silicate / carbonaceous [cm^2]	1.02×10^{-21} / 0	0 / 5.45×10^{-22}

4.1 FUV-to-visual extinction ratio and A_V per H

`dustPerCell` (`pdr_mod.f90:1363-1436`) computes

$$\kappa_V=\kappa_V^{\rm abs}+\kappa_V^{\rm sca},\qquad \kappa_{\rm FUV}=\frac{\sum_{k\in{\rm FUV}}w_k\,(\kappa_k^{\rm abs}+\kappa_k^{\rm sca})}{\sum_{k\in{\rm FUV}}w_k},$$
$$R_{\rm FUV}=\frac{\kappa_{\rm FUV}}{\kappa_V},\qquad \frac{A_V}{N_{\rm H}}=\frac{1.0857\,\kappa_V}{n_{\rm H}}\ [{\rm mag\,cm^2}], \tag{10}$$

stored as `grid%dustRfuv` and `grid%dustAVfac`. The weight w_k is the fraction of source 1's emitted energy in bin k (the increment of its cumulative distribution), so that $\kappa_{\rm FUV}$ is the mean extinction coefficient of the band-integrated field as the source emits it; a mean weighted by bin width would over-weight the 912 Å end of the band, where the extinction is steepest. The extinction includes scattering, as in the definition of A_V, so $e^{-R_{\rm FUV}\tau_V}$ is the transmission of the source's FUV beam with every scattered photon counted as lost.

Two approximations limit what $R_{\rm FUV}$ represents. It is weighted by the spectrum of source 1 for every source, and by the spectrum as emitted, not as it arrives: with depth the field is depleted first at the blue end of the band (D-349), so the band-mean extinction of the local field is lower than $R_{\rm FUV}\kappa_V$ and $R_{\rm FUV}$ overstates it increasingly with depth. In the 3D build $R_{\rm FUV}$ does not enter any photorate. It enters the combination of the A_V of several sources (Equation 16, where it cancels for a single source), the grey extinction of the diffuse-site weights (Section 5.4), the fallback field $\chi\,e^{-R_{\rm FUV}A_V/1.0857}$ of a cell the MC field did not reach (`pdr_mod.f90:3326`), and the seed.

`calibratePDRdust` (`pdr_mod.f90:1439-1662`) runs once, at the first PDR iteration. From the cell whose $A_V/N_{\rm H}$ is closest to the density-weighted mean it derives the model-wide fallbacks for cells without dust, `pdrAVfac` and `pdrRfuv`, and 3D-PDR's `UV_fac` $=R_{\rm FUV}/1.0857$, which the build does not read (it is used only under `GRAINRECOMB`, which is not compiled; `pdr_mod.f90:3308-3310`). It reports the energy-weighted FUV albedo and the scattering-weighted asymmetry parameter (Table 4), and (D-108) overwrites the extinction curve that 3D-PDR's `XLAMBDA` splines (`X_GRID`, the Savage and Mathis (1979) table) with this model's $\kappa(\lambda)/\kappa_V$ at the bin nearest each tabulated wavelength.

3D-PDR's grain attenuation factor. `SCATTER` (`pdr/shield.F90:449-497`) is the fit of Wagenblast and Hartquist (1989, their eqs. 1a–1d) to the attenuation of an FUV field incident on one side of a plane-parallel slab of absorbing and scattering grains, which they derived with the spherical-harmonics method of Flannery et al. (1980) for grains of albedo 0.3 and asymmetry parameter 0.8:

$$f_g(\tau_\lambda)=\begin{cases}e^{-0.7514\,\tau_\lambda}, & \tau_\lambda<1,\\ \sum_{i=1}^{5}a_i\,e^{-k_i\tau_\lambda}, & \tau_\lambda\ge1,\end{cases}\qquad \tau_\lambda=\frac{A_V}{1.086}\,X(\lambda), \tag{11}$$

with $(a_i) = (2.006, -1.438, 0.7364, -0.5076, -0.0592)$, $(k_i) = (0.8490, 1.013, 1.282, 2.005, 5.832)$ and $X(\lambda) = \tau_\lambda/\tau_V$ the extinction curve splined by `XLAMBDA`; the two branches agree at $\tau_\lambda = 1$ to 2×10^{-4}. The field thus falls as $e^{-0.75\tau_\lambda}$ to $e^{-0.85\tau_\lambda}$ rather than $e^{-\tau_\lambda}$, because forward scattering keeps photons in the beam. The MC transfer does this exactly, with the model's own albedo and phase function, which differ from 0.3 and 0.8 (Table 4). In the 3D build the photorates are evaluated at $A_V = 0$ (Section 7.1), where $f_g = 1$, so `SCATTER` matters only for the 1D build, the seed (Section 8.2) and diagnostics.

4.2 Grain area per H nucleus for H_2 formation

The formation rate of Cazaux and Tielens (2002, 2004) (Section 7.8) is linear in the geometric grain cross section per H nucleus of each material, for which 3D-PDR ships diffuse-ISM constants (8.473×10^{-22} and $7.908 \times 10^{-22}\,\mathrm{cm}^2$ per H for silicate and graphite, `pdr/h2_form.F90:37-39`). `measureH2grainArea` (`pdr_mod.f90:1266-1338`) computes instead, once, the mean geometric cross section per grain of each dust component, split by material,

$$\bar{\sigma}_{\mathrm{mat}} = \sum_{s \in \mathrm{mat}} X_s \sum_{a=a_{\mathrm{min},s}}^{a_{\mathrm{max},s}} \pi a^2 \, w(a) \quad [\mathrm{cm}^2], \tag{12}$$

with X_s the number abundance of species s within the component and $w(a) = n(a)\Delta a / \sum n(a)\Delta a$ the number fraction of grains in size bin a, normalised over the whole size grid (`ph_mod.f90:999-1012`) and summed over the species' own size range. This is the same weighting from which the opacity is built, so the area and the extinction describe the same grains. Species are classified by their label (`isCarbonGrain`, `isSilicateGrain`, `pdr_mod.f90:1228-1247, 1341-1360`): carbonaceous if it contains `graph`, `grph`, `amc`, `carb`, `pah`, `sic`, `gr1` or `gr2`, silicate if it contains `sil`, `oliv`, `pyr`, `enst`, `forst`, `quartz` or `oss`; an unrecognised label goes to the silicate channel with a warning. `dustPerCell` then stores per cell $\sigma_{\mathrm{H}}^{\mathrm{sil}} = (N_d/n_{\mathrm{H}})\,\bar{\sigma}_{\mathrm{sil}}$ and $\sigma_{\mathrm{H}}^{\mathrm{gra}} = (N_d/n_{\mathrm{H}})\,\bar{\sigma}_{\mathrm{gra}}$ [cm^2 per H] (`grid%dustSigSil`, `grid%dustSigGra`); a dust-free cell gets zero, not the upstream constant (D-195). On the fixtures the area is 1.2 times the upstream silicate constant on `hiipdr` and 0.69 times the graphite one on `pn150k`, each with no grains of the other material (Table 4); the formation rate follows linearly. The geometric area is the right weight for the collision rate of H atoms with grains; the efficiency per unit area is applied at one mean grain temperature (Section 4.4).

4.3 Grain albedo in the Lyman–Werner band

Cosmic-ray-induced photoreactions are divided by $1-\omega$ (Section 7.6), with ω the grain albedo at the wavelengths of the H_2 Lyman–Werner emission that cosmic rays excite: a photon is then scattered on average $1/(1-\omega)$ times before a grain absorbs it, and can photodissociate or photoionise at each step. Instead of 3D-PDR's constant $\omega = 0.42$ (`pdr_mod.f90:354`), each cell uses (`grainAlbedoLW`, `update_mod.f90:4658-4673`; handed over at `pdr_mod.f90:3691`, D-280)

$$\omega = \frac{\sum_{k \in \mathrm{LW}} \kappa_k^{\mathrm{sca}} \, \Delta\nu_k}{\sum_{k \in \mathrm{LW}} (\kappa_k^{\mathrm{abs}} + \kappa_k^{\mathrm{sca}}) \, \Delta\nu_k}, \tag{13}$$

an average flat in frequency over the window. MOCASSIN's own cosmic-ray-induced ionisation of the linked neutrals uses the same ω (`update_mod.f90:2916-2930`).

4.4 Dust temperature

MOCASSIN solves the radiative-equilibrium temperature $T_d(s, a)$ of every species s and size a in every cell from the local MC field, by equating the absorbed power (plus the heating by resonance lines and gas collisions, less the photoelectric losses) to $4\pi \int C_{\mathrm{abs}} B_\nu(T_d)\,\mathrm{d}\nu$; stochastic heating is off

by default (keyword quantumHeatGrain; L-8). The chemistry and the level model use one mean temperature per cell,

$$T_d \equiv \texttt{Tdust(0,0,cell)} = \sum_s X_s \sum_{a=a_{\min,s}}^{a_{\max,s}} w(a)\, T_d(s,a) \tag{14}$$

(update_mod.f90:4546-4555), floored at $T_{\rm CMB} = 2.725\,\mathrm{K}$ because MOCASSIN transports no cosmic microwave background (pdr_mod.f90:3338; L-30). Three properties of this mean matter for the molecules:

- it is weighted by grain *number*, which favours the smallest grains, whereas the formation rate is a sum over grains weighted by *area*;
- the size weights are those of the whole size grid, not renormalised to a species' size range, so a grid wider than the species' range enters with zero-temperature bins: on test_Problem's silicates (0.07–5 µm on a 10^{-4}–5 µm grid) the mean came out at (2–8) $\times\, 10^{-6}$ K and no H_2 formed anywhere (H-035). After 8fc1512 the user decided this: D-414 renormalises each species' mean to its own bins and weights the species by the grains they have, which is exact (byte-identical outputs) where a species fills the grid;
- one efficiency $\epsilon(T_d)$ is applied to the area of all sizes, where the grains have a spread of temperatures; since ϵ is non-linear, $\epsilon(\langle T_d\rangle) \neq \langle\epsilon(T_d)\rangle$.

The temperature is the second route by which the field acts on H_2, and a steep one. Evaluating the efficiency of Section 7.8 with 3D-PDR's surface energies: on silicate $\epsilon \approx 1$ for $6 \lesssim T_d \lesssim 15\,\mathrm{K}$, 0.75 at 25 K, 0.59 at 30 K, 0.37 at 40 K and 0.25 at 50 K; on graphite $\epsilon \approx 1$ for $10 \lesssim T_d \lesssim 40\,\mathrm{K}$, 0.91 at 50 K and 0.43 at 100 K. Below about 5.5 K (silicate) and 9 K (graphite) ϵ collapses, because newly formed H_2 no longer desorbs (the term F_1 of Section 7.8 exceeds unity); the floor at $T_{\rm CMB}$ does not prevent this. On hiipdr, whose silicate grains sit at 26–37 K in the PDR (L-33), $\mathrm{d}\ln\epsilon/\mathrm{d}\ln T_d \approx -1.3$ to -1.7: a 10% error in T_d is a 13–17% error in the formation rate, and the formation rate sets the depth of the H/H_2 transition together with the shielding. The grains of pn150k are 0.92–0.93 of Cloudy's temperature through the PDR (L-19), where graphite's efficiency is flat.

5 Geometry: rays, extinction and shielding columns

The MC field already contains the attenuation by dust (absorption and scattering) and by the gas continua. What it does not contain is the absorption in the H_2 and CO lines, which the MC does not transport. That absorption is applied through column densities integrated along the paths the photons took: the straight line from each source for the direct beam, and the path source → scattering point → cell for the scattered light. This section describes how those paths and columns are built; Sections 6 and 7 apply them.

5.1 Exact ray–cell traversal

Every ray in the bridge is traced by rayCells (pdr_mod.f90:1710-1833, D-397). MOCASSIN's cells are centred on the axis values, so a cell's faces are the midpoints between neighbouring axis values and the grid ends at the first and last values (the boundary cells are half-cells). From a start point inside the grid and a unit vector $\hat{u}$, the routine computes for each axis the distance along $\hat{u}$ to the next face and repeatedly steps through the nearest one (a three-dimensional digital differential analyser in the manner of Amanatides and Woo 1987), returning the list of cells crossed and the exact length ℓ of the ray in each, at a cost proportional to the number of cells crossed. The first entry is the cell the ray starts in; inactive cells (outside the nebula or in a cavity) are returned with

index zero and contribute nothing. The ray ends after a given length or at the edge of the grid. For a `symmetricXYZ` octant the lower faces are planes of symmetry: when `lgMirror` is set the ray is reflected there ($u_q \to -u_q$) and continues, a segment that re-enters the cell it just left being merged with it; otherwise the ray ends at the edge. This replaced fixed-step marches that sampled each cell an integer number of times and missed the part of the neighbouring cell nearest the target, which had made the source-ward $N(\mathrm{H_2})$ through `hiipdr`'s transition 0.46–0.75 of its exact value; with the exact paths that transition moved 0.25 pc towards the star (D-397, H-025, L-45).

5.2 Source-ward extinction and molecular columns

`computeAVradial` (`pdr_mod.f90:1859-2015`), called from `computeAVdust` (`:2410-2523`) for point sources, traces for every active cell and every source s a ray from the cell centre towards the source (no mirror) and accumulates

$$\tau_{V,s} = \sum_{\text{cells}} (\kappa_V^{\text{abs}} + \kappa_V^{\text{sca}})\,\ell, \qquad N_{X,s} = \sum_{\text{cells}\neq\text{target}} n_{\mathrm{H}}\, x_X\, \ell, \quad X \in \{\mathrm{H_2},\ \mathrm{C},\ \mathrm{CO}\}, \tag{15}$$

with the abundances left by the previous chemistry. $\tau_{V,s}$ runs to the cell centre, the target's half-chord included. The target cell is excluded from $N_{X,s}$ (D-088): including its near half made its H_2 shield itself, a positive feedback (more H_2, less dissociation, more H_2) that, with MOCASSIN's large cells, drove the surface cell to complete self-shielding. The column is therefore the one to the cell's near face. It misses the cell's own contribution, an error that shrinks with the cell size instead of feeding back: on `hiipdr`'s 31^3 grid a cell is 0.648 pc, 0.104 mag of A_V (H-026) and 2×10^{20} H nuclei $\mathrm{cm^{-2}}$, so half a cell can hold $5 \times 10^{19}\,\mathrm{cm^{-2}}$ of H_2 and, at a CO abundance of 10^{-4}, $10^{16}\,\mathrm{cm^{-2}}$ of CO, columns at which both self-shield strongly (the CO factor of van Dishoeck and Black 1988 is ≈ 0.05 at $N(\mathrm{CO}) = 10^{16}\,\mathrm{cm^{-2}}$). The level model averages the transmission through the cell's own column instead (Section 6.6); the CO photorate and the chemistry's own H_2 diagnostic do not: their column lags the column to the cell centre by half a cell (D-088), so they are under-shielded in the cells of their own transitions on coarse grids, which places those transitions deeper by up to about half a cell. The length of the source-ward ray inside the target cell, from its near face to its centre, is kept separately as ℓ_{own} for the scattering sites (D-398).

The sources are combined with the geometric weights $f_s = r_s^{-2}$ (dilution only, `:1986`):

$$A_V = -\frac{1.0857}{R_{\text{FUV}}}\,\ln\left[\frac{\sum_s f_s\, e^{-R_{\text{FUV}}\tau_{V,s}}}{\sum_s f_s}\right], \qquad N_X = \frac{\sum_s f_s N_{X,s}}{\sum_s f_s}, \qquad \ell_{\text{own}} = \frac{\sum_s f_s\, \ell_{\text{own},s}}{\sum_s f_s} \tag{16}$$

(`:1996-2009`), stored as `AVeff`, `pdrColH2`, `pdrColC`, `pdrColCO` and `difSrc(:,6)`. For one source these are the plain column and A_V, and all the fixtures have one source. For several, the A_V is the one that reproduces the summed transmitted FUV of sources of *equal* luminosity, and the columns are means weighted by geometry alone (neither the luminosities nor the attenuation of the sources enter f_s), applied to the summed direct field through shielding functions that are not linear in the column; the error of this is not measured. An outward ray along the same direction (with mirrors) gives the dust column to the edge, `AVback`.

A_V is a coordinate and a diagnostic in the 3D build: the photorates do not use it (Section 7.1). It is a geometric dust column, not an inversion of the MC field, which would under-read it by a factor 1.8 because the field carries scattered light (D-086, D-087); TORUS-3DPDR also derives A_V from the column density along rays (Bisbas et al., 2015, their Eq. 2). It is measured from the source and so includes the dust of the ionised gas: `hiipdr`'s ionisation front sits at $A_V = 0.48$ (D-260). A 1D PDR model, whose A_V starts at its illuminated face, has to be compared at A_V measured from the front.

Under plane-parallel illumination `computeAVdust` integrates A_V and the columns along the beam direction itself (`:2432-2521`); there the columns run to the cell centre, the cell's near half included (`:2492-2507`), unlike the point-source branch. The level model does not run in that geometry.

5.3 Snapshot for the diffuse walk

`snapDiffuseSources` (`pdr_mod.f90:2082-2121`) is called by `computeAVdust` before any cell is solved and stores per cell (`difSrc`):

$$n(\mathrm{H_2}),\ n(\mathrm{C}),\ n(\mathrm{CO})\ [\mathrm{cm^{-3}}]; \qquad \kappa_{\rm ext} = R_{\rm FUV}\,\kappa_V\ [\mathrm{cm^{-1}}]; \qquad \epsilon = \sum_{k\in\mathrm{FUV}} \kappa_k^{\rm sca}\, J_k^{\rm ste}\ [\mathrm{erg\,s^{-1}\,cm^{-3}}], \tag{17}$$

the number densities that shield, a grey FUV extinction coefficient, and the FUV power scattered per unit volume in the band (`:2099-2117`). For isotropic scattering $\epsilon/4\pi$ is the band-integrated scattering emissivity. It is computed from the full local MC field, direct and diffuse, so it contains every order of scattering.

5.4 Scattering sites of the diffuse field

The diffuse FUV field is produced by scattering inside the cloud. A photon scattered at a point P and arriving at the cell C has crossed the column $N(\star \to P) + N(P \to C)$, not the column from the star to C (D-347, D-348). `traceDiffuseSites` (`pdr_mod.f90:2136-2276`) represents the diffuse field arriving at the target cell by a set of *scattering sites*, built as the discretised formal solution of the transfer equation for the intensity arriving at C from each of a set of directions $\hat{k}$,

$$4\pi I_{\hat{k}} = \int_0^\infty \epsilon(s)\, e^{-\tau(s)}\, \mathrm{d}s, \qquad \tau(s) = \int_0^s \kappa_{\rm ext}\, \mathrm{d}s', \tag{18}$$

with s the distance from C along $\hat{k}$ (the direction the photons come from):

1. **Directions.** The twelve HEALPix $N_{\rm side} = 1$ pixel centres (Górski et al., 2005), which divide the sphere into twelve equal solid angles of $\pi/3$ sr, built directly in RING order: rays 0–3 at $\mu = +2/3$, $\phi = (k + \frac{1}{2})\pi/2$; rays 4–7 at $\mu = 0$, $\phi = k\pi/2$; rays 8–11 at $\mu = -2/3$, $\phi = (k + \frac{1}{2})\pi/2$ (`:2152-2165`), with μ measured along the grid's z axis. The directions are fixed in the grid frame.

2. **Walk.** Each ray is traced with `rayCells` from the target's centre, with mirror reflection at the octant's lower faces, to the edge of the model. Every cell crossed that scatters ($\epsilon > 0$, $\kappa_{\rm ext} > 0$) is a site, the target cell itself included (with the half-chord the ray crosses); the walk is stored (`wkCell`, `wkLen`, `siteRay`, `siteIst`) for the level model.

3. **Weight.** With τ the grey FUV optical depth between the target's centre and the entry of the site cell and $\Delta\tau = \kappa_{\rm ext}\ell$ across it,

$$w_{\rm site} = \epsilon\, \frac{1 - e^{-\Delta\tau}}{\kappa_{\rm ext}}\, e^{-\tau} \qquad \left(\tfrac{1-e^{-\Delta\tau}}{\kappa_{\rm ext}} \to \ell(1 - \tfrac{1}{2}\Delta\tau) \text{ for } \Delta\tau \le 10^{-4}\right) \tag{19}$$

   (`:2207-2217`): Equation (18) integrated over the site's segment with ϵ and $\kappa_{\rm ext}$ constant in the cell. Since the directions subtend equal solid angles, $4\pi J_{\rm formal} = \frac{1}{12}\sum_{\rm sites} w_{\rm site}$.

4. **Two-leg column.** For each species X (`:2219-2239`),

$$N_{X,\rm site} = N_X^\star(P) + n_X(P)\,\ell_{\rm own}(P) + \sum_{\rm cells\ between} n_X\,\ell + \tfrac{1}{2}\,n_X(P)\,\ell_P, \tag{20}$$

   the source-ward column of the site's cell to its near face, plus the site cell's own gas from its near face to its centre (D-398), plus the cells crossed between the target and the site, plus half of the site cell along the walk. The target cell is excluded from both legs, so a site in the target cell carries the direct beam's column.

Only the *relative* weights are used: the magnitude of the diffuse field is the MC's $\chi_{\rm dif}$ (Equation 5), which the sites distribute over directions and columns, each site becoming a ray of the chemistry with field $\chi_{\rm dif}\, w_{\rm site}/\sum w$ (Section 7.1). The distribution is the formal solution only under the assumptions built into Equation (19): isotropic scattering, whereas the grains of the fixtures have $g = 0.61$ and 0.29 (Table 4), so the real diffuse field arrives preferentially from the source side; a grey extinction equal to the source-weighted band mean, whereas the window in which H_2 and CO dissociate has a larger extinction and the local field at depth a smaller one (Section 4); an emissivity made by dust scattering alone, whereas $\chi_{\rm dif}$ also holds gas emission; uniform emissivity across a cell; and the twelve-direction quadrature. As a check of these assumptions, the formal solution's own magnitude, $4\pi J_{\rm formal}/(c\, u_{\rm D})$, recovers 52–130% of the MC's $\chi_{\rm dif}$, falling with depth, for reasons not yet identified (gas two-photon emission, the phase function and the quadrature were named as candidates; the source-weighted grey extinction, which overstates the extinction at depth, is another; L-4). The field arrives markedly anisotropically: the brightest of the twelve directions carries 1.9–3.1 times the mean (D-348). What the assumptions affect is the distribution of the diffuse field over the sites' columns, not its magnitude, and that distribution acts on the photorates only through the line shielding: at fixed abundances the two-leg columns took back 5–32% of the increase of the H_2 photodissociation rate that the diffuse field's magnitude produced (D-348).

5.5 Angular quadrature

The twelve directions of the sites and of the line rays are the coarsest HEALPix tessellation. Their moments are not isotropic: the mean fourth power of a direction's component is 0.132 along the ring axis (z) against 0.218 across it, where an isotropic set gives 0.2 (D-410). The result is a pattern fixed in the grid frame: of the cells that the octant's symmetry makes equivalent, the one nearest the plane $z = 0$ holds about 11% more H_2 before and across the H/H_2 transition, on both fixtures, independently of the packet count (L-53). Rotating the set moves the excess to the new ring axis, and 48 directions ($N_{\rm side} = 2$, fourth moments 0.188 and 0.201) remove it at 1.48 times the run time, moving every integrated quantity by less than 1%; the twelve were kept because no line map carries the pattern (D-410). 3D-PDR and TORUS-3DPDR also use HEALPix rays, twelve ($\ell = 0$) in TORUS-3DPDR's production runs (Bisbas et al., 2012, 2015).

5.6 Line-escape rays

`traceLineRays` (`pdr_mod.f90:2327-2376`, D-377) traces the same twelve directions from the target cell to the edge of the model, with mirrors at the lower faces, skipping inactive cells, and returns for each ray the cells crossed and the path length in each; the first entry is the cell itself with the half of it the ray crosses. These rays give the escape probabilities of the coolant lines (Section 9.3) and of the H_2 lines (Section 6.8), each crossed cell contributing with its own level populations. Before D-377 the 3D flavour's line escape was a slab with four of the twelve rays shut, which capped the escape probability of an optically thin line at 2/3 (L-27).

5.7 Comparison with 3D-PDR and TORUS-3DPDR

In 3D-PDR (Bisbas et al., 2012) the FUV field is a boundary condition: at every point the field incident on the cloud surface is attenuated along HEALPix rays traced to the boundary, $\chi(p) = N_\ell^{-1} \sum_q \chi_0(q)\, e^{-\tau_{\rm UV} A_V(q)}$ with $\tau_{\rm UV} = 3.02$ in the benchmark comparisons (Bisbas et al., 2015, their Eq. 1), and the shielding columns are integrated along the same rays. For a source inside the cloud these rays are the wrong paths: a ray pointing away from the source would deliver unshielded radiation from outside a nebula that has no outside source (D-342). TORUS-3DPDR (Bisbas et al., 2015) takes the FUV field of every cell from its MC path-length estimator (912–2400 Å, stored as two vectors for the positive and negative directions), derives A_V and the column densities along HEALPix rays from the density field, and leaves the dust's contribution to the diffuse field to

later work (their Sect. 3.1). The present coupling differs from both in that the dust attenuation and scattering are the MC's, the field is split into its direct and scattered parts, and the line-shielding columns follow the photons' paths: source-ward for the direct beam and through the scattering sites for the diffuse field.

6 The H_2 level model

The level model computes, in every cell, the fractional populations of the 302 rovibrational levels of the electronic ground state $X\,^1\Sigma_g^+$ of H_2 under the cell's own radiation field, gas temperature and colliders, and from them the Solomon photodissociation rate per molecule that the chemistry uses (Section 7) and three H_2 heating terms (Section 9.3). The physics is in `h2_mod.f90` (1570 lines); the driver that supplies the field, the columns, the colliders and the formation distribution is `h2LevelModel` with its internal routine `solveCell` (`pdr_mod.f90:5201-6014`). It is a re-implementation, from the data and conventions of Cloudy's large H_2 model (Shaw et al., 2005), of the line-by-line treatment of ultraviolet pumping and self-shielding that the Meudon PDR code (Le Petit et al., 2006) and the models of Draine and Bertoldi (1996) also use, in place of a shielding function fitted to such models. It was validated level by level against Cloudy on one-zone and slab models (H-003–H-006). Those comparisons isolate three deliberate departures from Cloudy, each discussed below: the closure of the level balance (Section 6.2), the damping width of the lines (Section 6.5) and the treatment of line overlap (Section 6.7).

Physical picture. A molecule in level l of X absorbs a far-ultraviolet photon in a Lyman ($B\,^1\Sigma_u^+$–X) or Werner ($C\,^1\Pi_u$–X) line, or in one of the weaker $B'\,^1\Sigma_u^+$–X and $D\,^1\Pi_u$–X lines; out of $v = 0$ the Lyman and Werner lines lie between 912 and 1110 Å (11.2–13.6 eV), and the B' and D systems reach below 912 Å. Within about a nanosecond the excited level u decays, either into the vibrational continuum of X, which dissociates the molecule into two H(1s) atoms (the Solomon process), or into a bound level of X, mostly vibrationally excited, from which a slow electric-quadrupole cascade ($A = 3\times10^{-11}$–$6\times10^{-6}\,\mathrm{s^{-1}}$, median $2\times10^{-8}\,\mathrm{s^{-1}}$ in `transprob_X.dat`) competes with collisions. In an unshielded field of the Draine shape the model dissociates one molecule per 8.1 absorptions, at $6.36\times10^{-11}\,\mathrm{s^{-1}}$ per unit Draine field (H-007); the mean dissociation probability per absorption is 0.12 in the computations of Sternberg et al. (2014) and about 0.15 when averaged through complete photodissociation fronts (Draine and Bertoldi, 1996). The strong lines out of a level become optically thick at level columns of order $10^{14}\,\mathrm{cm^{-2}}$, saturate, and above $\sim 10^{17}$–$10^{18}\,\mathrm{cm^{-2}}$ absorb mainly in their Lorentzian wings, where above $N(H_2) \sim 10^{20}\,\mathrm{cm^{-2}}$ the lines of different levels overlap (Draine and Bertoldi, 1996). The dissociation rate in a cell therefore depends on the field at every line, on the column of every level along the paths of the photons, on the line widths, and on the populations of the absorbing levels, which the pumping itself changes. The model treats all of these line by line.

6.1 Data

All data are Cloudy c25.00's (`dataPDR/h2/`, copied unmodified; provenance in `dataPDR/h2/README.md`). The reading rules are Cloudy's: a line starting with `#` is a comment, and the data of the excited-state files end at the first empty line. The sources in Table 5 are those that the files' own headers give, completed where the header is not the primary source.

Table 5: Data of the H_2 level model. Temperature ranges are those of the tables; coverage is given as the upper levels present.

Content	File(s)	Source
energies of the X rovibrational levels (302 levels, $v \leq 14$, $J \leq 31$; sorted by energy)	`energy_X.dat`	Komasa et al. (2011, their Table 1), compiled by K. Kaplan (header)
radiative transition probabilities within X: electric quadrupole, plus magnetic-dipole components of 30 $v = 1 \to 0$, $\Delta J = 0$ transitions listed separately and summed	`transprob_X.dat`	Wolniewicz et al. (1998); magnetic dipole Pachucki and Komasa (2011)
$H + H_2$: 1432 transitions, $v \leq 3$, $J \leq 17$, 728 of them ortho↔para (reactive); 100–5000 K	`coll_rates_H_15.dat`	Lique (2015)
$He + H_2$: 3251 transitions, $v \leq 14$, $J \leq 30$, no spin change; 1–10^4 K	`coll_rates_He_ORNL.dat`	ORNL data set; the header cites "Lee, T. G. et al. 2007, ApJ, in preparation"; no published source for this table was identified
ortho- and para-$H_2 + H_2$: 240 and 241 transitions, $v = 0$, $J \leq 31$; 1–10^4 K	`coll_rates_H2ortho_ ORNL.dat`, `coll_rates_ H2para_ORNL.dat`	Wan et al. (2018)
$H^+ + H_2$: 44 transitions, $v = 0$, $J \leq 9$, 24 of them ortho↔para (proton exchange); 3–6000 K	`coll_rates_Hp.dat`	Gerlich (1990)
energies of the rovibrational levels of B, C^+, C^-, B', D^+, D^- (1592 levels)	`energy_{B,C_ plus,...}.dat`	derived from Abgrall et al. (2000) (headers)
transition probabilities of the 32 453 lines to X	`transprob_{...}.dat`	Abgrall et al. (2000), from the band-system calculations of Abgrall et al. (1993) (B–X, C–X) and Abgrall et al. (1994) (B'–X, D–X)
spontaneous radiative dissociation rate D_u [s^{-1}] and mean kinetic energy of the atoms [eV] per excited level	`dissprob_{...}.dat`	Abgrall et al. (2000)
dissociation energy of X, $D = 36\,118.11\ cm^{-1}$ (4.478 eV; the only value of the file the model uses)	`energy_dissoc.dat`	Sharp (1970) per the header (which dates it 1971), with values revised in Cloudy in 2002

Reading: `h2ReadX` (`h2_mod.f90:206-334`) and `h2ReadElec` (`:366-516`). Statistical weights are $g = (2J + 1)(2I + 1)$, with nuclear-spin factor $2I + 1 = 3$ for ortho and 1 for para. In X ortho levels have odd J; in the excited states B, B' and the Π^+ components C^+, D^+ ortho levels have even J, and in the Π^- components C^-, D^- odd J, following Cloudy (Shaw et al., 2005): this is the assignment under which every allowed transition ($\Delta J = \pm 1$ to the Σ_u^+ states and the Π^+ components, $\Delta J = 0$ to the Π^- components) conserves the nuclear spin, and the reader counts

the lines that would not (none, H-004). The Π states have no $J = 0$ level. The D^- state stops at $v = 2$ because Cloudy's file has an empty line before the levels it marks "above the dissociation limit"; 3138 D^- lines are dropped with them, and 392 lines whose lower level lies beyond the 302 levels of X (H-004). The dissociation rates D_u are those of spontaneous emission into the vibrational continuum of X (Abgrall et al., 2000). Apart from 45 high-J levels of D^- ($v \le 2$), all the excited levels lie below the $\mathrm{H(1s)} + \mathrm{H}(n = 2)$ limit ($118\,376\,\mathrm{cm}^{-1}$ in `energy_dissoc.dat`), where this radiative channel is the only way they dissociate; predissociation, which acts above that limit, is not in the data. A transition listed twice adds its components, as Cloudy does; reading the X file without adding the magnetic-dipole components had doubled the $v = 1$ populations at low density (H-003). The lines are stored sorted by upper level (counting sort) and indexed by wavenumber (heapsort) for the overlap calculation.

6.2 Statistical equilibrium and the formation closure

Let n_i be the fraction of the cell's H_2 molecules in level i of X ($\sum_i n_i = 1$). The populations solve (`balance`, `h2_mod.f90:1456-1477`)

$$\sum_{j\neq i} n_j R_{ji} - n_i\Big(\sum_{j\neq i} R_{ij} + k_i\Big) + \phi_i \sum_j n_j k_j + s_i = 0, \qquad i = 1, \ldots, 302, \tag{21}$$

where R_{ij} [s^{-1}] is the total rate per molecule from level i to level j (collisions, spontaneous decay within X and the fluorescent transfer through the excited states; Sections 6.3 and 6.4), k_i [s^{-1}] the Solomon dissociation rate out of level i, ϕ_i the fraction of newly formed molecules that enter level i ($\sum_i \phi_i = 1$; Section 6.9) and s_i [s^{-1}] an optional further source per molecule, not used by `solveCell`. Fluorescent returns to the absorbing level (R_{ii}) drop out. Summed over i, the transfer terms cancel and so do the dissociation and formation terms, so the system conserves $\sum_i n_i$: the term $\phi_i \sum_j n_j k_j$ is the closure *every molecule destroyed by the Solomon process is replaced by one formed*. The level model determines how the molecules are distributed over the levels; how many there are is the chemistry's (Section 8).

Numerics. The equation of level 1 is replaced by $\sum_i n_i = 1$ and the 302×302 system is solved directly by Gaussian elimination with partial pivoting (`gaussSolve`, `:1536-1568`; about $n^3/3 \approx 9 \times 10^6$ operations per solution). `h2Solve` (`:1516-1533`) assembles $R = C + A + W$ (collisions, radiative decay, fluorescent transfer) and the k_i, and calls it. The populations of the validation runs were never negative and summed to unity to the printed precision (H-003). Cloudy instead solves its lowest 70 levels by matrix inversion and the others one by one (H-010).

Validity of the closure. In steady state the chemistry's formation rate per molecule, $R_\mathrm{f}\, n_\mathrm{H}\, n(\mathrm{H})/n(\mathrm{H_2})$, equals the sum of all destruction rates per molecule, of which k_Sol is one; the closure sets the formation flux into the levels equal to k_Sol alone. It is therefore exact where photodissociation dominates the destruction (93–100% of the H_2 destruction out to 0.43 pc in `pn150k`, H-009), and underestimates the formation flux by the factor $k_\mathrm{Sol}/(k_\mathrm{Sol} + k_\mathrm{other})$ elsewhere. The network's three $\mathrm{H_2} + \mathrm{CRP}$ channels alone give $k_\mathrm{other} = 0.99\,\zeta$, $1.0 \times 10^{-16}\,\mathrm{s}^{-1}$ at the default `pdrZeta`, so the closure fails where the Solomon rate is shielded below a few $10^{-16}\,\mathrm{s}^{-1}$, a reduction by more than 10^5 from its unshielded value at $\chi \sim 1$. There the formation pumping of the $v \ge 1$ levels, the infrared emission it produces and the part of the formation energy that reaches the gas by collisional de-excitation are underestimated; the Solomon rate itself is affected only through the populations of the absorbing levels, and the kinetic part of the formation energy follows the chemistry's own formation rate (Section 9.3; Section 12). Cloudy closes its level system differently: it applies its network's Solomon rate a second time as a uniform sink on the levels below $4100\,\mathrm{cm}^{-1}$, solves without a sum rule and rescales all populations by one factor (1.06–1.11 in one-zone models with $n_\mathrm{H} \le 10^4\,\mathrm{cm}^{-3}$ and fields up to 100 Habing). Put through that scheme, this

model's rates reproduce Cloudy's populations to 0.002–0.018 dex in 15 of 16 such models, against a median difference of 0.07 dex with its own closure (H-004); Cloudy's closure raises the pumped line intensities by about 1.2 (H-009).

6.3 Collisional and radiative rates within X

`h2CollRad` (`h2_mod.f90:1369-1415`) adds, for every pair of levels $u > l$ (energies E in cm^{-1}),

$$R_{ul} \mathrel{+}= \sum_c n_c\, k^c_{ul}(T) + A_{ul}, \qquad R_{lu} \mathrel{+}= \sum_c n_c\, k^c_{ul}(T)\, \frac{g_u}{g_l}\, e^{-hc(E_u-E_l)/kT}, \tag{22}$$

with n_c [cm^{-3}] the density of collider $c \in \{\mathrm{H}, \mathrm{He}, o\text{-}\mathrm{H_2}, p\text{-}\mathrm{H_2}, \mathrm{H^+}\}$, k^c_{ul} [$\mathrm{cm^3\,s^{-1}}$] its de-excitation rate coefficient, A_{ul} [$\mathrm{s^{-1}}$] the radiative transition probability, and $T = \max(T_e, 1\,\mathrm{K})$ (`pdr_mod.f90:5667`):

- downward coefficients from the tables of Table 5, interpolated linearly in T and held at the end values outside the tabulated range (`tabRate`, `h2_mod.f90:642-660`; Cloudy's `InterpCollRate` convention). Below 100 K the $\mathrm{H} + \mathrm{H_2}$ rates are therefore those of 100 K, which overestimates the reactive (ortho–para) channels, whose barrier makes them fall steeply with temperature; above 5000 K (H), 6000 K ($\mathrm{H^+}$) and 10^4 K (He, $\mathrm{H_2}$) the rates are those of the table's end;
- for a pair of levels of the same nuclear spin that a collider's table does not contain, the "g-bar" estimate of Shaw et al. (2005, their eq. 1 and Table 2),

$$\log_{10}(k^c_{ul}/\mathrm{cm^3\,s^{-1}}) = y_0 + a_g\,[\max(\Delta E,\, 100)]^{b_g}, \qquad \Delta E = E_u - E_l\ [\mathrm{cm^{-1}}], \tag{23}$$

  a fit, per collider, of the rate coefficients then available as a function of the energy gap (coefficients in Table 6; Cloudy's `GbarRateCoeff`), which Shaw et al. (2005) found to lie within 1 dex of 60–70% of the rates it was fitted to. A pair that changes the nuclear spin and is not in a table has no collisional rate;
- upward rates by detailed balance at the gas temperature, so that collisions alone relax the populations to a Boltzmann distribution at T;
- spontaneous radiative decay A_{ul} (quadrupole plus magnetic dipole) from `transprob_X.dat`.

Table 6: g-bar coefficients of Equation (23) (`h2_mod.f90:96-101`), identical to Table 2 of Shaw et al. (2005).

Collider	y_0	a_g	b_g
H	−9.9265	−0.1048	0.456
He	−8.281	−0.1303	0.4931
o-$\mathrm{H_2}$	−10.0357	−0.0243	0.67
p-$\mathrm{H_2}$	−8.6213	−0.1004	0.5291
$\mathrm{H^+}$	−9.2719	−0.0001	1.0391

Coverage. Of the 22 675 pairs of levels of the same nuclear spin, the tables contain 703 for H, 3250 for He, 240 and 239 for o- and p-$\mathrm{H_2}$ and 20 for $\mathrm{H^+}$ (counted from the data files); the remaining 86–99.9% take Equation (23). Ortho and para are coupled by collisions only through the 728 reactive $\mathrm{H} + \mathrm{H_2}$ transitions ($v \le 3$) and the 24 proton-exchange $\mathrm{H^+} + \mathrm{H_2}$ transitions ($v = 0$, $J \le 9$), the latter being by far the faster (Shaw et al., 2005). The tabulated pairs include the low-J levels of

$v = 0$ that carry most of the column; the g-bar rates matter where collisions compete with radiative decay in levels without data, i.e. high-v or high-J levels at high density, and Shaw et al. (2005) found the populations of the tabulated low levels nearly insensitive to them.

Colliders. The collider densities (`pdr_mod.f90:5887-5888`) are taken from the chemistry's abundances of the previous iteration, $n(\mathrm{H}) = n_{\mathrm{H}} x(\mathrm{H})$, $n(\mathrm{He})$ and $n(\mathrm{H}^+)$, and from the cell's own solution for the spin split, $n(o\text{-}\mathrm{H}_2) = f_o\, n(\mathrm{H}_2)$, $n(p\text{-}\mathrm{H}_2) = (1 - f_o)\, n(\mathrm{H}_2)$, with $f_o = \sum_{i \in \mathrm{ortho}} n_i$ iterated to self-consistency (Section 6.10). Electrons are not a collider, as in Cloudy; Aleman and Gruenwald (2011) include them in their models of the H_2 emission of the ionised region of planetary nebulae. Where electrons are a large fraction of the colliders, in the ionised gas and at the ionisation front, the collisional excitation of H_2 is therefore underestimated, in gas whose H_2 abundance is in any case uncertain (`L-43`). The rate coefficients and Boltzmann factors are cached for the last temperature (`crKd`, `crBf`), so the iterations over the colliders' densities at fixed T rebuild only R.

6.4 Ultraviolet pumping, Solomon dissociation and trapping

A molecule in X level l is excited in the line $l \to u$ at the rate

$$P_{lu} = B_{lu}\, \bar{J}_{lu} = A_{ul}\, \frac{g_u}{g_l}\, \eta_{lu} \quad [\mathrm{s}^{-1}], \qquad \eta_{lu} = \frac{c^2}{2h\nu_{lu}^3}\, J_\nu(\nu_{lu}), \tag{24}$$

where $\bar{J}_{lu}$ is the mean intensity averaged over the line profile and η_{lu} the photon occupation number at the line after shielding (Sections 3.6 and 6.5); the second form follows from the Einstein relations for a field constant across the line. In the unshielded Draine field $\eta \approx 2 \times 10^{-19}$ near 12 eV.

The excited levels have radiative lifetimes of 0.5–9 ns in these data (total decay rates $\Gamma_u = 1.1 \times 10^8$–$1.9 \times 10^9\ \mathrm{s}^{-1}$, median $\approx 10^9\ \mathrm{s}^{-1}$ for B and C). Collisions cannot compete with that below $\sim 10^{14}\ \mathrm{cm}^{-3}$ (Shaw et al., 2005), and the excited populations, of order $\sum_l P_{lu}/\Gamma_u \sim 10^{-18}\chi$ per molecule, are never a reservoir: every absorption is followed at once by a decay, into the continuum of X (dissociation) or into a bound level of X. The excited states are therefore not solved for. With the effective decay rates

$$A_t^{\mathrm{eff}} = A_t\,(1 + \eta_t)\,\beta_t, \qquad \Gamma_u^{\mathrm{eff}} = D_u + \sum_{t \in u} A_t^{\mathrm{eff}}, \tag{25}$$

where t runs over the lines from u to bound levels of X, $1 + \eta_t$ accounts for stimulated emission (negligible, kept for completeness), β_t is the escape probability of a photon emitted in line t (Section 6.8) and D_u the spontaneous radiative dissociation rate of u, `h2PumpRates` (`h2_mod.f90:540-572`) returns

$$k_l = \sum_u P_{lu}\, \frac{D_u}{\Gamma_u^{\mathrm{eff}}} \qquad \text{(Solomon dissociation out of } l\text{)}, \tag{26}$$

$$W_{ll'} = \sum_u P_{lu}\, \frac{A_{ul'}^{\mathrm{eff}}}{\Gamma_u^{\mathrm{eff}}} \qquad \text{(fluorescent transfer } l \to l'\text{)}, \tag{27}$$

$$H_l^{\mathrm{dis}} = \sum_u P_{lu}\, \frac{D_u}{\Gamma_u^{\mathrm{eff}}}\, E_u^{\mathrm{kin}} \qquad \text{(kinetic energy given to the atoms [eV s}^{-1}\text{])}, \tag{28}$$

and the total absorption rate $p_l = \sum_u P_{lu}$, with E_u^{kin} the mean kinetic energy released per dissociation of u (`dissprob_....dat`). The first step of the fluorescent cascade is W; the quadrupole cascade that follows is carried by A in Equation (21). The Solomon rate per molecule handed to the chemistry is

$$k_{\mathrm{Sol}} = \sum_l n_l\, k_l \qquad (\texttt{pdr_mod.f90:5918}). \tag{29}$$

Trapping. A photon emitted in a line $t = u \to l'$ that is optically thick is absorbed again close by, by a molecule in l', which it re-excites to the same level u. In the escape-probability ("on-the-spot") approximation the net effect is that the decay rate of the line is reduced to $A_t\beta_t$, and the level's other decays and its dissociation take its share in $\Gamma_u^{\rm eff}$. Only lines that end in populated low levels are thick, so trapping closes the decays to $v = 0$ and raises the dissociation probability $D_u/\Gamma_u^{\rm eff}$ in thick gas. It is needed: without it the rate into $v = 0$, $J = 0$–5 from the excited states was up to 2200 times Cloudy's in the shielded zones of the slab of H-005 and the Solomon rate 10–20% low; with it the Solomon rate comes within 3–8% of Cloudy's (H-005). The total absorption rate p_l counts only the external field. Photons that dust or an overlapping H_2 line absorb before they escape are not counted as lost (Cloudy adds such a destruction probability to the escape probability, Shaw et al. 2005); this slightly overestimates the dissociation probability of levels whose decays are trapped.

6.5 Self-shielding of the pumping lines

Line parameters. For line lu of wavelength $\lambda = 1/\tilde{\nu}_{lu}$ [cm] (`h2LineParams`, `h2_mod.f90: 1328-1336`),

$$\sigma_{lu} = \frac{g_u}{g_l}\,\frac{A_{ul}\,\lambda^3}{8\pi^{3/2}\,b}\ [\mathrm{cm}^2], \qquad \tau_0 = \sigma_{lu} N_l, \qquad a = \frac{\Gamma_u\,\lambda}{4\pi\,b}, \qquad \Gamma_u = D_u + \sum_{t\in u} A_t, \tag{30}$$

where σ_{lu} is the line-centre cross section of a pure Doppler profile of Doppler width $\Delta\nu_D = b/\lambda$, $(\pi e^2/m_e c) f_{lu}/(\sqrt{\pi}\Delta\nu_D)$ written with A_{ul}; N_l [cm^{-2}] is the column of molecules in level l; $b = \sqrt{2kT/m_{\mathrm{H}_2} + v_{\rm turb}^2}$ [$\mathrm{cm\,s}^{-1}$] the Doppler parameter of the target cell (`pdr_mod.f90:5668`; $v_{\rm turb}$ is `pdrVturb`, default $1\,\mathrm{km\,s}^{-1}$); and a the Voigt damping parameter, the ratio of the Lorentzian half width $\Gamma_u/4\pi$ to $\Delta\nu_D$. Γ_u is the total decay rate of the upper level, bound–bound plus bound–free (spontaneous radiative dissociation), i.e. the inverse radiative lifetime; since the lower level of every line is a metastable level of X, Γ_u is the natural width of the line. With these data $a \le 0.03$ for $b \ge 1\,\mathrm{km\,s}^{-1}$ (median 0.006 at 100 K and the default $v_{\rm turb}$).

The damping width. Γ_u exceeds a line's own A_{ul} by a factor 20 in the median over the thick lines of the H-005 slab (16–84%: 7.8–157), and the shielding in the damping wings scales as $a^{1/2}$ (Equation 33). Cloudy damps each H_2 line with its own A_{ul} (`dampXvel` $= A_{ul}\lambda/4\pi$ in `mole_h2_create.cpp` of c25.00, and in its saved damping constants, H-005), whereas its hydrogenic lines use the lifetime of the upper level (`iso_create.cpp`). The natural width is the physical choice. Cloudy's lowers the transmission in the damping regime by up to $\sqrt{20} \approx 4.5$ (pumped lines behind more than $10^{16}\,\mathrm{cm}^{-2}$ of H_2 fall to 0.18–0.41 of their intensity, H-008) and moves the H/H_2 crossing 0.21 mag shallower on `hiipdr` and 0.10 mag on `pn150k` (H-026; Section 12). The same b (the target cell's) is used for the whole column in front of the cell and along the escape rays; this matters in the Doppler-saturated regime, where $\tau_0 \propto b^{-1}$, and not in the damping regime, where the equivalent width in frequency does not depend on b.

Transmission and curve of growth. The rate at which a molecule behind the column N_l absorbs in the line, relative to the unshielded rate for a continuum flat across the line, and its integral over τ_0 are

$$S(\tau_0, a) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} H(a,x)\,\exp\!\left[-\tau_0 \frac{H(a,x)}{H(a,0)}\right] \mathrm{d}x, \tag{31}$$

$$W(\tau_0, a) = \int_0^{\tau_0} S\,\mathrm{d}\tau = \frac{2H(a,0)}{\sqrt{\pi}} \int_0^{\infty} \left\{1 - \exp\!\left[-\tau_0 \frac{H(a,x)}{H(a,0)}\right]\right\} \mathrm{d}x, \tag{32}$$

with $x = (\nu - \nu_{lu})/\Delta\nu_D$ and $H(a,x) = (a/\pi)\int e^{-y^2}[(x-y)^2 + a^2]^{-1}\mathrm{d}y$ the Voigt function ($\int H\,\mathrm{d}x = \sqrt{\pi}$), evaluated with the rational approximation of Humlíček (1982) (relative accuracy 10^{-4};

`h2Voigt`, `h2_mod.f90:664-686`). S is the single-line self-shielding function: the profile-weighted transmission, $S(0,a)=1$. W is the equivalent width in units of $\sqrt{\pi}\Delta\nu_D/H(a,0)$, normalised so that $W\to\tau_0$ for $\tau_0\ll 1$, and $\mathrm{d}W/\mathrm{d}\tau_0 = S$. The three regimes are

$$S \simeq \begin{cases} 1-\tau_0/\sqrt{2} & (\tau_0 \ll 1), \\ \left[\sqrt{\pi}\,\tau_0\,(\ln\tau_0)^{1/2}\right]^{-1} & \text{(saturated Doppler core)}, \\ \pi^{-1/4}\left[H(a,0)\,a/\tau_0\right]^{1/2} & \text{(damping wings)}, \end{cases} \tag{33}$$

the first two for $a\to 0$, the third when the absorption is dominated by the Lorentzian wings $H\simeq a/(\sqrt{\pi}x^2)$.

Profile normalisation. $\tau_0=\sigma_{lu}N_l$ is the line-centre optical depth of the Doppler core, and S normalises the profile to its own centre, $\tau(x)=\tau_0 H(a,x)/H(a,0)$; with this convention τ_0 agrees with Cloudy's line optical depths to 0.04% (H-005). The area-normalised Voigt profile gives $\tau(x)=\tau_0 H(a,x)$, smaller by the factor $H(a,0)\simeq 1-2a/\sqrt{\pi}$ at every frequency. For these data the difference is at most 3% in τ (0.8% weighted by the unshielded pumping, and half that in S in the damping regime), negligible against the other uncertainties.

Numerical strategy. `shieldQuad` (`:692-746`) integrates one side of the profile with the trapezoidal rule on 1200 uniform points to $x=6$ and logarithmically (60 points per decade) beyond, to $x_{\max}=\max(60,\,300\sqrt{\tau_0 a})$; beyond $x_{\max}$ the Lorentzian wing $H\simeq a/(\sqrt{\pi}x^2)$ is optically thin ($\tau\le 10^{-5}$ at $x_{\max}=300\sqrt{\tau_0 a}$) and is added analytically. It also records the offsets x_q at which 1/6, 1/2 and 5/6 of one side's absorption, $\int He^{-\tau}\mathrm{d}x$, has accumulated, for the overlap quadrature. Against an independent quadrature (the Faddeeva function of SciPy) S and W agree to 0.04% and 0.03% over $\tau_0=10^{-2}$–10^{11}, $a=10^{-6}$–10^{-1} (H-005). `shieldTable` (`:748-760`) tabulates $\log S$, $\log W$ and $\log x_q$ once on a 161×29 grid, $\log_{10}\tau_0\in[-4,16]$ (step 0.125) and $\log_{10}a\in[-7,0]$ (step 0.25). Look-up (`tabShieldJ`, `:833-851`) is bilinear in $(\log\tau_0,\log a)$ on the logarithms; below the table $S=1$ and $W=\tau_0$; above it the damping-wing asymptote, $S\propto\tau_0^{-1/2}$, $W\propto\tau_0^{1/2}$. The table's S is within 1.2% of the exact single-line integral (H-006). The damping parameter's column and fraction in the table are found once per line (`h2ShieldAIndex`, `:822-830`).

Comparison with shielding functions. A shielding-function treatment writes the Solomon rate as $k_{\rm Sol}\simeq k_{\rm Sol}(0)\,f_{\rm shield}(N_{\rm H_2})\,e^{-\tau_{d,1000}}$ (Draine and Bertoldi, 1996, their eq. 40), with $\tau_{d,1000}$ the dust optical depth at 1000 Å and $f_{\rm shield}$ a fit to multi-line models for given populations, temperature and field shape. The fit of Draine and Bertoldi (1996, their eq. 37),

$$\begin{aligned} f_{\rm shield} &= \frac{0.965}{(1+x/b_5)^2} + \frac{0.035}{(1+x)^{1/2}}\,\exp\!\left[-8.5\times10^{-4}(1+x)^{1/2}\right], \\ x &= \frac{N(\mathrm{H_2})}{5\times10^{14}\,\mathrm{cm^{-2}}}, \qquad b_5 = \frac{b}{\mathrm{km\,s^{-1}}}, \end{aligned} \tag{34}$$

describes the saturation of the lines in its first two terms and their overlap in the exponential. 3D-PDR's own rate uses the single-line function of Federman et al. (1979) for one representative line (Section 7.3). Here the equivalent of $f_{\rm shield}$ follows from the level columns, the line widths, the populations and the field at each line, and the dust attenuation is in the MC field. On the Cloudy slab of H-005 (100 K, $n_{\rm H}=10^4\,\mathrm{cm^{-3}}$, 100 Habing, $b=0.91\,\mathrm{km\,s^{-1}}$) the model without overlap followed Equation (34) without its overlap factor to 0.79–1.00 at $N(\mathrm{H_2})=10^{18}$–$10^{21}\,\mathrm{cm^{-2}}$, and with overlap it lies at 0.70–0.95 of the full fit there and 20–50% above it at 10^{14}–$10^{17}\,\mathrm{cm^{-2}}$ (H-005, H-006). The analytic theory of the H I-to-H_2 transition of Sternberg et al. (2014), in which the total atomic column of a slab in a beamed field is $N_{\rm H\,I}=\sigma_g^{-1}\ln(\alpha G/2+1)$ (their eq. 40), with σ_g the far-ultraviolet dust absorption cross section per H nucleus, α the ratio of the free-space

dissociation rate to the formation rate $R_{\rm f} n$ and G a cloud-averaged self-shielding factor, rests on the same kind of line-by-line computation (an "effective dissociation bandwidth" computed with the Meudon code).

6.6 Averaging through the cell's own column

The pumping rate that the level balance of a cell needs is its mean over the cell, whose populations are uniform. If the cell adds $\Delta\tau = \sigma_{lu}\Delta N_l$ to the column τ_0 in front of it, the mean of S over a uniform layer traversed along one chord is the mean slope of the curve of growth across it (`h2Shield`, `h2ShieldJ`, `h2_mod.f90:768-784, 854-866`; `h2LineShieldPrep/Eval`, `:876-907`):

$$\langle S\rangle = \frac{W(\tau_0+\Delta\tau,a)-W(\tau_0,a)}{\Delta\tau} \quad \text{if } \Delta\tau > 10^{-3}\max(\tau_0,10^{-3}), \qquad \text{else } S(\tau_0+\tfrac{1}{2}\Delta\tau,a), \tag{35}$$

clamped to $[0,1]$; the second form avoids the cancellation in the difference of W for thin layers, with an error of order $\Delta\tau^2 S''/24$. A cell whose own column takes its lines from thin to saturated absorbs the band in its front part, and $\langle S\rangle$ is then much smaller than $S(\tau_0)$. The cell's own column of level l is $\Delta N_l = n_l\, n(\mathrm{H_2})\,\ell$ with two path lengths (`pdr_mod.f90:5609-5615, 5697-5706, 5804-5819`):

$$\ell_{\rm dir} = \frac{\Delta x\,\Delta y\,\Delta z}{\Delta y\Delta z\,|u_x| + \Delta x\Delta z\,|u_y| + \Delta x\Delta y\,|u_z|}, \tag{36}$$

$$\ell_{\rm iso} = \frac{4V}{A} = \frac{2\,\Delta x\,\Delta y\,\Delta z}{\Delta x\Delta y + \Delta y\Delta z + \Delta x\Delta z}, \tag{37}$$

where $\ell_{\rm dir}$ is the mean chord of the cell along the unit vector $\hat{u}$ towards the source (the volume over the area projected on a plane normal to $\hat{u}$), averaged over sources with weights r_s^{-2} and used for the direct beam, and $\ell_{\rm iso}$ is Cauchy's mean chord of a convex body for isotropically distributed lines, used for the diffuse light. The widths $\Delta x, \Delta y, \Delta z$ are centred differences of the axis values (`cWid`), one-sided at the first and last axis value; there the full spacing is used, although the outermost cell of a grid that ends at its last axis value is half as wide. One representative chord stands for the distribution of chord lengths through the cell. The direct-path column uses the populations of the current pass of the cell's iteration, the diffuse one those of the previous sweep.

This is how the level model includes the self-shielding of the cell's own $\mathrm{H_2}$, which the chemistry's columns leave out (Section 5.2). Averaging the transmission through the cell is a different operation from adding the cell's near half to the shielding column, which made the chemistry's surface cell shield itself completely and settle on a self-shielded branch (D-088). The size of the effect shows how thin the pumping layer is: when the diffuse part was first computed without the cell's own column, the Solomon rate where $\mathrm{H_2}$ turns on was 4–12 times the chemistry's, and averaging through the mean chord lowered the 1–0 S(1) luminosity of the test model by a factor 4.6 (H-007). The average is exact for a uniform cell; cells across the H/$\mathrm{H_2}$ transition are not, and on `hiipdr` the 1–0 S(1) luminosity changed by +35% from 21^3 to 31^3 and by +3% from 31^3 to 41^3 cells while the $\mathrm{H_2}$ profile of the transition was still moving (H-007).

6.7 Line overlap

The $\mathrm{H_2}$ lines shade one another. Each line's transmission is its own S times the mean transmission of all the *other* lines over the frequencies where it absorbs (`lineOverlap`, `h2_mod.f90:1018-1289`, H-006):

$$\mathcal{O}_i = \frac{1}{6}\sum_{k=1}^{3}\sum_{s=\pm1}\exp\!\Big[-\tau_{\rm oth}\big(\tilde{\nu}_i + s\,x_{q,k}\,\Delta\tilde{\nu}_{D,i}\big)\Big], \qquad \tau_{\rm oth}(\tilde{\nu}) = \sum_{j\neq i,\ \tau_{0,j}\geq 10^{-2}} \tau_{0,j}\,\frac{H(a_j,x_j)}{H(a_j,0)}, \tag{38}$$

with $x_j = (\tilde{\nu}-\tilde{\nu}_j)/\Delta\tilde{\nu}_{D,j}$ and $\Delta\tilde{\nu}_{D,j} = \tilde{\nu}_j b/c$ [$\mathrm{cm^{-1}}$]. $\mathcal{O}_i$ approximates $\int \phi_i e^{-\tau_i} e^{-\tau_{\rm oth}}\mathrm{d}\nu / \int \phi_i e^{-\tau_i}\mathrm{d}\nu$, the mean of $e^{-\tau_{\rm oth}}$ weighted by line i's own absorption, by a midpoint rule in the cumulative

absorption: each side of the line is cut into three parts that carry equal shares of its absorption, and each part is represented by the offset $x_{q,k}$ that halves its share (1/6, 1/2 and 5/6 of the side). A thin line thus samples its core and a damped line its wings, hundreds of Doppler widths out. Three points per side were chosen by test: against an exact reference the factor on the Solomon rate is 0.86–1.03 with one point per side, 0.99–1.02 with two and 0.98–1.04 with three, and the line-by-line rms falls from 0.03–0.45 to 0.03–0.21 (H-006). Numerically:

- only lines with $\tau_0 \geq 10^{-2}$ shade; a shading line's reach is $\Delta\tilde{\nu}_D \max(12, \sqrt{\tau_0 a/(\sqrt{\pi} H(a,0)\,\varepsilon)})$ with $\varepsilon = 10^{-4}$, the offset where its wing optical depth falls to 10^{-4};
- the full Voigt function is used within 12 Doppler widths and the Lorentzian wing $\tau_0 a/(\sqrt{\pi} H(a,0) x^2)$ beyond;
- lines whose reach exceeds $100\,\mathrm{cm}^{-1}$ ("long") shade every line through their Lorentzian wings, summed as $\sum_j C_j/(\tilde{\nu} - \tilde{\nu}_j)^2$ with four interleaved partial sums; other lines ("short") are added where they reach, within $100\,\mathrm{cm}^{-1}$; lines within 40 Doppler widths of line i ("near") are evaluated term by term;
- the lines are processed in wavenumber order so that the windows only move forward; a thin line ($\tau_0 < 1$) takes the far contribution once at its centre; lines that no shading line reaches keep $\mathcal{O} = 1$.

In `solveCell` the factor is computed once per cell and path, at the columns in front of the cell plus half the cell's own (Section 6.10), and multiplies the cell-averaged $\langle S \rangle$ of Equation (35): a product of means stands for the mean of the product, adequate where the other lines' shading changes little across the cell. The result was checked against the same integral on a $0.05\,\mathrm{cm}^{-1}$ grid across $84\,000$–$119\,500\,\mathrm{cm}^{-1}$ (every line with $\tau_0 > 10^{-3}$ a Voigt profile). The exact factor on the Solomon rate of the H-005 slab's ground-state levels is 0.96, 0.93, 0.86, 0.65, 0.49 and 0.32 at $N(\mathrm{H_2}) = 10^{17}$, 10^{18}, 10^{19}, 10^{20}, 3×10^{20} and $10^{21}\,\mathrm{cm}^{-2}$; the model reproduces it to 0.2–1.6% up to $3 \times 10^{20}\,\mathrm{cm}^{-2}$ and to 4.5% at $10^{21}\,\mathrm{cm}^{-2}$, and the exact factor is within 5% of the overlap term of Equation (34) (H-006). Overlap lowered the total pumping through that slab by 33%.

Not included: a photon absorbed in another line's wing is counted as lost rather than as a pump of that line (cross-pumping), which underestimates the pumping of the weaker lines where the band is saturated; and the H I Lyman-series lines, which the MC field does not contain either (Section 3.6), do not shade the band. Their damping wings remove continuum around Lyβ, Lyγ and Lyδ (1026, 973 and 950 Å) over equivalent widths of about 4, 1.4 and 0.7 Å at $N(\mathrm{H\,I}) = 10^{21}\,\mathrm{cm}^{-2}$, growing as $N(\mathrm{H\,I})^{1/2}$ (an estimate of this document from the damping-wing curve of growth, i.e. a few per cent of the 912–1110 Å band); pumping and dissociation behind such columns are overestimated (Section 12). The Meudon code includes them in its exact transfer (Le Petit et al., 2006).

6.8 Escape probabilities

The probability that a photon emitted in line t ($u \to l$) at the cell centre leaves the cloud is (`h2LineEscape`, `h2_mod.f90:1344-1361`; `pdr_mod.f90:5821-5835`)

$$\beta_t = \frac{1}{12} \sum_{k=1}^{12} S(\sigma_t N_{l,k},\, a_t), \tag{39}$$

with $N_{l,k}$ the column of the line's lower level from the cell centre to the edge of the model along line ray k (Section 5.6), the cell's own half included, from the populations of the previous sweep. With complete redistribution the photon is emitted with the absorption profile, so the probability that it crosses a column in one direction, $\int \phi\, e^{-\tau} \mathrm{d}x / \int \phi\, \mathrm{d}x$, is the function S of Equation (31); the mean over the twelve HEALPix directions is the mean over the sphere. The medium is static (no

velocity field, so no Sobolev-type escape), the escape probability is that of the cell centre rather than a mean over the cell, and each line is treated alone (no overlap in the trapping). The Meudon code makes the same on-the-spot approximation for the ultraviolet photons of the H_2 cascades (Le Petit et al., 2006).

6.9 Formation on grains: the nascent distribution

Molecules formed on grains are placed in the (v, J) distribution of Takahashi and Uehara (2001), as Cloudy applies it (Shaw et al., 2005) (`h2FormDist`, `h2_mod.f90:589-639`). With D the dissociation energy of X from its $(0,0)$ level ($36\,118.11\,\mathrm{cm}^{-1}$, 4.478 eV; their $D_e = 4.48\,\mathrm{eV}$), $E_{\rm off} = 0.273\,\mathrm{eV}$ the energy of $(0,0)$ above the bottom of the potential well (their zero-point energy $E_{v0} = 0.27\,\mathrm{eV}$; 0.273 eV is Cloudy's value), $E(v,J)$ the level energies above $(0,0)$, and per surface the parameters of Table 7:

$$E_{vm} = X_{\rm vib} D + E_{\rm off}, \qquad E_v = E(v,0) + E_{\rm off}, \tag{40}$$

$$E_{\rm dust}(v) = D\,X_{\rm dust}\Big[1 - \frac{E_v - E_{vm}}{D + E_{\rm off} - E_{vm}}\,\frac{1 - X_{\rm dust}}{2}\Big], \qquad E_{\rm H_2}(v) = D + E_{\rm off} - E_{\rm dust}(v), \tag{41}$$

$$F(v) = e^{-x_v^2}, \quad x_v = \begin{cases} \dfrac{E_v - E_{vm}}{g_1 E_{vm}} & E_v \le E_{vm} \\[2ex] \dfrac{E_v - E_{vm}}{g_2\,(E_{\rm H_2}(v) - E_{vm})} & E_v > E_{vm} \end{cases}, \tag{42}$$

$$E_{\rm rot}(v) = (E_{\rm H_2}(v) - E_v)\,\frac{X_{\rm rot}}{X_{\rm rot} + X_{\rm trans}}, \qquad \phi(v,J) \propto g_{\rm ns}\,\frac{F(v)}{\sum_v F}\,(2J+1)\,\tfrac{1}{2}\Big[e^{-y^2} + e^{-\Delta E/E_{\rm rot}}\Big], \tag{43}$$

with $\Delta E = E(v,J) - E(v,0)$, $y = (\Delta E - E_{\rm rot})/(E_{\rm rot}/2)$, $g_{\rm ns} = 3$ (ortho) or 1 (para), $\phi = 0$ where $E_{\rm rot} \le 0$ (high v; J. Takahashi's prescription as quoted in Cloudy's source), and ϕ normalised to one. These are equations 1–7 of Takahashi and Uehara (2001): the energy left on the grain $E_{\rm dust}(v)$ (their eq. 1), the energy that stays in the molecule (eq. 2) and its share in rotation (eq. 3), and a vibrational distribution peaked at E_{vm} with different widths below and above it (eqs. 5–7). The rotational distribution is the mean of their two models, a Gaussian about $E_{\rm rot}$ of width $E_{\rm rot}/2$ (model A, eq. 12) and a thermal distribution at $E_{\rm rot}$ (model B, eq. 13); the mean is applied before one overall normalisation, as in Cloudy. Carbonaceous, silicate and icy surfaces put the new molecules mainly near $v = 2$, 6 and 7 respectively (Shaw et al., 2005). The fraction formed above $4100\,\mathrm{cm}^{-1}$ agrees with Cloudy's network to four digits (0.9771, H-004).

The energy budget of a formation follows by conservation: $E_{\rm grain} = \langle E_{\rm dust}\rangle_\phi$ is left on the grain, $E_{\rm kin} = D - E_{\rm grain} - \langle E(v,J)\rangle_\phi$ is the translational energy of the departing molecule, which heats the gas directly, and the internal energy $\langle E(v,J)\rangle_\phi$ is either radiated in H_2 lines or returned to the gas by collisional de-excitation, i.e. through $\mathcal{H}_{\rm coll}$ (Section 6.10). $E_{\rm kin}$ is 0.834 eV for silicate and 1.142 eV for carbonaceous surfaces, against 3D-PDR's 1.5 eV (H-013). $E_{\rm grain}$ is a diagnostic only (`h2Cells.out`); it is not added to MOCASSIN's grain energy balance.

Table 7: Takahashi and Uehara (2001, their Table 2) parameters as used (`h2_mod.f90:132-139`); the ice set is not used by `solveCell`.

Surface	$X_{\rm vib}$	$X_{\rm dust}$	$X_{\rm rot}$	$X_{\rm trans}$	g_1	g_2
ice	0.70	0.04	0.14	0.12	0.3	0.6
silicate	0.60	0.10	0.15	0.15	0.4	0.6
carbonaceous	0.20	0.40	0.15	0.25	0.9	0.4

In each cell the silicate and carbonaceous distributions are mixed in the proportion of the chemistry's

own formation rate on each material: $w_{\rm sil}$ and $w_{\rm gra}$ are `H2_FORMATION_RATE` (Section 7.8) evaluated with only that material's grain area (`pdr_mod.f90:5839-5876`). A cell with no grain formation (no dust, or $T_d = 0$) takes a Boltzmann distribution at T_e, i.e. no formation pumping, and is counted. Formation through H^- and $H_2{}^+$, which populates specific levels, is not distributed over the levels: every molecule the closure forms enters in ϕ.

6.10 The per-cell algorithm

`solveCell` (`pdr_mod.f90:5665-5954`) does, for one cell:

1. **Field** (:5667–5695): b from T_e; $J_\nu^{\rm dir}$, $J_\nu^{\rm dif}$ per bin and $\eta^{\rm dir}$, $\eta^{\rm dif}$ per line (Equation 9).
2. **Cell geometry** (:5697–5706): cell widths, $\ell_{\rm iso}$, and the cell's own column for the diffuse paths from its populations of the last sweep, $\Delta N_l^{\rm iso} = n_l n({\rm H_2}) \ell_{\rm iso}$.
3. **Diffuse shielding** (:5712–5802), if any line has $\eta^{\rm dif} > 0$: `traceDiffuseSites` gives the sites and their weights w (Section 5.4); walking the stored rays again, each site gets a column per level,

$$N_{l,\rm site} = N_l^\star(P) + n_l(P)\, n({\rm H_2}, P)\, \ell_{\rm own}(P) + \sum_{\rm between} n_l\, n({\rm H_2})\, \ell + \tfrac{1}{2}\, n_l(P)\, n({\rm H_2}, P)\, \ell_P \qquad (44)$$

   (the level-resolved form of Equation 20, with the source-ward level columns $N_l^\star$ of `starColumns`). Evaluating S for every line at every site would cost $32\,453 \times N_{\rm site}$ look-ups, and a cell has hundreds of sites, so the sites are binned:

   - four groups by total column $\sum_l N_{l,\rm site}$, with edges 10^{17}, 10^{19} and $3 \times 10^{20}\,{\rm cm^{-2}}$;
   - within a group and for each level, a histogram of the weights in $\log_{10} N_{l,\rm site}$, 0.1 dex bins from $10^9\,{\rm cm^{-2}}$ (150 bins, the end bins catching the columns beyond), plus a "thin" weight for sites where $\sigma_{{\rm max},l} N_l < 10^{-4}$, $\sigma_{{\rm max},l}$ being the largest cross section of the lines out of l;
   - the overlap factor (Equation 38) once per group with more than 10^{-4} of the total weight, at the group's weight-mean level columns plus $\frac{1}{2}\Delta N^{\rm iso}$ (none for the $< 10^{17}\,{\rm cm^{-2}}$ group);
   - per line, the diffuse shielding factor

$$\langle S\rangle_i^{\rm dif} = \frac{1}{\sum w} \sum_g \mathcal{O}_{i,g} \Big[w_{l,g}^{\rm thin}\, \langle S\rangle(0) + \sum_{\rm bins} w_{l,g,\rm bin}\, \langle S\rangle(\sigma_i N_{\rm bin}) \Big], \qquad (45)$$

     each $\langle S\rangle$ averaged through the cell's own column $\sigma_i \Delta N_l^{\rm iso}$ by Equation (35).

   The binning replaces each site's column by its bin centre (an error of at most ± 0.05 dex in N_l) and each group's overlap by that at the group's mean columns. If there is diffuse light but no site (no dust on any of the twelve rays) the diffuse part rides the direct path (:5799–5802).
4. **Direct path** (:5804–5819): source-ward level columns $N_l^\star$ and the chord $\ell_{\rm dir}$.
5. **Escape** (:5821–5835): β_t over the twelve line rays (Equation 39).
6. **Formation** (:5839–5876): mixture weights, ϕ, $E_{\rm kin}$, $E_{\rm grain}$.
7. **Solve** (:5879–5914). `h2LineShieldPrep` stores, for the direct column, each line's τ_0, $W(\tau_0)$ and damping index. Then up to 12 passes: colliders from the current ortho fraction; the cell's own direct column $\Delta N_l^{\rm dir} = n_l n({\rm H_2}) \ell_{\rm dir}$ from the current populations; on the first pass only, the direct overlap factor at $N^\star + \frac{1}{2}\Delta N^{\rm dir}$; the direct shielding $\langle S\rangle_i^{\rm dir} \mathcal{O}_i^{\rm dir}$ (`h2LineShieldEval`); the shielded occupation numbers

$$\eta_i = \eta_i^{\rm dir}\, \langle S\rangle_i^{\rm dir} \mathcal{O}_i^{\rm dir} + \eta_i^{\rm dif}\, \langle S\rangle_i^{\rm dif} \qquad (46)$$

(or $(\eta_i^{\rm dir}+\eta_i^{\rm dif})$ times the direct factor when there is no site); and the level balance (`h2Solve`). The loop stops when the ortho fraction changes by less than 10^{-4}; a cell that had settled in the previous sweep needs one pass, and the mean was 1.0–1.6 passes per cell (H-011).

8. **Products** (:5917–5952): $k_{\rm Sol}=\sum n_l k_l$; the absorption rate $\sum n_l \sum_u P_{lu}$; for comparison the chemistry's own single-line rate on the same field and columns (`h2SolChem`, 3D-PDR's `H2PDRATE`); diagnostics; and three heating terms per molecule: the atoms' kinetic energy from dissociation $\sum_l n_l H_l^{\rm dis}$, the net heating by collisions within X

$$\mathcal{H}_{\rm coll}=hc\sum_c\sum_{u>l}n_c k_{ul}^c\Big(n_u-n_l\frac{g_u}{g_l}e^{-hc(E_u-E_l)/kT}\Big)(E_u-E_l)\quad[\mathrm{erg\,s^{-1}}] \tag{47}$$

(`h2CollHeat`, `h2_mod.f90:1425-1448`; Cloudy's `HeatDexc`), and the kinetic energy per molecule formed, $E_{\rm kin}$. $\mathcal{H}_{\rm coll}$ is the energy that collisions take out of the levels populated above their Boltzmann ratio at T by pumping and formation; it is negative, i.e. H_2 line cooling, where the gas is hotter than the level populations. A pump deposits in the gas only what $\mathcal{H}_{\rm coll}$ recovers from the levels it feeds, and a dissociation the kinetic energy of the atoms, so the three terms do not count any energy twice.

6.11 Sweeps, parallel strategy and convergence

`h2LevelModel` reads the data once and, on every call:

- takes $n(\mathrm{H_2})=n_{\rm H}x(\mathrm{H_2})$ of every cell from the chemistry; on the first call it starts every cell from Boltzmann populations at T_e, or, on a warm start, from `output/h2Pop.bin` if that file describes the same grid and level set (`h2PopResume`, `pdr_mod.f90:5961-6012`, H-020); later calls start from the last call's populations. A cold start on a converged model overshoots the dissociations of its first sweep by 38% (H-016); a resumed one is within 0.6% of the checkpointed run (H-020);
- for each sweep, rebuilds the source-ward level columns of every cell, $N_l^\star=\sum n_l\, n(\mathrm{H_2})\,\ell$ along `rayCells` to each source, weighted over sources by r_s^{-2} and excluding the cell itself (`starColumns`, `:5621-5659`); the other cells' populations in all columns are those of the previous sweep, so a sweep is a Jacobi iteration over the cells, whose result does not depend on the order in which the cells are solved;
- distributes the cells dynamically: the ranks take cells one at a time from a shared counter in an MPI-3 one-sided window allocated on rank 0 (`MPI_Win_allocate`, `MPI_Fetch_and_op`, `:5398-5426`); the first sweep takes them in a fixed pseudo-random order (a Fisher–Yates shuffle driven by a linear congruential generator), later sweeps in decreasing cost of the previous sweep (`orderCells`, `:5538-5559`), so that cores of different speed finish together; a cell's result does not depend on the rank that solves it;
- reduces populations, rates and heating terms with `MPI_Allreduce` (in single precision);
- stops when the luminosities of 1–0 S(1), 2–1 S(1), 0–0 S(0) and 0–0 S(1), summed over the grid in the optically thin limit, all change by less than 1% between sweeps (`:5459-5495`), at most `nSweepMax` sweeps.

Inside the MC iteration `nSweepMax`= 1: one sweep per iteration, the level model converging together with the rest of the model. On the finished model `nSweepMax`= 12 (Section 10). On `pn150k` (41^3, 35 384 cells, 14 ranks) a call took 3736 s before and 1474 s after the optimisations of H-011; the overlap calculation was about half of the timed work (H-007).

7 Reaction rates handed to the chemistry

The chemistry's rate coefficients are computed per cell in `pdrStep` (`pdr_mod.f90:3180-4580`) by 3D-PDR's `CALCULATE_REACTION_RATES` (`pdr/calc_reac_rates.F90`), called at `pdr_mod.f90:3692` with MOCASSIN's inputs, and then partly overwritten with rates that MOCASSIN or the level model own. Table 8 summarises where each radiation-driven rate comes from; reaction numbers are those of `dataPDR/rates_reduced.d`.

Table 8: Origin of the radiation- and cosmic-ray-driven rates in the 3D build.

Reaction(s)	Number(s)	Rate used	Section
$H_2 + h\nu \rightarrow 2H$	329	the level model's $k_{\rm Sol}$	7.3
$CO + h\nu \rightarrow C + O$	290	$\alpha_{\rm CO} \sum_r \chi_r f_r \Theta$ (van Dishoeck and Black, 1988)	7.4
24 other photoreactions	266–268, 270–288, 291–292	$\alpha\chi$ (Draine spectral shape)	7.2
C, Mg $+ h\nu \rightarrow$ ion $+ e^-$	269, 289	set to zero	7.5
ionisation of H, He, C, O, Mg	293, 297, 298, 299, 312	cosmic-ray (or, Mg, induced-photon) rate plus MOCASSIN's photoionisation rate	7.5
$H_2 + \mathrm{CRP} \rightarrow H_2^+ + e^-$	296	cosmic-ray rate plus MOCASSIN's H_2 photoionisation	7.5
$X^+ + e^- \rightarrow X + h\nu$ (X = H, He, C, O, Mg)	242–246, 248, 249	MOCASSIN's total recombination coefficient	7.5
cosmic-ray ionisation	293–300	$\alpha\,\zeta/\zeta_0$	7.6
cosmic-ray-induced photoreactions	301–315	$\alpha(\zeta/\zeta_0)(T/300)^{\beta}\gamma/(1-\omega)$	7.6
charge transfer with H, H^+	156–157, 161–163, 167	`pdrCtCoef`	7.7
H + H (grain) $\rightarrow H_2$	328	Cazaux and Tielens (2002)	7.8

7.1 The field as a set of rays

3D-PDR evaluates each photoreaction as a sum over rays, each ray r carrying a field strength χ_r (in Draine units), a visual extinction $A_{V,r}$ and shielding columns. In the 3D build the bridge builds $n_{\rm chem} = 1 + N_{\rm site}$ (+1 fallback) rays (`pdr_mod.f90:3383-3455`, D-348, D-350), which represent the local MC field split by the paths its photons took (Sections 3.2 and 5.4):

Table 9: The rays handed to the chemistry in the 3D build.

Ray	χ_r	$A_{V,r}$	LW factor	Columns $N(H_2), N(CO)$
0 (direct)	$\chi_{\rm dir}$	0	$f_{\rm dir}$	source-ward, Eq. (16)
site j	$\chi_{\rm dif}\, w_j / \sum w$	0	$f_{\rm dif}$	two-leg, Eq. (20)
fallback	$\chi_{\rm dif}$	0	$f_{\rm dif}$	source-ward (only when $\chi_{\rm dif} > 0$ and no site exists)

$A_{V,r} = 0$ on every ray (`:3439`) because the MC field has already been attenuated by the dust; applying 3D-PDR's $e^{-\gamma A_V}$ again would count the dust twice (D-343, D-350). The C I column handed over is zero (`:3627`) because MOCASSIN's opacity already contains the C I photoionisation continuum; the HD and S columns are zero (neither species is in the network). If a source-ward column is missing (negative), it falls back to $(A_V/(A_V/N_{\rm H}))\,x_X$ (`:3458`, `:3603-3605`); zero is a valid column and is kept (D-088). The columns exclude the target cell (Section 5.2); only the level model averages the shielding through the cell's own gas (Section 6.6).

7.2 Generic photoreactions

For every reaction $X + \mathrm{PHOTON} \rightarrow \ldots$ without a dedicated treatment, 3D-PDR's rate is $k = \sum_r \alpha\,\chi_r\,e^{-\gamma A_{V,r}}$ [s^{-1}] (`pdr/calc_reac_rates.F90:321-347`), with α the rate in the unattenuated Draine field and γ the dust-attenuation exponent of the UMIST format; with $A_{V,r} = 0$,

$$k = \alpha\,(\chi_{\rm dir} + \chi_{\rm dif}) = \alpha\,\chi, \tag{48}$$

the coefficient α of the rate file times the local MC field in the 6–13.6 eV band. This applies to the photodissociation and photoionisation of H_2^+, H_3^+, CH, CH^+, CH_2, CH_2^+, CH_3, CH_3^+, CH_4, OH, OH^+, H_2O and O_2 (24 of the network's 28 photoreactions). The coefficients α were computed for the Draine spectral shape, while the MC field reddens with depth: its LW share falls from 0.30 to 0.008 through `hiipdr`'s PDR (D-349). Reactions whose cross sections lie mainly above 11.2 eV are therefore overestimated at depth, and those driven by the 6–11 eV part underestimated; none of these reactions is shielded by H_2 or CO lines. Rates computed from cross sections folded over the actual field would remove the assumption; Heays et al. (2017) provide cross sections for 102 species for that purpose, and find that their rates for the interstellar field agree with earlier compilations to about 30% in general.

7.3 H_2 photodissociation: the level model's rate

3D-PDR first computes its own rate on the rays (`H2PDRATE`, `H2SHIELD1`; `pdr/shield.F90:27-55, 231-281`; `pdr/calc_reac_rates.F90:259-268`),

$$k_{\rm H_2}^{\rm net} = \sum_r \alpha_{\rm H_2}\,\chi_r f_r\,J_{\rm FGK}(N_{{\rm H_2},r})\,\texttt{SCATTER}(0, 1000\,\text{Å}), \qquad \alpha_{\rm H_2} = 5.18\times10^{-11}\,\mathrm{s}^{-1}, \tag{49}$$

where $J_{\rm FGK} = J_D + J_R$ is the single-line self-shielding function of Federman et al. (1979, their eqs. A7–A9) for one representative line at 1000 Å with oscillator strength 0.01, half the H_2 in its lower level, a Doppler width $v_{\rm turb}/\lambda$ and a radiative width of $8\times10^7\,\mathrm{s}^{-1}$ (`pdr/shield.F90:46-51, 249-252`). The bridge then replaces it with the level model's Solomon rate (H-012):

$$k(\mathrm{H_2} + h\nu \rightarrow 2\mathrm{H}) = k_{\rm Sol} \qquad (\texttt{pdr_mod.f90:3736}). \tag{50}$$

The same rate enters 3D-PDR's heating routine, so the chemistry and the heating use one H_2 photodissociation rate. The network's own rate is kept as a diagnostic (`h2SolChem`, on the same field split and columns). On `hiipdr` the level model's rate is 2.4–2.6 times the network's in the ionised gas, 1.5–1.8 times in cold unshielded gas, and 0.6–0.7 and 0.4–0.6 times behind source-ward columns of 10^{15}–$10^{17}\,\mathrm{cm}^{-2}$ and more (H-007). Three things separate them: the level model's rate in the unshielded Draine field is 1.23 times $\alpha_{\rm H_2}$ and the stellar field is harder; in warm gas the molecules sit in excited levels that dissociate faster; and the network shields one line of fixed radiative width at the near face of the cell, whereas the level model averages every line through the cell and includes overlap.

7.4 CO photodissociation

`COPDRATE` (`pdr/shield.F90:83-106`; called at `pdr/calc_reac_rates.F90:283-292`):

$$k_{\rm CO} = \alpha_{\rm CO} \sum_r \chi_r \, f_r \, \Theta(N_{{\rm CO},r}, N_{{\rm H}_2,r}) \; \texttt{SCATTER}(0, \bar{\lambda}), \qquad \alpha_{\rm CO} = 2.0 \times 10^{-10}\,{\rm s}^{-1}, \tag{51}$$

with $\alpha_{\rm CO}$ from the rate file (reaction 290). Θ is the combined CO self-shielding and H_2 screening factor for ^{12}CO of van Dishoeck and Black (1988, their Table 5), interpolated by a two-dimensional cubic spline in $\log_{10}\Theta$ over $(\log_{10} N_{\rm CO}, \log_{10} N_{\rm H_2})$ and clamped at the table edges (`COSHIELD`, `pdr/shield.F90:375-416`). The table spans $N({\rm CO}) = 10^{13}$–10^{19} and $N({\rm H_2}) = 10^{19}$–$10^{23}\,{\rm cm}^{-2}$ plus the unshielded entries, which 3D-PDR places at 10^{12} and $10^{18}\,{\rm cm}^{-2}$ (`pdr/modules.F90:353-356`, `pdr/allocations.F90:9-14`); smaller columns are treated as these. $\bar{\lambda}$ is the mean wavelength of the 33 dissociating bands weighted by their contributions to the total rate at the given columns, at which the dust attenuation is evaluated (`LBAR`, their eq. 4, `pdr/shield.F90:570-599`):

$$\bar{\lambda} = \left[(5675 - 200.6\,w) - (571.6 - 24.09\,w)\,u + (18.22 - 0.7664\,w)\,u^2\right] \text{ Å}, \tag{52}$$

with $u = \log_{10} N_{\rm CO}$ and $w = \log_{10} N_{\rm H_2}$ [cm^{-2}], clamped to 913.6–1076.1 Å. With $A_V = 0$, `SCATTER` $= 1$ and $\bar{\lambda}$ has no effect. The window factor f_r corrects for the spectral shape of the MC field in the LW window (Section 3.5); CO is the one photoreaction in this branch that uses it. The columns are those of the chemistry's own $n({\rm H_2})$ and $n({\rm CO})$, on the source-ward or two-leg paths, and exclude the target cell, so CO is under-shielded in the cells of its own transition (Section 12).

The shielding functions of van Dishoeck and Black (1988) were computed for the unmodified Draine field, cool gas ($T \lesssim 100$ K, CO excitation temperature 10 K), Doppler parameters of $1\,{\rm km\,s^{-1}}$ for CO and $3\,{\rm km\,s^{-1}}$ for H_2 and a fixed H I column of $5 \times 10^{20}\,{\rm cm}^{-2}$ (notes to their Table 5). Visser et al. (2009) recomputed them from newer spectroscopic data and find an unattenuated rate of $2.6 \times 10^{-10}\,{\rm s}^{-1}$, 30% above the 2.0×10^{-10} used here, and depth-dependent rates that change by up to a factor of three when the excitation temperature or the Doppler width is raised, for temperatures above 100 K. The warm CO of the illuminated PDR is therefore outside the conditions of the table, and the H_2 screening uses the chemistry's total H_2 column rather than the level model's line-by-line transmission (Section 12).

7.5 Photoionisation and recombination of the linked atoms

Five elements are *linked*: their neutral and first ion are species of the network and stages of MOCASSIN (H, He, C, O, Mg; `pdr_mod.f90:553-564`). For them the network's own photoionisation is replaced by MOCASSIN's, computed on the full MC field with the real cross sections (D-358; `pdr_mod.f90:3706-3722`; slots found by structure at `:743-801`):

- the network reaction $X + {\rm PHOTON} \to X^+ + e^-$ is set to zero where it exists (C, reaction 269; Mg, 289; the network has none for H, He and O);
- MOCASSIN's photoionisation rate from the ground state

$$\Gamma_X = \sum_k \frac{J_k^{\rm ste}\,\sigma_X(\nu_k)}{h\nu_k} \quad [{\rm s}^{-1}] \tag{53}$$

  (`update_mod.f90:303-311`; outer-shell cross sections of Verner et al. 1996 and Verner and Yakovlev 1995, `ph_mod.f90:473-476`) is added to the reaction's first-order cosmic-ray slot $X + {\rm CRP} \to X^+ + e^-$ (H, He, C, O: reactions 293, 297, 298, 299), on top of the cosmic rays; Mg has no such reaction and its cosmic-ray-induced photoionisation (312) takes the rate instead;

- for hydrogen, MOCASSIN's collisional ionisation term $2.75 \times 10^{-16}\,T^{3/2}(157\,894/T + 2)\,e^{-157\,894/T}$ is added as well (`update_mod.f90:2620-2630`, stored at `:2859`). It has the form of a rate coefficient [$\mathrm{cm^3\,s^{-1}}$]: at 10^4 K it is 6.8×10^{-16}, the magnitude of the H I collisional ionisation coefficient, but it is added to the photoionisation rate [$\mathrm{s^{-1}}$] without the factor n_e, here and in MOCASSIN's own ionisation balance (`:2769-2770`), so collisional ionisation is underestimated by the factor n_e. With that factor it would reach 10% of a photoionisation rate of $10^{-8}\,\mathrm{s^{-1}}$ at about 1.5×10^4 K for $n_e = 10^4\,\mathrm{cm^{-3}}$; the omission matters in hot, dense ionised gas and not in the PDR. The source comment attributes the formula to Drake and Ulrich (1980); that paper takes its collisional ionisation rates from Johnson (1972) and tabulates $4.88 \times 10^{-16}\,\mathrm{cm^3\,s^{-1}}$ for the ground level at 10^4 K (their Table 3), so the formula's origin is not established (Section 12);
- the radiative recombination $X^+ + e^- \rightarrow X + \mathrm{PHOTON}$ takes MOCASSIN's total recombination coefficient (radiative, from the fits of Verner and Ferland 1996 and the others listed at `update_mod.f90:4101-4115`, plus dielectronic) in the first of its temperature ranges, the other ranges set to zero;
- H_2 photoionisation, $H_2 + h\nu \rightarrow H_2^+ + e^-$, is added to the slot $H_2 + \mathrm{CRP} \rightarrow H_2^+ + e^-$ (296) with $\Gamma_{H_2} = \sum_{h\nu_k \geq 15.4\,\mathrm{eV}} J_k^{\mathrm{ste}} \sigma_{H_2}(\nu_k)/h\nu_k$ and the fit of Yan et al. (1998), as revised in their erratum (Yan et al., 2001), to the cross section (`update_mod.f90:2863-2877`; `yanH2xSec`, `ionization_mod.f90:385-411`).

These rates are stored by `updateCell` at every trial temperature of its search, so the chemistry reads those of the temperature it settled on (`update_mod.f90:2838-2878`). In particular, **carbon is photoionised at MOCASSIN's rate**: the MC field contains dust and C I continuum absorption but no H_2 line absorption, so the C I rate is not reduced by the H_2 lines that overlap the C I continuum (Section 12). The C I treatment of Kamp and Bertoldi (2000) in 3D-PDR (`CIPDRATE`, `pdr/shield.F90:132-157`), $k = \alpha\chi \exp[-\gamma A_V - 1.1 \times 10^{-17} N_{\mathrm{C}} - 0.9\,T^{0.27}(N_{H_2}/1.59 \times 10^{21})^{0.45}]$, is computed with $A_V = 0$ and $N_{\mathrm{C}} = 0$, i.e. with its H_2 term only, and then discarded. (Equation 8 of Kamp and Bertoldi 2000, which the code cites, normalises the H_2 term at $10^{22}\,\mathrm{cm^{-2}}$, not 1.59×10^{21}; the vendored term is 2.3 times larger at a given column.) The ionisation balance of the linked elements is moreover imposed from MOCASSIN before and after the solve (Section 8.3).

7.6 Cosmic rays and cosmic-ray-induced photons

The cosmic-ray ionisation rate is the input `pdrZeta` (default $10^{-16}\,\mathrm{s^{-1}}$, `common_mod.f90:902`), stored normalised to the UMIST reference rate $\zeta_0 = 1.3 \times 10^{-17}\,\mathrm{s^{-1}}$ (`pdr_mod.f90:360`), and is the same in every cell (3D-PDR's attenuation option `CRATTENUATION` is not compiled). Direct cosmic-ray reactions have $k = \alpha\,\zeta/\zeta_0$, with α the rate at ζ_0 (`pdr/calc_reac_rates.F90:353-370`). Reactions driven by the ultraviolet photons that the secondary electrons of cosmic-ray ionisation excite in H_2 (Prasad and Tarafdar, 1983) have

$$k = \alpha\,\frac{\zeta}{\zeta_0}\left(\frac{T}{300\,\mathrm{K}}\right)^{\beta}\frac{\gamma}{1-\omega} \quad [\mathrm{s^{-1}}] \tag{54}$$

(`pdr/calc_reac_rates.F90:381-401`), the UMIST form of the rate of Gredel et al. (1989, their eq. 7), $\zeta\,p_M/(1-\omega)$ per molecule: $\alpha = \zeta_0$ for all 15 such reactions of the network, $\gamma = p_M$ is the efficiency of the photoreaction per ionisation, and ω the grain albedo, which enters because the induced photons are removed by grain absorption, $\sigma_g(1-\omega)$, rather than by extinction. That approximation holds when grains dominate the absorption of the induced photons, which Gredel et al. (1989) note is the case for all molecules except CO; the temperature factor is used only by CO ($\beta = 1.17$, reaction 313). Here ω is MOCASSIN's grain albedo in the LW band (Section 4). The network has 8 `CRP` and 15 `CRPHOT` reactions. Heays et al. (2017) find newly calculated cosmic-ray-induced rates that differ from the standard values by up to a factor of five.

7.7 Charge transfer with hydrogen

The charge-transfer reactions of the linked elements with H and H^+ (reactions 156–157, $He^+ + H$; 161–163, $O^+ + H$ and $H^+ + O$; 167, $H^+ + Mg$; the network has none for carbon) use one coefficient per reaction in both the chemistry and MOCASSIN's ionisation balance (`pdrCtCoef`, `common_mod.f90:1383-1436`; `pdr_mod.f90:3742-3747`; D-403): the network's $\alpha(T/300)^{\beta}e^{-\gamma/T}$ in its temperature ranges, replaced by the fits of Kingdon and Ferland (1996), $k = a\,t_4^{b}(1 + c\,e^{d\,t_4})$ with $t_4 = T/10^4\,\mathrm{K}$, inside their ranges for $He^+ + H$ (6000–10^5 K) and $H^+ + Mg$ (5000–30 000 K) (`pdr_mod.f90:848-855`). For oxygen the network's rates are fits to Stancil et al. (1999) and are kept. The coefficients are evaluated at the cell's own T_e, not at the clamped temperature of the other rates.

7.8 H_2 formation on grains

The reaction $H + H \rightarrow H_2$ on grains (reaction 328) takes the rate of Cazaux and Tielens (2002, their eqs. 15–18), whose model Cazaux and Tielens (2004) develop in full (`H2_FORMATION_RATE`, `pdr/h2_form.F90:9-134`, flag `H2FORM_CT02`; called at `pdr/calc_reac_rates.F90:116`):

$$R_{\mathrm{f}} = \tfrac{1}{2}\,v_{\mathrm{th}}\left(\sigma_{\mathrm{H}}^{\mathrm{sil}}\,\epsilon_{\mathrm{sil}} + \sigma_{\mathrm{H}}^{\mathrm{gra}}\,\epsilon_{\mathrm{gra}}\right) S_{\mathrm{H}}(T, T_d) \quad [\mathrm{cm^3\,s^{-1}}], \tag{55}$$

$$v_{\mathrm{th}} = 1.45\times10^{5}\sqrt{T/100\,\mathrm{K}}\,\mathrm{cm\,s^{-1}}, \qquad S_{\mathrm{H}} = \left[1 + 0.04\sqrt{T+T_d} + 0.002\,T + 8\times10^{-6}\,T^2\right]^{-1}, \tag{56}$$

$$\epsilon = \frac{\xi}{1+F_1+F_2}, \quad F_1 = \frac{\mu F}{2\nu_{\mathrm{H_2}}e^{-E_{\mathrm{H_2}}/T_d}}, \quad F_2 = \frac{\left(1+\sqrt{(E_{\mathrm{HC}}-E_{\mathrm{S}})/(E_{\mathrm{HP}}-E_{\mathrm{S}})}\right)^2}{4}\,e^{-E_{\mathrm{S}}/T_d}, \tag{57}$$

$$\xi = \left[1 + \frac{\nu_{\mathrm{HC}}}{2F}\,e^{-1.5E_{\mathrm{HC}}/T_d}\left(1+\sqrt{(E_{\mathrm{HC}}-E_{\mathrm{S}})/(E_{\mathrm{HP}}-E_{\mathrm{S}})}\right)^2\right]^{-1}, \tag{58}$$

with temperatures in K. v_{th} is the mean speed of H atoms, the factor $\frac{1}{2}$ makes one molecule of two atoms, and σ_{H} [$\mathrm{cm^2}$] is the geometric grain cross section per H nucleus. S_{H} is the sticking coefficient of Hollenbach and McKee (1979, eq. 3.7), which they take from calculations by Burke and Hollenbach; it is 0.58, 0.33 and 0.08 at $T = 100$, 300 and 1000 K (for $T_d = 20$ K). The recombination efficiency ϵ is limited by three processes (Cazaux and Tielens, 2002): F_1, newly formed molecules that stay on a very cold surface; F_2, the evaporation of physisorbed atoms before they meet a chemisorbed one, which controls the decline above a few tens of K; and ξ, the evaporation of chemisorbed atoms at several hundred K. The constants are $F = 10^{-10}$ monolayers $\mathrm{s^{-1}}$ (the accretion rate), $\mu = 0.005$, $\nu_{\mathrm{H_2}} = 3\times10^{12}\,\mathrm{s^{-1}}$, $\nu_{\mathrm{HC}} = 1.3\times10^{13}\,\mathrm{s^{-1}}$, $E_{\mathrm{HC}} = 3\times10^{4}$ K, and $(E_{\mathrm{S}}, E_{\mathrm{H_2}}, E_{\mathrm{HP}}) = (110, 320, 450)$ K for silicate and $(260, 520, 800)$ K for graphite. The functional form and μ are those of Cazaux and Tielens (2002); the per-material energies are those tabulated by Röllig et al. (2013, Table C.1) from the work of Cazaux and Tielens, whereas Cazaux and Tielens (2002, Table 1) list $(E_{\mathrm{S}}, E_{\mathrm{HP}}, E_{\mathrm{HC}}) = (200, 600, 10^4)$ K for their silicate surface, the values Cloudy uses; Röllig et al. (2013) also set the F_1 term to zero. With the vendored parameters (Equation 55 evaluated for this document) ϵ_{sil} is close to 1 between 6 and 15 K, 0.90 at 20 K, 0.25 at 50 K and 0.10 at 100 K, and ϵ_{gra} close to 1 between 10 and 40 K, 0.43 at 100 K and 0.12 at 300 K; both fall to zero below about 6 and 9 K (F_1) and above about 800 K (ξ). The silicate energies are the main reason why the silicate coefficient is 0.50–0.91 of Cloudy's on `hiipdr` (L-33, H-022).

The grain areas per H $\sigma_{\mathrm{H}}^{\mathrm{sil}}$, $\sigma_{\mathrm{H}}^{\mathrm{gra}}$ are MOCASSIN's (Section 4.2), handed over per cell at `pdr_mod.f90:3295-3307`; T_d is MOCASSIN's mean grain temperature (Section 4.4), floored at T_{CMB}; the metallicity factor is 1 (`pdr_mod.f90:347`). The formation rate per unit volume is $R_{\mathrm{f}}\,n_{\mathrm{H}}\,n(\mathrm{H})$ [$\mathrm{cm^{-3}\,s^{-1}}$]. The same function, evaluated with one material's area at a time, weights the two formation distributions of the level model (Section 6.9).

8 The chemical solve

8.1 Network and rate law

The network is 3D-PDR's REDUCED network (files `dataPDR/species_reduced.d` and `dataPDR/rates_reduced.d`), in the format of the UMIST Database for Astrochemistry. Bisbas et al. (2012) introduced its 33 species with 320 reactions drawn from UMIST 2006 (Woodall et al., 2007); Bisbas et al. (2015) describe the 33-species network then in use as a 330-reaction subset of UMIST 2012 (McElroy et al., 2013). The vendored file, from upstream commit `fabcdc6`, has 331 reactions and records neither its UMIST release nor the references of its coefficients, so the release cannot be determined from the files. It has 33 species,

$$\mathrm{H},\ \mathrm{H_2},\ \mathrm{H^+},\ \mathrm{H_2^+},\ \mathrm{H_3^+},\ \mathrm{He},\ \mathrm{He^+},\ \mathrm{C},\ \mathrm{C^+},\ \mathrm{CH},\ \mathrm{CH^+},\ \mathrm{CH_2},\ \mathrm{CH_2^+},\ \mathrm{CH_3},\ \mathrm{CH_3^+},\ \mathrm{CH_4},\ \mathrm{CH_4^+},\ \mathrm{CH_5^+},\ \mathrm{O},\ \mathrm{O^+},\ \mathrm{OH},\ \mathrm{OH^+},\ \mathrm{H_2O},\ \mathrm{H_2O^+},\ \mathrm{H_3O^+},\ \mathrm{O_2},\ \mathrm{O_2^+},\ \mathrm{CO},\ \mathrm{CO^+},\ \mathrm{HCO^+},\ \mathrm{Mg},\ \mathrm{Mg^+},\ \mathrm{e^-},$$

and 332 reactions: 3D-PDR's 331 plus reaction 332, $e^- + \mathrm{H_2^+} \rightarrow e^- + \mathrm{H} + \mathrm{H^+}$ ($2 \times 10^{-7} e^{-30720/T}\,\mathrm{cm^3\,s^{-1}}$), appended in this project because the network had no electron-impact dissociation of $\mathrm{H_2}^+$, Cloudy's main $\mathrm{H_2}^+$ sink in ionised gas (`pdr/UPSTREAM.md`, D-395); the Arrhenius form is the one of Cloudy's `mole_co_umisthack.dat`, which attributes it to Janev et al. (1987). Of the 332, 28 are photoreactions, 8 cosmic-ray and 15 cosmic-ray-induced photoreactions (Section 7), and three carry the grain marker `#`: grain formation of $\mathrm{H_2}$ (328) and the grain-assisted recombinations of $\mathrm{He^+}$ and $\mathrm{C^+}$ (330, 331). These two have no term in the generated right-hand side, so their coefficients are computed but unused and 330 reactions act in the solve. The other 278 are two-body reactions: ion–molecule, neutral–neutral and charge-transfer reactions, dissociative and radiative recombination, radiative association and collisional dissociation. The ODE right-hand side is generated C code selected at compile time (`pdr/odes_reduced.c`, flag `-DREDUCED`); the keyword `pdrNetwork` only validates that the requested network is the compiled one.

Two-body coefficients follow the modified Arrhenius form

$$k(T) = \alpha \left(\frac{T}{300\,\mathrm{K}}\right)^{\beta} e^{-\gamma/T} \quad [\mathrm{cm^3\,s^{-1}}], \tag{59}$$

with γ in K (`pdr/calc_reac_rates.F90:172-252`). Where a reaction has several entries for different temperature ranges, the first whose upper limit is above T (else the last) is used and the others are set to zero; a fit with $\gamma < 0$ is evaluated at the lower limit of its range below it, and one with $\gamma < -200\,\mathrm{K}$ is set to zero there. The chemistry is evaluated at MOCASSIN's T_e clamped to [`pdrTmin`, `pdrTmax`] $= [1,\ 2 \times 10^4]\,\mathrm{K}$ by default (`pdr_mod.f90:3346`), the range over which the fits are meant to be used; MOCASSIN's own floor `teFloor`, 2.725 K by default, keeps T_e above T_{CMB}. The vendored bound $\max(\texttt{pdrTmin}, T_{\mathrm{CMB}})$ (`:364`) is read only by 3D-PDR's own temperature iteration (`pdr/changetemperature.F90`), which the bridge does not call. In terms of fractional abundances $x_i = n_i/n_{\mathrm{H}}$ the system is

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = \sum_{j,k} k_{jk}\, x_j x_k\, n_{\mathrm{H}} + \sum_j k_j\, x_j - x_i\Big(\sum_j k_{ij}\, x_j\, n_{\mathrm{H}} + \sum k_i\Big), \tag{60}$$

with two-body coefficients k_{jk}, k_{ij} [$\mathrm{cm^3\,s^{-1}}$] and first-order rates k_j, k_i [$\mathrm{s^{-1}}$] (photoreactions and cosmic rays, Section 7). Grain formation enters as $R_{\mathrm{f}}\, x(\mathrm{H})\, n_{\mathrm{H}}$ in $\mathrm{d}x(\mathrm{H_2})/\mathrm{d}t$ and twice that as a loss of H. Inside the right-hand side the electron abundance is not a free variable: it is the sum of the network's cations plus `xe_extra`, the electrons of the ions the network has no species for (higher stages, other elements), supplied by the bridge (D-402; `pdr/odes_reduced.c:58`).

8.2 Initial conditions and seeding

First visit of a cell. `seedAbunFromMocassin` (`pdr_mod.f90:3123-3157`, D-208): every molecule starts at zero; each linked element's neutral and first ion take MOCASSIN's stage-1 and stage-2

fractions times the element abundance; $x_e = n_e/n_{\rm H}$. Nothing is read from the network file's dark-cloud defaults.

First PDR iteration. `seedMolecules` (`pdr_mod.f90:2855-3023`, D-209) places a first H_2 and CO content so that the first shielding columns are not zero. H_2 comes from the balance of grain formation and unshielded photodissociation, with the neutral hydrogen of MOCASSIN, $x_{\rm n} n_{\rm H}$, shared between H and H_2:

$$R_{\rm f}\, n_{\rm H}\, n({\rm H}) = D\, n({\rm H_2}), \qquad n({\rm H}) + 2n({\rm H_2}) = x_{\rm n} n_{\rm H} \quad \Longrightarrow \quad x({\rm H_2}) = \frac{R_{\rm f}\, x_{\rm n}\, n_{\rm H}}{D + 2R_{\rm f} n_{\rm H}}, \tag{61}$$

$$D = \alpha_{\rm H_2}\, \chi\, \mathrm{SCATTER}(A_V, 1000\,\text{Å}), \tag{62}$$

with $\alpha_{\rm H_2}$ the network's coefficient ($5.18 \times 10^{-11}\,{\rm s^{-1}}$, read from the rate file) and χ the local MC field of Equation (3); $x({\rm H_2})$ is capped at $x_{\rm n}/2$. CO follows the ansatz $x({\rm CO}) = x({\rm C})\,[2x({\rm H_2})/x_{\rm n}]$, capped by the oxygen, with carbon and oxygen conserved. Steps 4–5 of Table 3 are then repeated on the seeded abundances. The seed attenuates the already attenuated local field χ once more with `SCATTER` and ignores self-shielding; it only provides the starting point, which the iteration then forgets.

8.3 Imposition of MOCASSIN's ionisation before the solve

MOCASSIN owns the ionisation state; the chemistry owns what is locked in molecules (D-153, D-193, D-402; `pdr_mod.f90:3503-3584`). For each linked element X with abundance A_X and molecular content $M_X = \sum_i \nu_{X,i} x_i$ (atoms of X in each species, from the species' formulae, `speciesAtoms`):

$$x(X) = (A_X - M_X)\,\frac{f_1}{\sum_s f_s}, \qquad x(X^+) = (A_X - M_X)\,\frac{f_2}{\sum_s f_s}, \tag{63}$$

where f_s are MOCASSIN's stage fractions; what MOCASSIN holds in stages above the second stays MOCASSIN's. The electron abundance is set to $n_e/n_{\rm H}$ from MOCASSIN, and `xe_extra` to x_e minus the linked first ions (`:3565-3581`). The imposition fixes the starting point of the integration; inside it, the linked atoms are ionised and recombined at MOCASSIN's own rates (Section 7.5), so the solver's neutral/ion ratio should stay close to MOCASSIN's, and the remaining difference is carried by the network's other channels (charge transfer, ion–molecule reactions).

8.4 Integration to steady state

`calculate_abundances` (`pdr/calculate_abundances.c`), called at `pdr_mod.f90:3749`, integrates Equation (60) with CVODE from SUNDIALS 7.8.0 (D-015; Gardner et al. 2022; Hindmarsh et al. 2005). The line numbers in this list are those of `pdr/calculate_abundances.c` unless a file is named:

- variable-order backward differentiation formulae (`CV_BDF`, :196), the stiff integrator, with Newton iteration on a dense direct linear solver (`SUNDenseMatrix`, `SUNLinSol_Dense`, :229–238) and CVODE's difference-quotient Jacobian (the analytic Jacobian is not used);
- relative and absolute tolerances 10^{-8} and 10^{-30} on the fractional abundances (set at `pdr_mod.f90:369-370`, applied at :212); at most 10^7 internal steps per output interval (:115);
- output times from $10^{-4}\,{\rm yr}$, multiplied by 10 until `pdrEndTime` (default $10^7\,{\rm yr}$) (:246, :364);
- `CHEMRETRY` (:248–384): a failed integration is restarted from the initial state up to four more times with a forced first step of $10^{-2}\,{\rm s}$, divided by 100 at each retry;

- `CHEMSTEADY` (:387–437): the integration then continues in blocks that each double the elapsed time, at most seven (128×`pdrEndTime`, 1.3×10^9 yr by default), until no species above the absolute tolerance changes by more than 1% over a block;
- abundances below the absolute tolerance are set to zero and the electron entry is restored (:440–452).

The result is meant to be the steady state of the network at fixed temperature, field, columns and imposed ionisation; there is no time dependence carried between MC iterations. A species that relaxes exponentially on a time scale τ from an offset of order unity passes the 1% test over a doubling once the elapsed time exceeds about $4.6\,\tau$, so species that relax more slowly than about 10^8 yr cannot pass it by the last block; if the seventh block does not pass, the abundances of that time are returned without a flag. An unchanged abundance vector is counted (`pdrNstuck`, `pdr_mod.f90:3860`) but not treated as a failure, since it is also the signature of a converged cell. The slowest chemical process in molecular gas is usually H_2 formation, whose time scale $1/(R_f n_H)$ is $\sim 10^7$ yr at $n_H = 100\,\mathrm{cm}^{-3}$; where the age of the object is shorter the steady state overstates the molecular content (Section 11).

8.5 After the solve

1. The solver's own neutral/ion split of each linked element is overwritten with MOCASSIN's stage-1:stage-2 ratio, keeping the solver's total $x(X) + x(X^+)$ (`pdr_mod.f90:3770-3784`, D-162); the electron abundance is reset to MOCASSIN's (`:3801-3804`, D-231). The solver's own ratio is kept for diagnostics (D-402, D-404).
2. The abundances are stored (`:3922`) and $n(H_2) = x(H_2) n_H$ is written for MOCASSIN's opacity (`:4573`).
3. What the network does to each linked element's X and X^+ beyond the channels `ionBalance` computes itself (photoionisation, cosmic rays, radiative recombination, charge transfer with H) is classified reaction by reaction at the solver's state, as net conversions $X \leftrightarrow X^+$ and as sources and sinks to and from molecules, and stored for MOCASSIN's next ionisation solve (`pdrIonRate`, `:3924-4054`, D-402). The classified totals reproduce the right-hand side's own formation and destruction totals to 4×10^{-16} (D-402).

9 Feedback from the molecules on MOCASSIN

The molecular content computed in Section 8 acts back on the radiation field, the ionisation and the temperature, which closes the loop of Figure 1. It does so through three channels only: the fraction of each element held in molecules, the H_2 photoionisation continuum, and the heating and cooling terms. No molecular line enters the MC transfer.

9.1 Elements locked in molecules

The whole handback for an element is the fraction of it *not* in molecules (`setElemAvail`, `pdr_mod.f90:5120-5144`, called by `pdrToMocassin`, `:5038-5104`):

$$\texttt{elemAvail}_X = \min\Big[\max\Big(1 - \frac{M_X}{A_X},\, 0\Big),\, 1\Big], \qquad M_X = \sum_i \nu_{X,i}\, x_i, \tag{64}$$

for the five linked elements H, He, C, O and Mg, with A_X MOCASSIN's own abundance ($A_H = 1$) and $\nu_{X,i}$ the number of atoms of X in species i, read from the species' formula for every species other than X and X^+ themselves (`:585-603`, D-402). MOCASSIN's `ionBalance` constrains the sum

of the element's ion stages to it (update_mod.f90:3130-3134): carbon in CO therefore appears in no atomic stage, emits no atomic line and gives no free electron. The chemistry writes no ionisation state and no temperature (D-193). After each iteration rescaleIonToAvail (pdr_mod.f90:2764-2803) rescales the stages of every element to elemAvail, keeping MOCASSIN's ratios between them, so that the hydrogen budget $x(\mathrm{H}^0) + x(\mathrm{H}^+) + 2x(\mathrm{H}_2) = 1$ holds in every cell (up to the hydrogen in other molecules) before the next transfer builds its opacity (D-220). It prints the size of the pre-rescale defect ([D220]), which is the informative test of the budget: since the kept second solve (below) no cell of either fixture is off by more than 1% in the written model (L-29).

The network channels classified in Section 8 (sources and sinks of X and X^+ to and from molecules, and net conversions between them) enter ionBalance's rate matrix as sinks on the diagonal and sources on the right-hand side, the sum still constrained to elemAvail (update_mod.f90:3255-3300, D-402). They are those of the previous chemistry, i.e. one iteration behind (L-47). Cosmic-ray ionisation, which MOCASSIN otherwise lacks, is added as MOCASSIN's own physics with the network's coefficients: $\alpha\,\zeta/\zeta_0$ for direct ionisation and $\alpha\,(\zeta/\zeta_0)(T/300\,\mathrm{K})^{\beta}\gamma/(1-\omega)$ for the cosmic-ray-induced photons, $\zeta_0 = 1.3 \times 10^{-17}\,\mathrm{s}^{-1}$, for every linked neutral (update_mod.f90:2916-2930; D-193, D-280).

9.2 H_2 in the gas opacity

$n(\mathrm{H}_2)$ (H2den) adds the H_2 photoionisation continuum above 15.4 eV to the gas opacity of the next transfer (addOpacity, ionization_mod.f90:435-472, D-107),

$$\kappa_\nu \mathrel{+}= n(\mathrm{H}_2)\,\sigma_{\mathrm{H}_2}(\nu), \qquad h\nu \geq 15.4\,\mathrm{eV}, \tag{65}$$

with the cross section fitted by Yan et al. (1998), as revised in their erratum (Yan et al., 2001) (yanH2xSec, ionization_mod.f90:385-411; zero below 15.4 eV, as in Cloudy, and clipped at zero just above threshold where the fit goes negative). The comment above the opacity loop (:444-453) still describes "a single power law anchored at threshold"; the function actually used is the piecewise fit of Yan et al. (1998). This continuum matters only for hydrogen-ionising photons that reach molecular gas. No H_2, CO or H I Lyman-series *line* opacity is added to the MC transfer: line absorption is represented only through the shielding columns.

9.3 Heating and cooling

MOCASSIN solves the thermal balance of every cell, heating equal to cooling per H nucleus, with thermBalance (update_mod.f90:1734-2456) inside its temperature search iterateT (:609-1191). Its own terms are the photoionisation heating by every ion on the MC field; the grain photoelectric heating (setPhotoelHeatCool, update_mod.f90:1460-1556), summed over every grain species and size from the MC field above the photoelectric threshold (work function plus grain potential), with the yield of Weingartner and Draine (2001) (D-289) and the grain charge of MOCASSIN's own charge balance, whose electron sticking was set to the same paper's value (D-326); the recombination, free–free and collisionally excited line cooling of every ion; and the gas–grain collisional exchange (setDustGasCollHeatCool, :1562-1731), which omits collisions with H_2 (L-9). The molecules enter it in three ways.

1. **Coolants.** The coolants named by pdrCoolants (in the fixtures C^+, C, O and ^{12}CO, LAMDA-format files, Schöier et al. 2005) are treated in two ways.

   - C^+, C and O are cooled by MOCASSIN's own model atoms (its electron collision data and all its levels), to which the coolant files' rates for collisions with H, H^+, He and H_2 and the chemistry's escape probability of each transition are added (emission_

`mod.f90:1861-1887`; `common_mod.f90:1489-1635`; D-363). They are solved at every trial temperature, and the same atoms give the reported line fluxes, so the thermal balance and the lines share one atom per ion.

- CO, which MOCASSIN has no atom for, is solved by 3D-PDR's escape-probability level-population solver in `pdrStep` at 16 temperatures log-spaced over [`pdrTmin`, `pdrTmax`], a factor 1.94 apart for the defaults (`initPDRcoolTab`, `pdr_mod.f90:2713-2732`, D-215). The temperature search reads this table by log–log interpolation, clamped at its ends (`pdrCoolAt`, `common_mod.f90:1445-1486`), and adds it to MOCASSIN's cooling (`update_mod.f90:2232-2249`). For CO the table misses an exact solve at the cell's temperature by a mean $|\Delta \ln \Lambda|$ of 0.02 (`hiipdr`) and 0.004 (`pn150k`) per cell (L-36).

In both the level populations are in statistical equilibrium with the colliders H, H^+, e^-, He and H_2 at the chemistry's abundances (the linked ions and n_e being MOCASSIN's). The escape probability of a transition $u \to l$ is the mean over the twelve line rays of Section 5.6 of $\beta = (1 - e^{-\tau})/\tau$, with

$$\tau = \frac{A_{ul}\, c^3}{8\pi \nu_{ul}^3\, b} \sum_{\text{cells}} \Big(n_l \frac{g_u}{g_l} - n_u \Big) \ell, \tag{66}$$

each crossed cell contributing its own level populations as last stored (`pdr/escape_probability.F90:121-137, 196-202`; D-377). With `-DSOBOLEV` the rate equations use the equivalent net-rate form, the spontaneous rate $A_{ul}\beta$ and the radiative rates driven by $\beta J_{\rm bg}$ (`:219-233`), where the background $J_{\rm bg}$ is MOCASSIN's own J_ν at the line plus the CMB (D-377). The cooling is the net escaping power, $\Lambda_{ul} = A_{ul} h\nu_{ul} n_u \beta\, (S_{ul} - J_{\rm bg})/S_{ul}$ with S_{ul} the line source function (`:209-211`).

2. **Extra heating.** 3D-PDR's heating routine `CALC_HEATING` (`pdr/heatingfunctions.F90`) is called at `pdr_mod.f90:4499` with the local field χ at MOCASSIN's T_e. The terms MOCASSIN computes itself are excluded: grain and PAH photoelectric heating (3D-PDR's Bakes and Tielens 1994 expression with the modifications of Wolfire et al. 2003, and its alternative Weingartner and Draine 2001 fit), carbon photoionisation heating and gas–grain collisions; these are the only terms that read χ. The rest, `pdrHeatXtra` (`pdr_mod.f90:4545-4546`), is added to MOCASSIN's heating (`update_mod.f90:2391`) and held fixed during the temperature search:

   - *H_2 terms from the level model* (`pdr_mod.f90:4504-4528`, H-013): formation, $R_{\rm f}\, n_{\rm H}\, n({\rm H})\, E_{\rm kin}$, with $E_{\rm kin}$ the kinetic energy of a newly formed molecule from the formation distribution; dissociation, $n(H_2) \sum_l n_l H_l^{\rm dis}$, the kinetic energy of the atoms; and the net collisional term $n(H_2)\, \mathcal{H}_{\rm coll}$, which is the heating by the de-excitation of pumped and newly formed molecules and, where the gas is warmer than the level populations, H_2's line cooling (Section 6.10). They replace 3D-PDR's fixed energies (1.5 eV per formation, which the vendored code attributes to Hollenbach and Tielens 1999; 0.4 eV per dissociation; 9×2.2 eV per dissociation for pumping, with the critical-density factor it attributes to Hollenbach and McKee 1979), which are kept beside them for comparison (`h2Cells.out`);
   - cosmic-ray heating, $\Gamma_{\rm CR} = 9.4\,{\rm eV} \times \zeta\, n(H_2)$, i.e. $1.5 \times 10^{-11} \zeta\, n(H_2)\, {\rm erg\,cm^{-3}\,s^{-1}}$, the expression of Tielens and Hollenbach (1985, their eq. A23): about 8 eV of heat per primary ionisation in molecular gas, helium included, with ζ the rate per H_2. Ionisations in atomic gas deposit no heat in this term, which Tielens and Hollenbach (1985) justify by cosmic-ray heating being unimportant there;
   - turbulent dissipation, $\Gamma_{\rm turb} = 3.5 \times 10^{-28}\, (v_{\rm turb}/{\rm km\,s^{-1}})^3\, (L_{\rm turb}/{\rm pc})^{-1}\, n_{\rm H}\, {\rm erg\,cm^{-3}\,s^{-1}}$ with $L_{\rm turb} = 5\,{\rm pc}$ fixed and $v_{\rm turb} =$ `pdrVturb`, which the vendored code attributes to Black (1987): $7 \times 10^{-29}\, {\rm erg\,s^{-1}}$ per H nucleus at the default $1\,{\rm km\,s^{-1}}$;

- the energy released by exothermic reactions (for example the dissociative recombination of HCO^+, H_3^+ and H_3O^+, and He^+ with H_2 and CO), the reactions being found automatically in the network, after Clavel et al. (1978), who found that such reactions could contribute significantly to the heating if their energy is released entirely as kinetic energy, as is assumed here.

3. **Photoelectric heating** is MOCASSIN's own, from its grains, as described above (D-289, D-326); 3D-PDR's PAH term is not used (D-134, D-352). The model grains include no PAHs (L-7), so the heating lacks the contribution of PAHs and very small grains that dominates in the Bakes and Tielens (1994) prescription of most PDR codes and of the code comparison (Röllig et al., 2007); the Cloudy reference models omit them too. Against Cloudy the photoelectric efficiency per H per unit field is 0.66–1.10 with no trend with depth on `pn150k` (D-304); on `hiipdr` the gas temperature at $A_V = 0.73$ and 0.97 is 1.07 and 1.01 times Cloudy's since the sticking fix, the deeper gas remaining colder (L-22).

The gas temperature found this way is the one the next chemistry runs at, and, one iteration later, the level model. It acts on the molecules through the rate coefficients, through the H_2 formation coefficient (the thermal speed of the H atoms and their sticking probability, Section 7.8), through the collisional excitation of H_2 and of the coolants, and through the CO cooling it controls. The dust temperature acts through the formation efficiency (Section 4.4). These are channels by which the field sets the molecular content even where no FUV photon reaches the molecules.

9.4 The second ionisation solve

After the chemistry, `ionBalance` is solved a second time in every cell, at the temperature of the first solve, on the channels and the molecular content this iteration's chemistry has just left; it is counted against the first and the model is put back (`ionSecondSolve`, `iteration_mod.f90:3361-3434`, called at `:2168`; D-404). A control pass on the old terms, before the chemistry (`:1274`), moves nothing. On the finished model the second solve's stages replace the first's (`ionKeepSecond`, `:3455-3521`), so the written ions are consistent with the last chemistry, and the level model's final call runs on them (H-029). The second solve removes the one-iteration delay of L-47: the neutral cells whose solver X^+/X is more than 10% from MOCASSIN's fall from 28.4% to 0.0% for hydrogen on `hiipdr`, and to 0.0% for every linked element except oxygen there (9.7%, L-50). Neither the temperature nor the chemistry is solved again on the second solve's ions, and in the iterations before the last the transfer, the thermal balance and the chemistry still run on the first solve's.

10 Iteration control and convergence

10.1 The fixed point

The model is converged when the field, the ionisation, the temperatures, the molecular abundances and the H_2 level populations reproduce one another from one MC iteration to the next. Because the columns and the level populations lag by one iteration (Section 2.1) and the level model makes one sweep per iteration, the shielding and the populations converge together with the rest of the model rather than inside each iteration. This is a Picard (successive-substitution) iteration on the coupled system; there is no acceleration of the outer iteration. Damping across iterations (`iterDamp`, `iteration_mod.f90:2424-2467`) is off by default (D-406). Because the MC field is redrawn at every iteration, the fixed point is reached only to within the MC noise of Section 10.3.

10.2 Tests

- **Integrated quantities** (`common_mod.f90:798-818`, test at `iteration_mod.f90:3243`; D-391, D-392): the emission measure $\int n_e n_i \, \mathrm{d}V$ of every ion stage holding at least 1% of its element, the H^0, H_2 and CO contents $\int n \, \mathrm{d}V$, and the gas temperature weighted by $n_e \, n(H^+)$, by $n(H^0)$ and by $n(H_2)$. The run stops when the largest $|q_n/q_{n-3} - 1| \leq$ `convTotals` (default 0.02), and not before the first iteration after `pdrStartIter`. The test does not see the packet count: on `hiipdr` it stops at iteration 14 or 15 from 6×10^6 to 3.84×10^8 packets while the carbon turn moves by 0.5 mag (L-52, D-408); and the temperature weighted by $n(H^0)$ is carried by the few hundred hot cells of the ionisation front, whose iteration-to-iteration scatter alone puts the 2% limit at only 1.4–1.5 standard deviations (L-52).

- **Per-cell tests** (MOCASSIN's own, on $x(H^0)$ and T_e), extended to $x(H_2)$: the abundance after the chemistry against the copy taken at the start of the iteration (`iteration_mod.f90:1173-1188, 2484-2512`, D-179, D-391). They are reported; the stop is the integrated test, since the converged-cell fraction plateaus at about 72% in the PDR fixtures from the per-cell MC noise (`iteration_mod.f90:3236-3241`).

- **Level model**: inside its final call, the luminosities of the four H_2 lines 1–0 S(1), 2–1 S(1), 0–0 S(0) and 0–0 S(1) to 1% between sweeps (Section 6.11). Each call also reports the median and 90th percentile of $|k_{\mathrm{Sol}}/k_{\mathrm{Sol}}^{\text{last call}} - 1|$ over cells with $x(H_2) \geq 10^{-6}$ (`pdr_mod.f90:5499-5520`).

10.3 Monte Carlo noise

Since every photorate reads its own cell's MC field (D-350), the chemistry of a cell inherits the cell's sampling noise, redrawn at every iteration because J is rebuilt from zero. On `hiipdr` at 6×10^6 packets the seed-to-seed scatter of the FUV field per cell is 2–4% at $A_V \sim 1$, 7% at 1.6–2.1, 14% at 2.1–2.6 and 22% (90th percentile 38%) at 2.6–3.05 (L-26). It follows $N^{-1/2}$ in the number of packet tracks per cell: on a 61^3 grid with 5×10^8 packets, 21 times the tracks per cell, the iteration-to-iteration noise is 0.8, 1.5, 2.8 and 4.8% at A_V = 1–1.5, 1.5–2, 2–2.5 and 2.5–3.5 (D-365). For the level model the noise is that of a single 2%-wide frequency bin per line rather than of the band: its per-cell Solomon rate scatters about 1.5 times as much as the chemistry's band-integrated rate on the same field, and moved by 50–60% (median) between iterations against 3.4% on a fixed field (H-012).

The noise has three consequences beyond scatter. Through the non-linear dependence of the photorates on the field it biases cell means, by an estimated +5% at 22% scatter (L-26), and the bias of the means is one way: fewer packets give a shallower carbon turn and colder deep gas on `hiipdr` (D-408). At the carbon turn the noise is not scatter about a mean but two states: 85–97% of the cells in each 0.1 mag bin between $A_V = 1.9$ and 2.6 are either ionised in carbon or in the state maintained by cosmic rays alone, depending on whether a packet of the 11.26–13.6 eV band crossed them in that iteration, so "C^+ =C" marks where half the cells are missed (D-405). And the noise is redrawn in full at every iteration: the spread of a cell's mean over seven to nine iterations among symmetry-equivalent cells is 0.33–0.41 of one iteration's (L-26). An estimator accumulated over iterations is registered for future development (D-411).

What converges at the default 6×10^6 packets was measured by quadrupling the count three times (D-408, H-032). The H_2 lines, the ionised gas's lines, [O I], Hβ and the H/H_2 crossing are converged at the default ($A_V = 1.395$ at 6×10^6 and 1.394 at 3.84×10^8 packets on `hiipdr`; 1–0 S(1) changes by 0.8%). The carbon turn and the [C I] lines need 16–64 times the default, and CO still moves at 64 times (by 5–10% per factor of four on `hiipdr`). Per-cell values of the deep chemistry are not converged at any of these counts.

10.4 End of the run

When the stop test passes, or at the maximum iteration count, the code keeps the second ionisation solve (`ionKeepSecond`), calls `h2LevelModel(grid,12)` (`iteration_mod.f90:3254, 3283`) and writes the outputs; the chemistry is not solved again on the final level populations (Section 12). The level model's outputs are written by `h2Lines_mod.f90`: `output/lineFluxH2.out` (every H_2 X-state line above 10^{-6} of the strongest, integrated over the grid, optically thin, no extinction), `output/h2Cells.out` (per cell: $n(H_2)$, T_e, $k_{\rm Sol}$ and the chemistry's own rate on the same field, pumping rate, ortho/para, fraction above 4100 cm^{-1}, the H_2 heating terms beside 3D-PDR's and MOCASSIN's heating and cooling) and `output/h2Pop.bin` (all populations, read back by a warm start).

11 Physical assumptions

The assumptions below are made, explicitly or implicitly, on the path from the field to the molecular content. Each names the section where it enters, says why it is made and where it fails, and gives the direction of the resulting bias and its measured size where the project has measured it. Section 12 lists the defects and open issues.

11.1 Time dependence and equilibrium

1. **Chemical steady state.** The network is integrated until no species changes by more than 1% over a doubling of the integration time, at fixed temperature, field, columns and imposed ionisation (Section 8). No chemical age, advection or time-dependent field is represented. The slowest process is H_2 formation, with a time scale $1/(R_{\rm f} n_{\rm H}) \approx 10^7\,(n_{\rm H}/100\,{\rm cm^{-3}})^{-1}$ yr for $R_{\rm f} \approx 3\times10^{-17}\,{\rm cm^3\,s^{-1}}$, close to the model's coefficient (`L-33`): 10^7 yr at `hiipdr`'s density and 10^6 yr at `pn150k`'s. This is comparable to or longer than the age of an H II region and much longer than that of a planetary nebula, so where molecules have to form from atomic gas the steady state overstates the molecular content, and it describes neither gas just uncovered by an advancing front nor molecules inherited from an earlier phase.

2. **Statistical equilibrium of the H_2 levels**, with the excited electronic states treated as instantaneous (radiative lifetimes 0.5–9 ns in the data; collisions compete only above $\sim 10^{14}$ cm^{-3}) (Sections 6.2, 6.4). The slowest radiative relaxation within X, the quadrupole decay of the lowest rotational levels ($A = 2.9\times10^{-11}\,{\rm s^{-1}}$ for $J = 2 \to 0$, about 1100 yr, in `dataPDR/h2/transprob_X.dat`), is short compared with the dynamical times of the modelled objects, and collisions shorten it further.

3. **Formation closure in the level model**: every molecule removed by the Solomon process is replaced by one formed on a grain, and the level populations are normalised to one. The formation flux into the levels therefore equals the Solomon flux; destruction by other channels and the corresponding formation are not in the level balance (Section 6.2). Where the Solomon rate has been shielded away and cosmic rays and ion–molecule reactions dominate the destruction, formation pumping, and the internal energy it brings, are underestimated (Section 12). With the network's H_2 + CRP channels alone the other destruction is $0.99\,\zeta$, $10^{-16}\,{\rm s^{-1}}$ at the default `pdrZeta`, so this happens where the Solomon rate is shielded below a few $10^{-16}\,{\rm s^{-1}}$ (Section 6.2).

4. **The coupling is a fixed-point iteration**: shielding columns, level populations of other cells, the chemistry's channels in `ionBalance` and the heating terms held in the temperature search are one or two iterations behind (Section 2.1). The lags vanish at a fixed point; because the MC field is redrawn at every iteration, the fixed point is statistical, and the noise biases cell means through the non-linear rates (Section 10.3).

5. **Homogeneous cells.** Every quantity is uniform within a cell, and the MC field is the cell average. A front thinner than a cell is not resolved: on `hiipdr`'s 31^3 grid a cell is 0.65 pc and 0.104 mag of A_V (H-026), so the H/H_2 transition is located only to a cell, and an ionisation-front cell spreads over its whole volume the warm, partly ionised gas of a layer about 0.18 pc thick, which then dominates [O I] 63 µm (L-37). The shielding columns of the chemistry run to the near face of the cell (Section 5.2), which under-shields CO in the cells of its transition by up to half a cell's column.

11.2 Radiation field

6. **Point sources only.** The field is that of the sources of the input and their reprocessing; no interstellar radiation field illuminates the outer boundary and the cosmic microwave background is not transported (Sections 3, 4.4). Where the attenuated field of the sources falls below the interstellar one, the model's outer gas is more molecular and colder, and its grains colder, than the same cloud immersed in the interstellar field would be. On `hiipdr` the source's field falls below $\chi = 1$ beyond about 0.6 of the outer radius and to a median $\chi = 0.08$ in the outermost shell; on `pn150k` it stays above 1.6 everywhere (`output/pdrFUV.out` of the fixtures' runs). The grain temperature is floored at $T_{\rm CMB}$ for the chemistry only (L-30); the line background of the coolants does include the CMB (D-377).

7. **The MC field is a continuum field.** It contains dust absorption and scattering and the gas continuum opacities (including C I and H_2 photoionisation), but no H_2, CO or H I Lyman-series *line* opacity (Section 3.6). The photons that the H_2 lines absorb remain in the transported field, where they heat dust, eject photoelectrons and photoionise C I. Behind the H/H_2 transition this overstates dust heating and photoelectric emission slightly, since the saturated line cores hold a small part of the band's energy (the whole window holds 15% of a Draine-shaped field's), and C I photoionisation, which happens only in the window, by more: a few tens of per cent at $A_V \sim 3$ by the project's estimate (Section 12). Unmeasured.

8. **Separability of continuum and line attenuation.** The rate seen by a molecule is the MC continuum at the line times a line transmission computed from column densities along a characteristic path: the straight line to each source for the direct beam, and the two-leg path through a scattering site for diffuse light. Dust is never applied twice: $A_V = 0$ on every ray (Section 7.1). Exact for the direct beam; for the diffuse light it assumes that a photon's history is represented by the one site it is assigned to.

9. **Direct beam geometry.** The direct field arrives along the straight line from each point source. With several sources the columns and the in-cell chord are averages weighted by r_s^{-2} alone, without the sources' luminosities or attenuation, applied to the summed direct field (Section 5.2). Exact for one source, as in all the fixtures; for several, the error is unmeasured and has no definite sign.

10. **Diffuse field geometry.** Its total is the MC's $\chi_{\rm dif}$; its distribution over twelve fixed directions and over sites along them is the formal solution with an isotropic emissivity $\sum \kappa^{\rm sca} J$ and a grey FUV extinction $R_{\rm FUV}\kappa_V$ (Section 5.4). The same window factor is applied at every site. Measured: the formal solution recovers 52–130% of $\chi_{\rm dif}$, falling with depth (L-4); the forward-peaked phase function ($g = 0.29$–0.61 on the fixtures) puts more of the real diffuse field on the source side than the weights do.

11. **Angular quadrature**: twelve HEALPix $N_{\rm side} = 1$ directions fixed in the grid frame, for the sites and for line escape (Sections 5.5, 5.6). Biased along the ring axis: cells nearest the plane $z = 0$ hold 5–11% more H_2 before and across the transition than their symmetry-equivalent cells; 48 directions remove this, moving the totals by under 1% (L-53, D-410).

12. **Spectral shape.** The generic photoreactions assume the Draine spectral shape across 6–13.6 eV (Section 7.2), while the MC field's LW share falls from 0.30 to 0.008 through `hiipdr`'s PDR (D-349); CO takes the ratio of the MC field's LW share to Draine's, with the shape inside the window assumed Draine's (Section 3.5); H_2 uses the MC specific intensity at each line, constant across the 2%-wide frequency bin that contains it (Section 3.6).

13. **Symmetry.** For a `symmetricXYZ` octant the lower faces are mirrors for every walk to the edge of the model; the walks towards the sources do not reflect (Section 5.1). Valid for a model symmetric in the three planes with its sources on them.

11.3 H_2 microphysics

14. **Line profile**: Voigt, with Doppler parameter $b^2 = 2kT_e/m_{H_2} + v_{turb}^2$ and damping from the upper level's total radiative decay rate (to the bound levels and the continuum of X; the data contain no predissociated levels) (Section 6.5). Damping each line by its own A, as Cloudy does, would put `hiipdr`'s H/H_2 crossing 0.21 mag and `pn150k`'s 0.10 mag shallower (H-026).

15. **Self-shielding** of each line by the column of its lower level, averaged through the cell's own column with the curve of growth; the cell's own column is taken over its mean chord along the source direction (direct) or over all directions (diffuse) (Section 6.6).

16. **Line overlap** by the other H_2 lines, as a six-point mean of their transmission over the line's absorption; photons absorbed in another line's wing are lost, not counted as pumping it; H I Lyman lines do not shade (Section 6.7).

17. **Trapping**: complete redistribution; a trapped photon re-excites the same upper level (on-the-spot); escape probability per line, without overlap, averaged over twelve directions (Section 6.8).

18. **Colliders**: H (including reactive collisions), He, ortho- and para-H_2 and H^+; no electrons, so the excitation of H_2 in ionised gas is underestimated. Missing same-spin pairs (86–99.9% of them, depending on the collider) take g-bar estimates; missing spin-changing pairs get no collisional rate; outside the tabulated temperatures (for H, 100–5000 K) the rates are held at the table's end values (Section 6.3).

19. **Formation distribution** of Takahashi and Uehara (2001), silicate and carbonaceous surfaces mixed in the proportion of the formation rates (Section 6.9). Formation through H^- and H_2^+, cosmic-ray excitation and destruction other than Solomon are not in the level model.

11.4 Grains

20. **H_2 formation** follows Cazaux and Tielens (2002, 2004) with the Hollenbach and McKee (1979) sticking coefficient and 3D-PDR's surface energies per material, on the geometric grain area per H nucleus of the model's own grains, split into silicate and carbonaceous by the species' labels (Sections 4.2, 7.8). On `hiipdr`'s silicates this gives 0.50–0.91 of Cloudy's coefficient, mainly through the surface energies and partly through the colder gas; on `pn150k`'s graphite 0.86–1.00 (L-33).

21. **One grain temperature per cell**: the number-weighted mean over species and sizes, the size weights normalised over the whole size grid, enters an efficiency that is steep in T_d; a size grid wider than a species' sizes pulls the mean towards zero and can switch formation off (H-035; Section 4.4).

22. **Grain temperature** from MOCASSIN's radiative equilibrium with no stochastic heating by default (L-8), floored at T_{CMB} because the CMB is not transported (Section 4.4).

23. **Extinction ratio** $R_{\rm FUV}$ weighted by source 1's emitted spectrum; it overstates the band extinction of the reddened local field at depth, which matters only for the diffuse-site weights (Section 4).

24. **Photoelectric heating** from the model's grains alone: no PAHs or very small grains (L-7), so the heating is lower than PDR models with the Bakes and Tielens (1994) prescription would have at the same field, and the gas colder (Section 9.3).

11.5 Ownership and coupling

25. **MOCASSIN owns the ionisation** of H, He, C, O and Mg and the electron density: imposed on the chemistry before and after the solve, with the photoionisation and recombination rates of these atoms taken from MOCASSIN inside the solve (Sections 7.5, 8.3). In particular the C^+/C ratio, which controls the carbon available to CO, is MOCASSIN's, and its C I photoionisation has no H_2-line shielding.

26. **MOCASSIN owns the temperatures**: the chemistry and the level model run at T_e (clamped to the fitted range of the rate data, while 3D-PDR's heating terms are evaluated at the unclamped T_e, L-30) and T_d (Section 8). The thermal balance is solved at fixed chemistry, with the extra heating terms held fixed during the temperature search (Section 9.3).

27. **The chemistry owns the molecular content**, which reaches MOCASSIN only as the fraction of each element locked in molecules, as $n(H_2)$ in the opacity, and through the coolants and heating terms (Section 9).

28. **Cosmic rays**: a uniform ionisation rate (`pdrZeta`), with no attenuation with column (Section 7.6); their heating, 9.4 eV per H_2 ionisation, is counted in molecular gas only (Tielens and Hollenbach, 1985).

11.6 Relation to the standard PDR benchmark

The PDR code comparison (Röllig et al., 2007) fixed, for its benchmark models, the field in Draine units with $\chi = 1.71\,G_0$, $A_V = 6.289 \times 10^{-22} N_{\rm H}$, an FUV attenuation $e^{-3.02 A_V}$ wherever a code did not treat it as a function of wavelength, an H_2 formation coefficient $3 \times 10^{-18} T^{1/2}\,{\rm cm^3\,s^{-1}}$, a cosmic-ray ionisation rate of $5 \times 10^{-17}\,{\rm s^{-1}}$, and photoelectric heating after Bakes and Tielens (1994), the only heating process it standardised; the H_2 shielding was left to each code (their Sect. 4.2.3 and Table 5). The present model departs from all of these choices by construction: its field is the transported MC field, scattering included (Section 3.2); its $A_V/N_{\rm H}$, $R_{\rm FUV}$ and grain area are its own grains' (Table 4); its formation coefficient follows the grains' temperature (Section 4.4); ζ defaults to $10^{-16}\,{\rm s^{-1}}$; its photoelectric heating comes from its own grains without PAHs; and its H_2 shielding is the level model's. A benchmark comparison therefore tests the coupling only after these inputs have been matched, as was done against the vendored 3D-PDR in D-257 (in ζ, composition, A_V conversion and dust); unmatched, it measures differences of physics rather than of numerics.

12 Known limitations and open issues

This section lists the limitations that bear on the path from the field to the molecules, with the direction of the bias each introduces and its measured size where one exists. The first group is recorded in the project's own logs. The second group was found while reading the code for this document and is not, as far as could be found, recorded in `DEVLOG.md`, `DEVLOG-H2.md` or `doc/pdr-status.md`; its sizes have not been measured.

12.1 Recorded in the project's logs

1. **Per-cell Monte Carlo noise in the photorates** (L-26, D-405, H-012). Every photorate reads its own cell's field, whose per-cell scatter reaches 14–22% beyond $A_V \sim 2$ at the default packet count; the level model's per-line field is noisier than the band integral. Besides scatter, the noise biases cell means through the non-linear rates (estimated +5% at 22% scatter) and, at the carbon turn, decides which of two states a cell is in (D-405). An estimator accumulated over iterations is registered for future development (D-411). Section 10.3 gives the numbers.

2. **Packet convergence of carbon and CO** (D-408, H-032). At the default 6×10^6 packets the H_2 lines and the H/H_2 crossing are converged, but the carbon turn is placed too shallow (C^+ =C moves by 0.5 mag on `hiipdr` up to 3.84×10^8 packets, towards Cloudy's), the deep gas is too cold, [C I] needs 16–64 times the packets and CO is still moving at 64 times. The stop test does not see this.

3. **The twelve-direction quadrature** (L-53, D-410): cells that the grid's symmetry makes equal differ in $n(H_2)$ by up to about 11% following the ring axis of the twelve directions, independently of the packet count; 48 directions remove the pattern at 1.48 times the run time, moving totals by under 1%. The twelve were kept by decision, since no line map carries the pattern (CO's abundance and the H_2 density do).

4. **Damping width of the H_2 lines** (H-008, H-026). The code damps each line with the upper level's total radiative decay rate (to the bound levels and the continuum of X), which is the physical natural width; damping each line with its own A, as Cloudy does, narrows the wings, strengthens the shielding and moves the H/H_2 crossing 0.21 mag shallower on `hiipdr` (from $A_V = 1.396$ to 1.188, Cloudy's being 1.280) and 0.10 mag shallower on `pn150k`. Between $A_V = 1.05$ and 1.2 on `hiipdr` the model's photodissociation rate per molecule is 5–7 times Cloudy's. On `hiipdr` the width accounts for +0.21 mag of the difference between the two codes' crossings and everything else for −0.09 mag (H-026); it is the largest single identified term and should accompany any comparison with Cloudy, which uses the total decay rate for every species but H_2 (H-008).

5. **No H i Lyman-series shading** of the H_2 band (`DEVLOG-H2.md`, H-006, "What is not yet known"). Neither the MC transfer nor the level model contains it, so pumping and dissociation are overestimated where a large H I column lies in front of the molecular gas, and the transition is placed too deep. The hydrogen column between the ionisation front and the H/H_2 crossing is about 1.8×10^{21} cm^{-2} on `hiipdr` (A_V 0.48 to 1.40 at 1.93×10^{21} cm^{-2} mag^{-1}, Table 4; D-260, H-027) and at most 1.1×10^{21} cm^{-2} on `pn150k` (crossing at $A_V = 0.39$, 2.79×10^{21} cm^{-2} mag^{-1}), part of it molecular. The damping wings of Lyβ, Lyγ and Lyδ alone cover about 4, 1.4 and 0.7 Å of the 912–1110 Å band at $N(\mathrm{H\,I}) = 10^{21}$ cm^{-2}, growing as $N^{1/2}$ (an estimate, Section 6.7). Unmeasured.

6. **No cross-pumping and no overlap in trapping** (H-006): a photon absorbed in another line's wing is lost rather than pumping that line, and the escape of each line ignores the others.

7. **Electrons are not colliders** of the H_2 levels, and the level model has no destruction other than Solomon (`h2_mod.f90` header); this matters in the partly ionised gas, where H_2 is already 14–230 times Cloudy's for chemical reasons (L-43).

8. **C i photoionisation without H_2-line shielding** (`DEVLOG.md`, item 8 of the audit list near line 28742): MOCASSIN's rate replaces `CIPDRATE`, and MOCASSIN does not transport LW lines, so the rate is overestimated behind the H/H_2 transition; estimated at a few tens of per cent at $A_V \sim 3$, with the opposite sign to the deep C^+ deficit (L-3).

9. **Chlorine is absent from the network**; in Cloudy it carries 25–50% of the H_2 sink from $A_V = 0.26$ on in the PN model, and Cloudy without chlorine turns molecular 0.042 mag earlier (L-46, H-026). Its absence moves the model's crossing shallower, by about 0.04 mag on `pn150k`, partly offsetting there the 0.10 mag of item 4 (L-46).

10. **Silicate formation efficiency**: the per-material Cazaux & Tielens surface energies of 3D-PDR give 0.50–0.91 of Cloudy's coefficient on `hiipdr` (L-33, H-022); reported, not tuned.

11. **The mean grain temperature can switch formation off** (H-035): the number-weighted mean over the whole size grid counts the bins outside a species' size range at zero temperature; on a grid wider than the grains present this put T_d at 10^{-6} K and no H_2 formed. Fixed after `8fc1512`: D-414 takes the mean over the bins that hold grains (Section 1).

12. **Diffuse-site weights** recover 52–130% of the MC's diffuse field (L-4); the magnitude is the MC's, the directional distribution, and so the columns assigned to the diffuse light, is approximate (Section 5.4).

13. **Unresolved fronts** (L-37, D-088): on uniform grids the ionisation front and the H/H_2 transition are thinner than a cell; the front cells, which carry the warm partly ionised gas through their whole volume, make 92% of `hiipdr`'s [O I] 63 µm, 9.3–9.5 times Cloudy's at 31^3 and 5.5 times at 61^3. Front-sensitive lines need a resolution check; a sub-cell estimate of [O I] 6300 Å is available offline (`frontLines`).

14. **The gas re-emits from the hydrogen and helium spectrum whatever absorbed** (L-51). A packet absorbed by neutral carbon in the PDR is re-emitted, some of the time, as a hydrogen- or helium-ionising photon: 1.9% of `hiipdr`'s cells per iteration at $A_V = 2.15$–2.43 are over-ionised, and their chemistry, solved to steady state on that iteration's ionisation, can lose more than half its H_2 (202 such drops in six iterations at the default packet count, 23 at four times, none at sixteen times).

15. **No PAHs or very small grains** (L-7): the photoelectric heating lacks the component that dominates in real PDRs, so the gas is colder than in PDR models with that component at the same field; common-mode with the Cloudy references.

16. **Gas–grain collisional coupling omits H_2** (L-9): 68 times weaker than Cloudy's dust cooling at $A_V = 0.65$ on `pn150k`, negligible at $n_H = 10^3$ cm^{-3} and decisive at high density.

17. **The H_2 line emission is computed optically thin and without dust extinction** in `lineFluxH2.out` (`h2Lines_mod.f90` header).

12.2 Found while preparing this document

18. **Formation flux in the level model is tied to the Solomon flux.** `solveCell` calls `h2Solve` without a sink or a source (`pdr_mod.f90:5904`), so the rate at which new molecules enter the levels in the formation distribution ϕ equals the Solomon dissociation rate. The chemistry's actual formation rate, $R_f n_H n(H)$, balances all destruction (cosmic rays, ion–molecule reactions, Solomon) in steady state. Where the Solomon rate has been shielded away, formation pumping is therefore underestimated. With the network's H_2 + CRP channels alone the other destruction is $0.99\,\zeta$, 10^{-16} s^{-1} at the default `pdrZeta`, so this happens where the Solomon rate is shielded below a few 10^{-16} s^{-1} (Section 6.2). The energy budget is split the same way: the internal energy that newly formed molecules bring to the levels (and that the collisional term returns to the gas) follows the Solomon flux, while the kinetic part of the formation energy (`heat(5)`) follows the chemistry's formation rate. Both numbers needed for a consistent closure are already available in `solveCell` ($w_{sil} + w_{gra}$, $n(H)$, $n(H_2)$): passing the non-Solomon destruction as `sink`, or the chemistry's formation flux per molecule as `src`,

would make the two agree. A per-cell print of formation flux over Solomon flux would size the effect first.

19. **The final output mixes two Solomon rates.** The last `h2LevelModel(grid,12)` (`iteration_mod.f90:3254, 3283`) runs after the last chemistry, which is not re-solved. The written level populations and $k_{\rm Sol}$ are those of up to twelve sweeps; the written $n(\mathrm{H_2})$ was obtained with the previous one-sweep rate. The size of the difference is printed by the call itself ("Solomon rate against the last call's").

20. **The $\mathrm{H_2}$ collisional heating term is frozen and two iterations old in the temperature search.** $\mathcal{H}_{\rm coll}$ changes sign and depends steeply on T (it is $\mathrm{H_2}$'s line cooling where the gas is warmer than the levels), but it enters `pdrHeatXtra`, which MOCASSIN holds constant in `iterateT`, and the value used to find $T_e^{(n)}$ was computed by the level model at $T_e^{(n-2)}$ (Section 2.1). H-030 associates this term, together with the damping, with the two derailed iterations at the start of the chemistry. Tabulating it against T, as the CO cooling is (D-215), would remove the freeze.

21. **Generic photoreactions keep the Draine spectral shape** (Section 7.2). L-24 corrected the LW-window reactions; the other 24 photoreactions still multiply the 6–13.6 eV band strength by coefficients computed for the Draine shape, although the MC field's shape changes strongly with depth (its LW share falls from 0.30 to 0.008 on `hiipdr`, D-349). Reactions whose thresholds lie in the window are overestimated at depth, those driven by the 6–11 eV part underestimated. Folding each reaction's cross section, or a few sub-bands, over the MC field would remove the assumption.

22. **The level model's $\mathrm{H_2}$ transmission is not shared.** The per-line $\mathrm{H_2}$ shielding computed in `solveCell` is exactly what is missing from MOCASSIN's C I rate (item 8) and what CO's van Dishoeck & Black factor approximates for $\mathrm{H_2}$ screening; neither uses it.

23. **CO shielding and the chemistry's own $\mathrm{H_2}$ diagnostic** exclude the target cell's own column (D-088); only the level model averages through it. On `hiipdr`'s grid half a cell can hold $10^{16}\,\mathrm{cm^{-2}}$ of CO, so CO is under-shielded in the cells of its own transition and the transition is placed up to about half a cell too deep (Section 5.2).

24. **Several sources are combined by geometry alone.** The effective A_V, the source-ward columns, the in-cell chord (`computeAVradial`, `pdr_mod.f90:1986`) and the level model's source-ward level columns (`starColumns`, `:5651`) weight the sources by r_s^{-2}, without their luminosities or their attenuation, and apply the mean column to the summed direct field through non-linear shielding functions. Exact for one source, which is every fixture; untested for several. Weighting each source by its own attenuated direct field, $L_{{\rm FUV},s} r_s^{-2} e^{-R_{\rm FUV}\tau_{V,s}}$, and evaluating the shielding per source would remove it.

25. $R_{\rm FUV}$ **is weighted by the emitted spectrum of source 1** (`pdr_mod.f90:1381-1390`). It overstates the band extinction of the local, reddened field at depth, and so the optical depth between the scattering sites and the target; this suppresses distant sites increasingly with depth and is a candidate for the fall with depth of the formal solution's recovery in L-4.

26. **The grain temperature is a number-weighted mean applied to an area-weighted rate** (Section 4.4). Besides the zero-bin trap of item 11, the mean favours the smallest grains and one efficiency $\epsilon(\langle T_d\rangle)$ replaces $\langle\epsilon(T_d)\rangle$ over the sizes. On silicate at 20–50 K, where ϵ falls by a factor of 3.6 (Section 4.4), the formation rate depends on how the temperatures of the sizes are spread. Evaluating the rate per size with πa^2 weights would remove both; a temperature per grain material is registered for future development as D-417 (L-54), after the commit described here.

27. **No interstellar field at the outer boundary** (Section 11, item 6). On `hiipdr` the source's field is below $\chi = 1$ in the outer 40% of the radius, so the outer PDR is more molecular and colder than the same cloud in the interstellar field.

28. **The coolant line escape depends weakly on the sweep order.** The optical depths of the coolant lines read the other cells' level populations as stored on the rank (`pdr_mod.f90:4315-4316`), which cells solved earlier on the same rank have already updated (`:4440-4442`); the shielding columns are snapshots and do not have this dependence.

29. **The seed double-attenuates.** `seedMolecules` multiplies the local, already attenuated χ by `SCATTER`(A_V) (`pdr_mod.f90:2937-2938`; Section 8.2). It affects only the starting point of the first PDR iteration.

30. **The plane-parallel branch includes the near half of the cell in its columns** (`pdr_mod.f90:2492-2507`), against D-088's rule in the point-source branch. The level model does not run in that geometry.

31. **Collisional ionisation of hydrogen lacks the electron density.** MOCASSIN adds $2.75\times10^{-16}T^{3/2}(157\,894/T+2)e^{-157\,894/T}$, a rate coefficient in $\mathrm{cm^3\,s^{-1}}$, to the photoionisation rate in $\mathrm{s^{-1}}$ without the factor n_e, in its ionisation balance (`update_mod.f90:2769-2770`) and in the rate handed to the chemistry (`:2859`). Collisional ionisation is therefore misscaled by a factor n_e (underestimated wherever $n_e > 1\,\mathrm{cm^{-3}}$); with the factor, the term would reach 10% of a photoionisation rate of $10^{-8}\,\mathrm{s^{-1}}$ at about $1.5\times10^4\,\mathrm{K}$ for $n_e = 10^4\,\mathrm{cm^{-3}}$, so hot dense ionised gas is affected and the PDR is not. Inherited from MOCASSIN (Section 7.5).

32. **Two reactions of the rate file are inactive.** The grain-assisted recombinations of $\mathrm{He^+}$ and $\mathrm{C^+}$ (reactions 330 and 331) have no term in `pdr/odes_reduced.c`; their coefficients are computed but unused (Section 8.1).

33. **$\mathbf{H + H_2}$ rates below 100 K** are those of 100 K, the first temperature of the Lique (2015) table, which overestimates the reactive ortho–para conversion by H in cold gas, where $\mathrm{H^+}$ exchange usually dominates it (Section 6.3).

34. **The $\mathbf{H_2}$ line profile is normalised to its own centre** ($\tau_0 H(a,x)/H(a,0)$ with Doppler-core τ_0), overstating the Voigt optical depth by $1/H(a,0)$, at most 3% for these data (Section 6.5).

35. **Stale or inaccurate comments.** `constants_mod.f90:18-20` labels the FUV lower edge 2400 Å (the value is 6.0 eV, 2068 Å); the comment in `addOpacity` (`ionization_mod.f90:444-453`) describes a power-law H_2 cross section, whereas `yanH2xSec` is the piecewise fit of Yan et al. (1998); and the comment of `draineWindowShare` (`pdr_mod.f90:2285-2286`), D-349 and L-24 give the Draine field's LW share of the continuous band as 0.1516 (0.152), whereas Equation (7) gives 0.148. The code uses the binned value (0.151–0.164 on the fixtures), consistently with the MC sums, so no result is affected. The comment of `h2FormDist` (`h2_mod.f90:578-579`) gives the vibrational distribution as "their eqs 4-5" about $E_{vm} = X_{\mathrm{vib}}D$ and the rotational one as "their eqs 6 and 7", whereas in Takahashi and Uehara (2001) the vibrational distribution is eqs. 5–7 (with $E_{vm} = X_{\mathrm{vib}}D + E_{\mathrm{off}}$) and the rotational models are eqs. 12 and 13 (the code is right); and `update_mod.f90:2621` attributes the collisional-ionisation formula to Drake and Ulrich (1980), which does not contain it (Section 7.5).

A Index of code locations

Files are under `source/` of the `h2-model` worktree at commit `8fc1512`; `pdr/` is the vendored 3D-PDR. A range is the whole routine unless the process column names a part of it.

Table 10: Index of code locations.

Process	Routine	File:lines
Iteration and transfer		
MC iteration, PDR block	`iterateMC`	`iteration_mod.f90: 56-3345`
packet transport	`energyPacketDriver`, `pathSegment`	`photon_mod.f90:37-3575, 1249-3344`
path-length estimator ($J^{\rm ste}$, $J^{\rm dif}$)	`pathSegment`	`photon_mod.f90: 1900-1924, 2236-2250`
scattering, re-flag of the packet as diffuse	`pathSegment`	`photon_mod.f90: 2025-2035, 2154-2165`
absorption by dust or gas	`pathSegment`	`photon_mod.f90:2050-2103`
$J^{\rm dif}$ always accumulated in PDR runs	`readInput`	`set_input_mod.f90: 775-778`
estimator units, octant	`iterateMC`	`iteration_mod.f90:924, 940`
FUV band, LW window, Draine unit, V band	constants	`constants_mod.f90:16-35`
FUV and V bin indices	`setContinuum`	`continuum_mod.f90:92-95`
Dust		
grain area per H by material	`measureH2grainArea`	`pdr_mod.f90:1266-1338`
material from the grain label	`isCarbonGrain`, `isSilicateGrain`	`pdr_mod.f90:1228-1247, 1341-1360`
$R_{\rm FUV}$, $A_V/N_{\rm H}$, area per cell	`dustPerCell`	`pdr_mod.f90:1363-1436`
model-wide fallbacks, own extinction curve	`calibratePDRdust`	`pdr_mod.f90:1439-1662`
grain LW albedo	`grainAlbedoLW`	`update_mod.f90:4658-4673`
grain temperatures; their mean	`getDustT`	`update_mod.f90: 4282-4562; 4546-4555`
size weights of the distribution	`makeDustXsec`	`ph_mod.f90:999-1012`
grain attenuation factor, extinction curve	`SCATTER`, `XLAMBDA`	`pdr/shield.F90:449-497, 525-549`
Geometry		
exact ray–cell traversal	`rayCells`	`pdr_mod.f90:1710-1833`
source-ward A_V and columns	`computeAVradial`	`pdr_mod.f90:1859-2015`
dispatch, plane-parallel case	`computeAVdust`	`pdr_mod.f90:2410-2523`
diffuse snapshot	`snapDiffuseSources`	`pdr_mod.f90:2082-2121`
scattering sites, two-leg columns	`traceDiffuseSites`	`pdr_mod.f90:2136-2276`
Draine LW share	`draineWindowShare`	`pdr_mod.f90:2288-2314`
line-escape rays	`traceLineRays`	`pdr_mod.f90:2327-2376`
χ, χ_0	`mocassinToPDR`	`pdr_mod.f90:2538-2677`
H_2 level model		
driver, sweeps, MPI scheduling	`h2LevelModel`	`pdr_mod.f90:5201-6014`
source-ward level columns	`starColumns`	`pdr_mod.f90:5621-5659`
per-cell solution	`solveCell`	`pdr_mod.f90:5665-5954`
warm-start populations	`h2PopResume`	`pdr_mod.f90:5961-6012`

Process	Routine	File:lines
1D-build stub	`h2LevelModel`	`pdr_mod.f90:6015-6027`
X levels, A values, collision tables	`h2ReadX`	`h2_mod.f90:206-334`
excited states, lines, dissociation	`h2ReadElec`	`h2_mod.f90:366-516`
UV pumping, Solomon rate, trapping	`h2PumpRates`	`h2_mod.f90:540-572`
formation distribution	`h2FormDist`	`h2_mod.f90:589-639`
collision-table interpolation	`tabRate`	`h2_mod.f90:642-660`
Voigt function	`h2Voigt`	`h2_mod.f90:664-686`
S, W, absorption quantiles	`shieldQuad`, `shieldTable`	`h2_mod.f90:692-760`
shielding look-up, cell average	`h2Shield`, `tabShieldJ`, `h2ShieldJ`	`h2_mod.f90:768-866`
direct-path shielding (prepare/evaluate)	`h2LineShieldPrep`, `h2LineShieldEval`	`h2_mod.f90:876-907`
Cloudy's FGK79 fit (comparison only)	`h2ShieldFederman`	`h2_mod.f90:915-934`
line overlap	`lineOverlap`	`h2_mod.f90:1018-1289`
line parameters σ, a	`h2LineParams`	`h2_mod.f90:1328-1336`
escape probabilities	`h2LineEscape`	`h2_mod.f90:1344-1361`
collisional and radiative rates	`h2CollRad`	`h2_mod.f90:1369-1415`
collisional heating	`h2CollHeat`	`h2_mod.f90:1425-1448`
statistical equilibrium	`balance`, `h2Solve`, `gaussSolve`	`h2_mod.f90:1456-1568`
outputs	`h2Lines_mod`	`h2Lines_mod.f90`
Chemistry		
network set-up, linked elements, replaced channels	`initPDRnetwork`	`pdr_mod.f90:243-948`
atoms of each element per species	`initPDRnetwork`	`pdr_mod.f90:585-603`
first-visit seed	`seedAbunFromMocassin`	`pdr_mod.f90:3123-3157`
first-iteration molecular seed	`seedMolecules`	`pdr_mod.f90:2855-3023`
one cell of chemistry	`pdrStep`	`pdr_mod.f90:3180-4580`
rays: $\chi_{\rm dir}$, $\chi_{\rm dif}$, sites, window factors	`pdrStep`	`pdr_mod.f90:3383-3455`
imposition before the solve	`pdrStep`	`pdr_mod.f90:3503-3584`
columns handed over	`pdrStep`	`pdr_mod.f90:3593-3628`
rate coefficients	`CALCULATE_REACTION_RATES`	`pdr/calc_reac_rates.F90`; call `pdr_mod.f90:3692`
MOCASSIN's atomic rates into the network	`pdrStep`	`pdr_mod.f90:3706-3722`
H_2 photodissociation $= k_{\rm Sol}$	`pdrStep`	`pdr_mod.f90:3736`
charge transfer coefficients	`pdrCtCoef`	`common_mod.f90:1383-1436`
H_2 formation rate	`H2_FORMATION_RATE`	`pdr/h2_form.F90:9-134`
H_2, CO, C I photorates and shielding	`H2PDRATE`, `COPDRATE`, `CIPDRATE`, `COSHIELD`, `SCATTER`, `LBAR`	`pdr/shield.F90:27-55, 83-106, 132-157, 231-281, 375-416, 449-497, 570-599`

Process	Routine	File:lines
steady-state integration	`calculate_abundances`	`pdr/calculate_abundances.c`; call `pdr_mod.f90:3749`
ODE right-hand side	generated	`pdr/odes_reduced.c`
post-solve imposition	`pdrStep`	`pdr_mod.f90:3770-3784, 3801-3804`
channels for `ionBalance`	`pdrStep`	`pdr_mod.f90:3924-4054`
coolant level populations, CO table	`pdrStep`	`pdr_mod.f90:4144-4490`
escape probabilities of the coolant lines	`escape_probability`	`pdr/escape_probability.F90:121-137, 196-233`
heating, H_2 terms from the level model	`pdrStep`	`pdr_mod.f90:4499-4546`
3D-PDR's heating terms	`CALC_HEATING`	`pdr/heatingfunctions.F90:7-375`
Feedback		
handback	`pdrToMocassin`, `setElemAvail`	`pdr_mod.f90:5038-5104, 5120-5144`
hydrogen budget	`rescaleIonToAvail`	`pdr_mod.f90:2764-2803`
molecular locking in the ion solver	`ionBalance`	`update_mod.f90:3130-3134, 3255-3300`
cosmic-ray ionisation in the ion solver	`ionBalance`	`update_mod.f90:2916-2930`
atomic photoionisation for the chemistry	`updateCell`	`update_mod.f90:303-311, 2838-2878`
H_2 continuum opacity	`addOpacity`, `yanH2xSec`	`ionization_mod.f90:385-472`
thermal balance, temperature search	`thermBalance`, `iterateT`	`update_mod.f90:1734-2456, 609-1191`
grain photoelectric heating	`setPhotoelHeatCool`	`update_mod.f90:1460-1556`
gas–grain collisional exchange	`setDustGasCollHeatCool`	`update_mod.f90:1562-1731`
CO table and look-up	`initPDRcoolTab`, `pdrCoolAt`	`pdr_mod.f90:2713-2732`; `common_mod.f90:1445-1486`
CO cooling added in the balance	`thermBalance`	`update_mod.f90:2232-2249`
C I, C II, O I atoms with the chemistry's colliders	atom solve; `pdrNeutralRates`	`emission_mod.f90:1861-1887`; `common_mod.f90:1489-1635`
extra heating into the thermal balance	`thermBalance`	`update_mod.f90:2391`
second ionisation solve, kept at the end	`ionSecondSolve`, `ionKeepSecond`	`iteration_mod.f90:3361-3434, 3455-3521`
Convergence		
integrated stop test	`cvTotalsUpdate`	`iteration_mod.f90:3579-3765`; test at `:3243`
per-cell $x(H_2)$ test	`iterateMC`	`iteration_mod.f90:2484-2512`

B Input keywords and defaults on this path

Keywords are read in readInput (set_input_mod.f90); defaults are set there or in common_mod.f90. Retired keywords that once selected ownership or prescriptions are of two kinds (set_input_mod.f90:731-757; the reason for each is printed by retiredKeyword, :1101-1172). Those whose choice is settled or whose value the model now measures (pdrAVfac, pdrUVfac, pdrAllCells, pdrCoolFeedback, pdrHealpix, pdrCoupleIter, pdrChemIter, ...) are still parsed and ignored with a message, so old inputs run. Those that selected alternative physics or stood in for what MOCASSIN knows (pdrZoneLimit, pdrFUVmode, pdrDustModel, pdrOwnDust, pdrPEfromPDR, pdrMetallicity, pdrGrainRadius, ...) are refused and the run stops.

Table 11: Input keywords on the path from the field to the molecules.

Keyword	Default	Effect	Read / default
pdr	off	switches the coupling on	set_input:638
pdrNetwork	reduced	validated against the compiled network; cannot switch it	set_input:645; common:889
pdrDataDir	install prefix	directory of dataPDR (network, coolants, h2/)	set_input:648
pdrStartIter	1	first MC iteration with the PDR coupling	set_input:654; common:893
pdrZeta	$10^{-16}\,\mathrm{s}^{-1}$	cosmic-ray ionisation rate per H_2 (uniform), in the network, in MOCASSIN's ion balance and in the cosmic-ray heating	set_input:651; common:902
pdrEndTime	$10^7\,\mathrm{yr}$	first integration span; steady state extends it up to 128×	set_input:670; common:903
pdrVturb	$1\,\mathrm{km\,s}^{-1}$	microturbulence in b (H_2 lines, coolant lines); velocity of the turbulent heating	set_input:673; common:907
pdrTmin, pdrTmax	$1\,\mathrm{K}$, $2\times10^4\,\mathrm{K}$	clamp of the chemistry's gas temperature (pdr_mod.f90:3346) and range of the CO cooling table (:2720); the vendored routines' own lower bound is raised to T_{CMB} (:364)	set_input:676, 679; common: 908-909
pdrCoolants	none	coolant data files (fixtures: 12co.dat 12c+.dat 12c.dat 16o.dat); CO is tabulated, the atoms' collision rates and escape probabilities go to MOCASSIN's own atoms	set_input:757
iterDamp	0	damping of abundances, ion stages and temperature across iterations	set_input:700; common:1193
teFloor	$2.725\,\mathrm{K}$	floor of MOCASSIN's T_e	set_input:691; common:1278
convTotals	0.02	integrated stop test	set_input:468; common:816
maxIterateMC	30	maximum number of MC iterations (its second number is reported, not used to stop)	set_input:413; : 75
nPhotons	input	packets per iteration; sets the MC noise of every photorate (Section 10.3)	set_input:422

Keyword	Default	Effect	Read / default
randomSeed	−1 (clock)	a non-negative seed makes a run repeatable at fixed rank count	set_input:403; common:734
pathStretch	off	composite path-length biasing (Baes et al., 2016)	set_input:719; common:1102
isotropic Scattering	off	isotropic instead of Henyey–Greenstein dust scattering	set_input:173; :40
quantumHeat Grain	off	stochastic heating of small grains (emitted spectrum only)	set_input:200
symmetricXYZ	off	octant symmetry; mirrors in the walks	set_input:359
nbins	600	frequency bins of the MC field	set_input:77, 350

There is no keyword for the H_2 level model: in the 3D build it always runs, and its Solomon rate always replaces the network's (H-012).

C Data files on this path

The table lists the data files that enter the path from the field to the molecules, with the source of their content. For the H_2 files the source is the one the file's own header gives, completed where the header is not the primary source (Section 6.1); dataPDR/h2/ is a copy of Cloudy c25.00's data/h2/, unmodified.

Table 12: Data files on the path from the field to the molecules.

File	Content and provenance
dataPDR/species_reduced.d	the 33 species, initial abundances (unused: seeding is MOCASSIN's) and masses; 3D-PDR (Bisbas et al., 2012)
dataPDR/rates_reduced.d	332 reactions in UMIST format: number, reactants, products, α, β, γ, source code, $T_{\min}$, $T_{\max}$, accuracy; 331 from 3D-PDR (a subset of UMIST, Woodall et al. 2007 or McElroy et al. 2013, release not recorded; Section 8.1); reaction 332 added in this project with the coefficient of Cloudy's mole_co_umisthack.dat, attributed there to Janev et al. (1987)
dataPDR/12c+.dat, 12c.dat, 16o.dat, 12co.dat	coolant level data in LAMDA format (Schöier et al., 2005); for 16o.dat and 12c.dat the $^1D/^1S \rightarrow {}^3P_J$ collision rates are split by $g_J/9$ (pdr/UPSTREAM.md, D-285)
dataPDR/h2/energy_X.dat	X level energies (302 levels), Komasa et al. (2011, their Table 1), compiled by K. Kaplan
dataPDR/h2/transprob_X.dat	X electric-quadrupole A values, Wolniewicz et al. (1998); magnetic-dipole components of 30 $v = 1 \rightarrow 0$, $\Delta J = 0$ transitions, Pachucki and Komasa (2011), listed separately and summed by the reader
dataPDR/h2/coll_rates_H_15.dat	$H + H_2$ de-excitation rate coefficients, reactive channels included, 100–5000 K, Lique (2015)
dataPDR/h2/coll_rates_He_ORNL.dat	$He + H_2$, 1–10^4 K, the ORNL data set; the header cites "Lee, T. G. et al. 2007, ApJ, in preparation", and no published source for the table was identified

File	Content and provenance
`dataPDR/h2/coll_rates_H2ortho_ORNL.dat`, `coll_rates_H2para_ORNL.dat`	*o*-H_2 and *p*-$H_2 + H_2$, $v = 0$, 1–10^4 K, Wan et al. (2018)
`dataPDR/h2/coll_rates_Hp.dat`	$H^+ + H_2$ including proton exchange, $v = 0$, $J \leq 9$, 3–6000 K, Gerlich (1990)
`dataPDR/h2/energy_{state}.dat`, `transprob_{state}.dat`, `dissprob_{state}.dat`	excited electronic states B, $C^{\pm}$, B', $D^{\pm}$: levels, lines to X, spontaneous radiative dissociation rates and kinetic energies, Abgrall et al. (2000), whose line transition probabilities rest on Abgrall et al. (1993) and Abgrall et al. (1994); the $J = 0$ energies of the Π components that the files carry as comments are extrapolations by Cloudy's authors and are not read
`dataPDR/h2/energy_dissoc.dat`	dissociation energies of the electronic states, Sharp (1970) per the header (which dates it 1971), with values revised in Cloudy in 2002; only the value for X, 36 118.11 cm^{-1}, is used
`dataPDR/h2/README.md`	provenance and Cloudy's licence notice (zlib)
MOCASSIN atomic data (`data/ph1.dat`, `ph2.dat`, `radrec.dat`)	photoionisation cross-section fits (Verner and Yakovlev, 1995; Verner et al., 1996) and radiative recombination fits (Verner and Ferland, 1996, and the sources listed at `update_mod.f90:4101-4115`), used for the linked atoms' rates; the H_2 photoionisation cross section is a fit in the code (Yan et al., 1998, 2001)

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